

Geometry and analysis

Facilitator guide and complete solutions / Remaining atlas entries / September 2026

Give one student page at a time; allow exploration before hints. The accompanying check programs verify finite instances and identities. General statements need the arguments or supplied facts in the keys. Preparation, timing and classroom engagement remain untested.

GA-01	Can a box keep a secret?	5
	Fixed-axis spatial tracking; shortest-word proof; no arithmetic needed	
GA-02	Design a fence that defeats a match	10
	Height comparison and continuity; exact fractions are optional	
GA-03	A cover with a tiny budget	15
	Fractions, geometric sums, and reasoning about an infinite list	
GA-04	A root takes a journey	20
	Half angles and continuous choices; complex notation optional	
GA-05	Can a network hide a high point?	24
	Averages and halves; proof by following a maximum through a graph	
GA-06	What does a real picture miss?	29
	Complex arithmetic and factorization; four-real-dimensional reasoning	
GA-07	Build an angle machine	34
	Polynomial algebra, unit-circle cosine, and cosine addition	
GA-08	Can a rate law choose a future?	38
	Calculus required: derivatives, piecewise joins, and square roots	

Print selected pages, single-sided, on US Letter at actual size. Keep student pages and solutions separate. The original random-ten book remains a separate volume.

Contents continued

Facilitator guide and complete solutions / Remaining atlas entries / September 2026

The listed tools are prerequisite gates, not age labels. More detailed gates appear in the facilitator guide.

GA-09	Choose a shape, then choose its future	44
	Partial derivatives, trigonometry and exponentials; calculus is the core	
GA-10	Program the folded interval	49
	Fractions, affine composition and a proof about repeating words	
GA-12	Can tiny payments break a budget?	54
	Fractions, doubling and a rule that defeats every finite target	
GA-13	How narrow can the error band be?	58
	Graphs, signed fractions, affine formulas and inequalities	
GA-14	Choose a photograph that reveals the motion	62
	Fractions, rotating angles, periodicity and integer remainders	
GA-15	Decode a picture from four sign patterns	67
	Signed arithmetic, four-entry arrays and weighted-sum cancellation	
GA-16	Predict what the sliding window draws	72
	Lengths, interval endpoints and piecewise graphs; integral meaning is supplied	
GA-17	A graph feeds back its own mean	78
	Definite integrals, functions and linear equations are the core	

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Contents continued

Facilitator guide and complete solutions / Remaining atlas entries / September 2026

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GA-18	Choose what best flat height means	82
	Weighted rectangle sums, absolute values and algebraic optimality proofs	
GA-19	Defeat a claimed derivative bound	86
	Polynomial derivatives, supremum norms and an every-bound counterexample	
GA-20	Choose a journey and its effort bill	90
	Weighted squares for schedules; calculus for the proof over all paths	
GA-21	A mirror that is too short	94
	Reflection and triangle inequality; convexity for a constrained optimum	
GA-22	Make a four-peg challenge	100
	Convex-hull pictures and complete cases; fractions for exact certificates	
GA-23	Let the route turn a tangent arrow	105
	Spatial tangent vectors and component tracking; trigonometry comes later	
GA-24	Test a reversible reshaping	112
	Connected pieces and a point-deletion proof; maps are defined explicitly	
GA-27	Find the flood's joining height	117
	Coordinate inequalities and paths; calculus only on the Morse continuation	

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Contents continued

Facilitator guide and complete solutions / Remaining atlas entries / September 2026

The listed tools are prerequisite gates, not age labels. More detailed gates appear in the facilitator guide.

GA-28	How much can one question guarantee?	123
	Fractions, comparing squares, and interval reasoning	
GA-30	Find every shortest route around a cylinder	127
	Spatial unrolling and the Pythagorean theorem	
GA-31	Can every pair agree and everyone disagree?	131
	Ordering endpoints and drawing intervals	
GA-32	Predict the cut before using scissors	136
	Tracing joined edges and counting circuits	
GA-33	Where did the starting value go?	140
	Derivatives, integration by parts, improper integrals and elementary exponential algebra.	
GA-34	Infinitely many doublings before the deadline	144
	Calculus is the core gate: derivatives, rational functions and limits	
GA-35	Can four samples fool the average?	148
	Coordinates, signed squares and algebra for the finite identity	
GA-36	Will feedback find the point that exists?	152
	Fractions, affine algebra and graph reasoning	

Print selected pages, single-sided, on US Letter at actual size. Keep student pages and solutions separate. The original random-ten book remains a separate volume.

Can a box keep a secret?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate with careful physical demonstration. The hidden-orientation task is stronger than merely seeing two different arrows. Spatial load is real; supply a partner recorder and allow three-face checking.

A witnessed order-sensitive pair, followed by a shortest three-turn word that restores the blue direction while changing the red direction.

Prerequisites and preparation

Core: Spatial reasoning: keep three room axes fixed, follow and reverse quarter-turns, compare two marked directions. No arithmetic is required for the physical counterexample. Signed coordinates support recording; proving shortest length needs a small exhaustive case split.

Extensions: Matrix multiplication for the optional complete commuting-powers classification; derivatives are required for any Lie algebra continuation, which is not assigned here.

Reading and representation: Turn cards can be read aloud. X means the room x-axis turn, never the current direction of a box edge. Red and blue marks need distinct shapes as well as colors.

Materials: One small rigid box per pair. Mark adjacent faces R (red) and B (blue); a face points in its outward perpendicular direction. Reset R toward +x and B toward +z. Printed room-axis mat, X/Y and inverse turn cards, pencil. No cutting is essential. Preparation: about 10 minutes.

5-10 minutes free box play; 5 demonstrate the two fixed-axis turns; 15-20 invent and reverse words; 5 movement/reset; 15 seek the hidden twist and justify shortest length. The plane comparison can be a later 15-minute continuation.

Launch and facilitation

With red pointing right (+x) and blue pointing up (+z), choose two turn cards. Predict both marks before moving. Reset all faces, reverse the card order, and try to catch a difference.

Student page 1: Choose two turns

Core: fixed axes and spatial tracking

Reset: the red R face points +x and the blue B face points +z. A face points outward, perpendicular to itself. Room axes stay still. X sends +y to +z and +z to -y; Y sends +z to +x and +x to -z. Both are quarter-turns. A minus card undoes its matching turn. Read a word left to right.

Can a box keep a secret?

1 / prompt and solution

Choose two cards from X, Y, X⁻, Y⁻. Predict where red and blue will point. Carry out your word, reset, then reverse its order. Record one comparison worth showing another pair.

Solution: Answers depend on the chosen word. X then Y sends (red,blue) to $(-z,-y)$; Y then X sends them to $(+y,+x)$. Opposite turns about one axis cancel in either order; repeating the same card is unchanged by reversal. Any two signed quarter-turns on different axes give different orientations. Validate predictions using the printed direction cycles and a complete reset.

1 / hints to offer only after exploration

1. Track blue first, then repeat with red.
2. An inverse card moves backward round the same four-direction cycle.

2 / prompt and solution

Give your partner an exact reset and two short words that look interchangeable but are not. What single observation disproves equality of the two motions?

Solution: For example use the reset above and X Y versus Y X. Blue finishes $-y$ versus $+x$, so the motions cannot be equal. A single differing image is enough to disprove equality; a matching image is not enough to prove equality. Other exact witnesses are valid if the reset and order are stated.

2 / hints to offer only after exploration

1. Compare final directions, not only where the box sits.

Student page 2: One mark returns

Core continuation: inverse moves and a shortest-word argument

Use only X, X⁻, Y and Y⁻. A word may repeat cards. Restore the reset position before every attempt. The challenge concerns the whole orientation, not the box location.

3 / prompt and solution

Find a nonempty word that returns blue to $+z$ but leaves red pointing somewhere else. Then try to use as few cards as possible. Draw or write your best word.

Solution: A shortest word is Y X Y⁻: blue follows $z \rightarrow x \rightarrow x \rightarrow z$ while red follows $x \rightarrow -z \rightarrow y \rightarrow y$. Thus blue returns and red moves to $+y$. Another is X Y X⁻, which also fixes z and sends x to $-y$. Three cards suffice.

Can a box keep a secret?

3 / hints to offer only after exploration

1. Put blue along an axis that the middle turn leaves fixed.
2. Undo your first turn after making a different middle turn.

4 / prompt and solution

Why can no one-card or two-card word win this challenge? Check kinds of words instead of writing all sixteen pairs.

Solution: One X-type or Y-type quarter-turn moves z to a horizontal direction. For two cards on the same axis, opposite signs give the identity (red returns too), and equal signs give a half-turn sending z to $-z$. For two cards on different axes, the first sends z to the other turn's axis, so the second leaves that horizontal direction fixed; blue cannot return. These exhaust one and two cards, proving length three is optimal.

4 / hints to offer only after exploration

1. Separate same-axis pairs from different-axis pairs.

5 / prompt and solution

A partner says, "Blue ended in the same place, so these motions are the same." Repair the test. How many marked directions are enough for a rigid box?

Solution: Track a second nonparallel direction, such as red. A proper rigid rotation is determined by the images of two nonparallel directions: they determine their plane and, by the right-hand orientation, the perpendicular third direction. For a concrete check, mark three distinguishable perpendicular box directions; in the challenge blue agrees but red does not. One direction is insufficient because the box can rotate about it.

5 / hints to offer only after exploration

1. Turn the box around the blue direction while watching red.

Student page 3: What if the turns are flat?

Continuation: angles; signed coordinates optional

Use the supplied flat compass for same-center turns and the line with centers $A=0$ and $B=2$ for half-turns. A half-turn about c sends a point x on the line to $2c-x$.

Can a box keep a secret?

6 / prompt and solution

Choose two angles about the same center, in either direction. Compare both orders. Explain what happens for any two angles, not just yours.

Solution: For angles a and b the final direction has angle $\theta+a+b$ or $\theta+b+a$, equal modulo 360° . The distance from the common center stays fixed. Thus all plane rotations about one center commute, including negative and non-quarter angles.

6 / hints to offer only after exploration

1. Record the total angle from the start.

7 / prompt and solution

Start a dot at $A=0$. Compare a half-turn about A followed by a half-turn about $B=2$, with the reverse order. Which word in your same-center claim matters?

Solution: A then B gives $0 \rightarrow 0 \rightarrow 4$. B then A gives $0 \rightarrow 4 \rightarrow -4$. The results differ. The common-center assumption matters; planar rotations about different centers need not commute.

7 / hints to offer only after exploration

1. Use the line and reflect through the named center each time.

Optional facilitator extension 1

Gate: Matrix multiplication or tracking all three coordinate arrows.

For a, b in $\{0, 1, 2, 3\}$, classify when X^a and Y^b commute.

Solution: They commute exactly when $a=0$, $b=0$, or $a=b=2$. Identity powers commute. The two half-turn matrices are diagonal and commute. Multiplying the remaining signed permutation matrices rules out all other pairs; `ga1-checks.py` exhausts the 16 pairs. This is a complete finite classification, not a classification of arbitrary axes and angles.

Mathematics and prior use

This is the actual action of rotations on a rigid frame. The quarter-turn examples prove noncommutativity in $SO(3)$; the same-axis angle argument reaches arbitrary real angles. No claim about Lie brackets is made.

Prior use: Week 3 explored permutation composition; this uses continuous physical rotations and fixed axes. Actual use of related instances remains unknown.

Can a box keep a secret?

Source: Peter Woit, Quantum Theory, Groups and Representations

§6.1, coordinate-axis rotation matrices, printed p.63 / PDF p.80 (October 2025 version).. Uses the rotation convention only. Hidden-twist minimality, room-mat rules and different-center half-turn comparison are newly derived. Evidence: Inspected the three matrices and the noncommutativity statement online on 2026-09-25; exact quarter-turn calculations checked independently in ga1-checks.py.

<https://www.math.columbia.edu/~woit/QMbook/qmbook-latest.pdf>

Design a fence that defeats a match

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for the core comparison argument, with close attention to opposite spacing and the seam. The exact fractions page is optional; doing its arithmetic is not the payoff.

A learner's own attempted counterexample leads to the argument that a continuous height difference changes sign when opposite markers exchange places.

Prerequisites and preparation

Core: Compare heights and track opposite points moving continuously. The satisfying proof requires reasoning about every intermediate position and the half-turn exchange. Exact crossing locations are a separate fractions/linear-interpolation continuation.

Extensions: Signed fractions for all ramp crossings or bisection error bounds. The real-analysis justification may use the intermediate value theorem; no higher-dimensional theorem is claimed.

Reading and representation: An adult can read all core questions. Use “same, red higher, blue higher” before introducing a signed difference. A jump means missing intermediate heights, not a steep or cornered ramp.

Materials: Printed 12-position repeated profile (two periods), ruler, two linked pencil marks exactly three post spacings apart, blank six-post profile grids. A strip can be left flat; no model fence construction is necessary. Preparation: about 8 minutes.

5-10 minutes move markers freely; 5 establish the seam and separation; 15-20 design attempted escapes; 5 physically exchange places; 15 develop the half-turn argument. Optional exact locations and broken rules add 15-25 minutes.

Launch and facilitation

Six equally spaced posts have heights 0,5,1,4,2,3 around a circle. Straight ramps join neighboring tops, including the last to the first. Choose where opposite red and blue markers begin. Can you move them together while always avoiding equal height?

Student page 1: Try to beat the markers

Core: comparison and opposite positions

The numbered posts are equally spaced. Join neighboring tops by straight ramps, and repeat the profile after six spacings. Red and blue are always exactly three spacings apart. Slide both in the same direction; corners are allowed.

Design a fence that defeats a match

1 / prompt and solution

On the supplied fence, choose a starting position that is not a match. Move the markers and mark a place where their heights seem equal. Explain how you kept them opposite.

Solution: Any nonmatching start is acceptable; the printed linked markers enforce separation three. There are three distinct matching antipodal pairs, located in the first half-period at post coordinates $4/7$, $8/5$, and $7/3$ (coordinate 0 at the height-0 post). Each has its partner three units later. A drawing or bracket is sufficient here; exact locations are reserved for prompt 6.

1 / hints to offer only after exploration

1. Compare only which marker is higher at first.
2. Mark both locations at once.

2 / prompt and solution

Choose six new heights on the blank grid. Make every pair of opposite posts unequal. Join all ramps, including the seam. Try to design a fence whose moving opposite markers never match. Keep one attempted design and mark what happened.

Solution: There are many legal designs, but none defeats a match after continuous ramps are added. For example heights 0,1,2,3,4,5 have unequal opposite posts; along the ramp from post 2 to 3 the opposing ramp goes from 5 to 0, so the difference changes from -3 to $+3$ and vanishes halfway. Validate that the repeated end height agrees and all ramps are present; do not present failed searches as proof.

2 / hints to offer only after exploration

1. A steep ramp is legal. A missing part of the ramp is a different rule.

Student page 2: Explain the exchange

Core proof: continuity and a half-turn

Imagine that red begins higher. The markers move half a turn, without lifting or jumping, until each reaches the other marker's starting place.

3 / prompt and solution

What must be true about which marker is higher at the finish? Explain why a match must occur somewhere on the way. Include the case in which the start is already a match.

Solution: After half a turn red is at blue's old height and blue at red's old height, so red is now lower. A continuously changing difference cannot go from positive to negative without being zero. If red begins lower, reverse the words; if they begin equal, the desired pair is already found. This proves existence for every continuous closed profile, not only straight ramps.

Design a fence that defeats a match

3 / hints to offer only after exploration

1. Follow places first, not numbers.
2. Imagine trying to change from “red higher” to “blue higher” without ever saying “same.”

4 / prompt and solution

Another group checked all six posts and found no equal opposite heights. Does that defeat your argument? Use the printed fence to settle the disagreement.

Solution: No. Opposite post differences for the printed fence are -4 , 3 , -2 and their negatives, none zero. But between posts the ramps contain all intermediate heights and the moving difference crosses zero. A problem allowing only the six isolated posts has no matching pair here; it is a different domain.

4 / hints to offer only after exploration

1. The markers are allowed between the posts.

Student page 3: Test the rules

Continuation: exact locations use fractions

For exact work let x measure post spacings, with $x=0$ at the height-0 post. Opposite locations differ by 3; x and $x+6$ name the same position. For the rule tests, give a precise counterexample or explanation.

5 / prompt and solution

Keep opposition but allow jumps. On one half of the circle set height 0 and on the other half set height 1: $h(t)=0$ for $0 \leq t < 1/2$ and $h(t)=1$ for $1/2 \leq t < 1$, repeated every turn. Can opposite heights match? Where did the proof break?

Solution: They never match: adding half a turn exchanges the two half-open intervals, so the values are always 0 and 1. There are jumps at $t=1/2$ and at the seam $t=0$. The difference can jump from -1 to $+1$ without passing through zero. Both endpoints are assigned by the half-open convention, so no position is omitted.

5 / hints to offer only after exploration

1. Check the seam as well as the middle.

Design a fence that defeats a match

6 / prompt and solution

Optional exact challenge: find every matching pair on the printed straight-ramp fence. Work over $0 \leq x < 3$ so you do not count the same opposite pair twice.

Solution: On the three unit intervals, let u be the distance from the left post. The differences are $-4+7u$, $3-5u$, and $-2+6u$. Their unique zeros are $u=4/7, 3/5, 1/3$; hence $x=4/7, 8/5, 7/3$, each paired with $x+3$. Their common heights are $20/7, 13/5, 2$ respectively. Each difference is a nonconstant affine function, so this list is exhaustive. In turns the positions are $2/21, 4/15, 7/18$, with opposite positions one-half turn later.

6 / hints to offer only after exploration

1. Write each height as start height plus u times its change.
2. Subtract the two ramp heights before solving.

7 / prompt and solution

Replace the circle by a straight trail: height $h(t)=t$ for $0 \leq t \leq 1$. Keep walkers $1/2$ unit apart, with the first at $0 \leq t \leq 1/2$. Must they ever match? Explain which part of the circle argument is lost.

Solution: No: their heights are t and $t+1/2$, always different. They never exchange places while staying in the permitted trail domain and order. Continuity alone is insufficient; the half-turn exchange on a closed circle supplied the sign reversal.

7 / hints to offer only after exploration

1. Compute the height difference once for a general t .

Optional facilitator extension 1

Gate: Fractions and geometric halving.

Starting with a bracket of width $1/6$ turn whose difference has opposite signs, what location guarantee follows from k bisections?

Solution: If an exact midpoint zero is found, its location is exact. Otherwise retain a subinterval with opposite endpoint signs at each step; continuity keeps at least one zero inside. After k steps the bracket width is $1/(6 \cdot 2^k)$ turn, and its midpoint is within $1/(12 \cdot 2^k)$ of some zero. This assumes exact sign readings; physical height uncertainty can prevent certified bisection.

Design a fence that defeats a match

Mathematics and prior use

The circle theorem is an exact consequence of the intermediate value theorem applied to a difference of two heights. Finite posts and discontinuous step profiles deliberately test the hypotheses.

Prior use: New continuous strand; no matching worksheet in Weeks 1-10. The advisory atlas example suggested this mechanism; this exact ramp instance is separately checked.

Source: Jiří Lebl, Basic Analysis I

§3.3, Lemma 3.3.7 and Theorem 3.3.8, PDF pp.132-133.. The source supplies intermediate value reasoning. The six-post profile, designer challenge, exact crossings and missing-hypothesis cases are newly constructed. Evidence: Inspected the sign-change lemma and bisection proof online on 2026-09-25. All piecewise-linear crossings are checked with rational arithmetic.

<https://www.jirka.org/ra/realanal.pdf>

A cover with a tiny budget

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with specific gates: finite entry is easy but the actual theorem requires geometric-series and infinite-list reasoning. Strong mathematical payoff for prepared learners; do not market paper cutting as a complete K-5 version.

A rule that covers every named point within any requested positive budget, followed by the surprising application to a dense list of rational numbers.

Prerequisites and preparation

Core: Halves, interval length, and geometric sums; the satisfying core includes an infinite list with a rule for its n th member. Finite paper play alone does not establish zero length.

Extensions: Enumerating fractions and distinguishing dense from all points for the rational-number continuation. The interval-cover lower bound is supplied explicitly, not proved by a picture.

Reading and representation: Short rules with point labels $p_1, p_2, \dots$; an adult may record chosen centers. Algebraic ε means any positive budget and may first be replaced by $1/100$.

Materials: Printed number line and rational-pair grid, pencil, ruler, optional paper strips for the first few intervals. The printed first few widths are illustrative; no demand to cut unmeasurably small strips.

Preparation: about 5 minutes.

10 minutes finite covering play; 5 state the never-ending opponent; 20 design and justify a budget rule; 5 movement/reset; 15-25 tackle the dense rational list. The final lower-bound contrast may be a separate continuation.

Launch and facilitation

An opponent names points one at a time. Every named point must lie strictly inside an interval you assign to it. You choose the widths and pay the sum of all assigned widths, even when intervals overlap. Can you promise to spend less than $1/100$ while covering an endless numbered list?

Student page 1: Name a point; buy a cover

Core entry: fractions and lengths

An interval has a center and a positive width. It covers its center even when the interval is open (endpoints excluded). You pay the sum of assigned widths; overlap does not refund your budget. The opponent starts with 0, 1, $1/2$, then chooses two more points in $[0, 1]$.

A cover with a tiny budget

1 / prompt and solution

Choose widths and cover these five points for a total cost less than $1/100$. Describe each interval by its center and width. Challenge a partner with your fourth and fifth points.

Solution: A valid universal example gives all five intervals width $1/1000$, centered at the named points; total $5/1000=1/200<1/100$. Center p and width w mean the open interval $(p-w/2, p+w/2)$. Other positive widths are valid when their sum is less than $1/100$. Chosen locations do not affect this arithmetic; intervals may extend beyond $[0, 1]$.

1 / hints to offer only after exploration

1. Choose the widths before worrying about where the centers lie.

2 / prompt and solution

The opponent now promises to keep naming points forever. Before hearing them, invent a rule giving a positive width to point number n . It must keep the total cost below $1/100$, no matter which points are named. Test several terms without pretending the test proves the whole rule.

Solution: One rule is $w_n=1/(100 \cdot 2^{(n+1)})$ for $n=1,2,\dots$. The first widths are $1/400, 1/800, 1/1600, \dots$; center each at p_n . All widths are positive and their infinite sum is $1/200<1/100$. Any rule of positive widths with a rigorously bounded sum below the budget is valid. A finite successful prefix alone does not certify the infinite sum.

2 / hints to offer only after exploration

1. Save a fixed fraction of the remaining money for later.
2. Try halving the next width each time.

Student page 2: Defend the promise

Core: a rule for every numbered point

Keep your rule visible. A list may repeat points and may be ordered by an opponent. Your explanation must work for a point very far down the list and for every positive budget, not merely for the first picture.

3 / prompt and solution

Prove both parts of your promise: every listed point is covered, and the total cost stays below $1/100$. Write a finite partial-sum formula or a geometric argument that also controls the whole tail.

Solution: For the displayed rule p_n is the center of its own positive-width interval, hence lies inside it for every n . The first N widths sum to $(1/200)(1-2^{-N})$, and the remaining tail sums to $1/(200 \cdot 2^N)$. The infinite total is $1/200$. The coverage statement and the sum bound are distinct; neither depends on point spacing. An alternative rule needs the same two justifications.

A cover with a tiny budget

3 / hints to offer only after exploration

1. Name an arbitrary point number n .
2. A diagram of a halved bar can explain why the missing tail shrinks to zero.

4 / prompt and solution

An opponent changes the budget to a positive number B as small as they like. Adapt your rule. Does “a cover for every $B > 0$ ” mean you have one cover whose intervals all have width zero?

Solution: Use $w_n = B/2^{(n+1)}$, giving total $B/2 < B$. Every B produces a different family of positive-width open intervals. It does not produce a single zero-width cover; an open interval of zero width contains no point. The infimum of possible total costs is zero because cost is nonnegative and can be below every positive B .

4 / hints to offer only after exploration

1. Scale every width by the same factor.

5 / prompt and solution

Suppose instead that you pay the same positive width w for each named point forever. Can that assigned-cost rule fit under a fixed finite budget? Would overlapping intervals change this answer?

Solution: No: after N points the assigned cost is Nw , which exceeds any fixed budget for sufficiently large N . Overlap does not reduce assigned cost under the stated rules. This says nothing against a different efficient cover of a special set, and the actual union of repeated overlapping intervals could have small length.

5 / hints to offer only after exploration

1. Keep “sum of purchases” separate from “length of their union.”

Student page 3: The points become dense

Advanced continuation: fractions, lists and a supplied theorem

List fractions row by row: for $q=1,2,3,\dots$ list $0/q, 1/q, \dots, q/q$. Repetitions are allowed. Supplied theorem: any countable collection of open intervals covering every real point of $[0,1]$ has summed lengths at least 1. We use this theorem; a finite drawing is not its proof.

A cover with a tiny budget

6 / prompt and solution

Choose three fractions in $[0,1]$, including one with a large denominator. Find the row and position at which each appears. Explain why every rational number in $[0,1]$ appears at a finite place. Can the tiny-budget rule cover this entire list?

Solution: Every rational in $[0,1]$ has a form p/q with integers $q \geq 1$ and $0 \leq p \leq q$, so it appears in row q after finitely many earlier rows. Row q begins after $(q-1)(q+2)/2$ entries; p/q is at index $(q-1)(q+2)/2 + p + 1$. For example $2/3$ is index 8, $4/7$ index 32, and $37/101$ index 5188. Apply $w_n = B/2^{(n+1)}$ to the list, including repetitions. Thus all rationals in $[0,1]$ are covered within B .

6 / hints to offer only after exploration

1. A large denominator still has a finite row number.

7 / prompt and solution

There are rationals between any two different real points. Does covering them all with total cost below $1/100$ force your intervals to cover every real point too? Use the supplied theorem. What can you say about a point that is left out?

Solution: No. If every real point in $[0,1]$ were covered, the supplied theorem would force total cost at least 1, contradicting a cost below $1/100$. Therefore at least one real point remains uncovered, and every such point in $[0,1]$ is irrational because all rationals are covered. Density does not mean equality with the whole interval. This proves existence, not a formula for a missing point.

7 / hints to offer only after exploration

1. Assume every real point is covered and compare the two bounds.

Optional facilitator extension 1

Gate: Archimedean property and floor/ceiling reasoning.

Why are the rationals dense, even though the covering cost can be tiny?

Solution: Given $a < b$, choose a positive integer q with $q(b-a) > 1$. An integer p exists strictly between qa and qb (choose $\text{floor}(qa) + 1$; the interval length exceeds 1), giving $a < p/q < b$. Density concerns whether every open interval contains a point; outer measure concerns arbitrarily cheap covers. These statements do not conflict.

A cover with a tiny budget

Optional facilitator extension 2

Gate: Use the GA-29 uncountability argument before asserting the example.

Does zero outer measure force a set to be countable?

Solution: No. The previous trial GA-29 constructs the uncountable middle-thirds Cantor set and gives covers of total length $(2/3)^n$ tending to zero. This is a reference to that separately proved construction, not an inference from today's countable-set argument.

Mathematics and prior use

The arbitrary-budget construction proves zero outer measure for countable sets. Applying it to all rationals in $[0,1]$ separates density, cardinality and length without claiming that every zero-measure set is countable.

Prior use: No matching countable-cover worksheet in Weeks 1-10. The accepted GA-29 trial proves the middle-thirds Cantor set uncountable with covers tending to zero; cross-link it to prevent the false converse "zero outer measure means countable." Here the new work is adversarial enumeration, an arbitrary-budget rule, and dense rational coverage. Actual classroom use remains unknown.

Source: Sheldon Axler, Measure, Integration & Real Analysis

§2A, result 2.4, PDF p.30; result 2.14 and its countable-cover lower bound (using Theorem 2.12), PDF pp.34-35.. Uses the countable-cover mechanism with a different strict budget allocation. The opponent protocol, denominator enumeration and contrast with covering every real point are original activity design. Evidence: Inspected results 2.4 and 2.14 and the Heine-Borel step online on 2026-09-25. The student page supplies the interval-cover lower bound as a theorem; it is not inferred from finite experiments.

<https://measure.axler.net/MIRA.pdf>

A root takes a journey

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with specific gates. Dials provide real exploration, but some learners need prolonged practice distinguishing target, both candidate roots and tracked root. Finite checkpoints alone cannot establish continuity.

A one-circuit obstruction justified for every continuous choice, and a parity rule for routes specified by a continuous running angle.

Prerequisites and preparation

Core: Track and halve a running angle, including turns beyond 360° , and reason about a continuous choice between two separated alternatives. Complex notation is optional for the circle model but the no-jump argument is essential.

Extensions: Complex division and continuity for the formal ratio proof; integer multiplication for m th-root circuits. No unrestricted monodromy theorem is assigned.

Reading and representation: A partner can read route instructions aloud. Every signed turn is explicit; “go to 90° ” alone does not specify the route.

Materials: Two large printed circle dials with 45° spokes, movable pencil arrows or fingers, colored/shaped root marks, blank route strip. Dials are sufficient without a fastener. Preparation: about 8 minutes.

10 minutes freely use the angle-doubling machine; 5 state no jumping; 15-20 compare chosen routes; 5 reset; 15 defend the obstruction. The m th-root page is a later continuation if needed.

Launch and facilitation

A root arrow is legal when doubling its angle points at the target arrow. Start both at 0° . Choose how to move the target once counterclockwise around its circle, changing speed or reversing temporarily if you like. Keep a legal root moving without jumps. Where does it end?

Student page 1: Choose a route

Core: angle doubling and halving

A root arrow is legal when doubling its angle gives the target angle, modulo one full turn. There are two legal unit root arrows for each unit target arrow. Start the target and chosen root at 0° . You may turn either way, but the chosen root must move continuously and remain legal at every instant.

A root takes a journey

1 / prompt and solution

Choose a target position that is not 0° or 180° . Find both legal roots and explain why there are exactly two. Test your pair by doubling the angles.

Solution: If the target angle is θ , roots have angles $\theta/2$ and $\theta/2+180^\circ$ modulo 360° . Doubling either gives θ modulo 360° , and solving $2\phi=\theta$ modulo 360° gives precisely these two residues. For $\theta=90^\circ$, roots are 45° and 225° ; any chosen target is checked by the same formula.

1 / hints to offer only after exploration

1. A half-turn of the root becomes a full turn of its square.

2 / prompt and solution

Invent a target route starting at running angle 0° and finishing at 360° . You may stop or reverse along the way. Record a few signed moves, then follow one root without jumping. Predict and check its final direction.

Solution: For any continuous running angle $\theta(s)$ with endpoints 0° and 360° , the root beginning at 0° can be tracked as $\theta(s)/2$, ending at 180° . Example signed target moves $+270^\circ, -180^\circ, +270^\circ$ total $+360^\circ$; corresponding root moves $+135^\circ, -90^\circ, +135^\circ$ total $+180^\circ$. Other paths with these endpoints have the same result. Check intermediate legality, not merely the endpoints.

2 / hints to offer only after exploration

1. Halve every signed move, including a backward move.

Student page 2: Can another choice escape?

Core proof: continuity

Compare two possible no-jump root choices along the same target route. At every moment they are either the same root or opposite roots. A drawing of selected moments suggests an answer; explain what happens between them.

3 / prompt and solution

Could a clever root follow your choice for a while, switch to the other root without a jump, and return to the start after one target turn? Explain why or construct such a motion.

Solution: It cannot switch continuously: the two legal unit roots are always a distance 2 apart. Formally, divide a competing root $v(s)$ by the tracked root $w(s)=\exp(i\pi\theta(s)/360)$, with θ still in degrees. The continuous quotient is always $+1$ or -1 , so it is constant on the time interval. If both start together the quotient stays $+1$. Thus every no-jump root beginning at 1 ends at -1 after one circuit. The constant-sign fact follows from the intermediate value property.

A root takes a journey

3 / hints to offer only after exploration

1. The two legal arrows never merge.
2. How could a continuous quantity change from $+1$ to -1 while taking only those two values?

4 / prompt and solution

Suppose someone claims to assign one root to every target position on the circle, with the assigned root varying continuously as the target moves. Use one circuit to test the claim.

Solution: At the repeated target 1 the single-valued assignment must give the same root at start and finish. But following a continuous choice once around gives the opposite root by prompt 3, whether the initial choice is 1 or -1 . This contradiction rules out a continuous single-valued square root on the entire circle. A rule with a cut may assign a root everywhere but jumps at the cut.

4 / hints to offer only after exploration

1. The target finishes where it started. Must a function give the same value there?

Student page 3: Count the net turns

Continuation: integer turns; optional m th roots

For these tasks routes are supplied by a continuous running angle θ , with $\theta(\text{start})=0$ and $\theta(\text{finish})=360^\circ k$ for an integer k . This is our definition of k net turns for these routes; no theorem about arbitrary loops is needed.

5 / prompt and solution

Choose one route with reversals that restores the root and one that does not. Give a rule using k that decides all these routes, including clockwise circuits.

Solution: The root's total angle is $180^\circ k$. It returns exactly when k is even. For example target moves $+540^\circ, -180^\circ, +360^\circ$ have $k=2$ and restore the root; moves $+540^\circ, -180^\circ$ have $k=1$ and reverse it. Negative k obeys the same parity rule; $k=0$ returns. The proof uses half of the continuous running angle plus the no-switch argument.

5 / hints to offer only after exploration

1. Only the total signed change matters for the endpoint.

A root takes a journey

6 / prompt and solution

Advanced: change the machine so that it triples a root angle. Start with the root at 0° . Find all legal roots for target 120° , and decide how many counterclockwise target circuits first restore your tracked root.

Solution: The three roots at target 120° have angles $40^\circ, 160^\circ, 280^\circ$. Tracking a root divides every running target angle by 3. One target circuit moves the root 120° , so exactly three circuits first restore it. More generally with m th roots, $m \geq 1$, legal roots are $\theta/m + 360^\circ j/m$ for $j=0, \dots, m-1$, and k circuits return the initial root exactly when m divides k . Continuity prevents switching among the finitely separated alternatives.

6 / hints to offer only after exploration

1. For three roots, equal spacing is 120° rather than 180° .

Optional facilitator extension 1

Gate: Limits on a circle or one-sided continuity.

Describe precisely what a principal-angle square-root rule loses.

Solution: Choose target angles $0 \leq \theta < 360^\circ$ and output angle $\theta/2$. It is a single-valued set-theoretic root on the entire circle. As the target approaches 1 from angles near 360° , outputs approach -1 , while the output at 1 is $+1$. Thus it is discontinuous at the cut. Restricting the target to the circle with 1 removed permits a continuous branch.

Mathematics and prior use

The unit-circle squaring map is a two-to-one covering. Explicit angle paths show monodromy and the impossibility of a global continuous square-root choice on the circle; the proof here stays on the circle.

Prior use: Week 3 cycles are an algebraic comparison; the new obstruction concerns continuous root selection around a loop.

Source: Jiří Lebl, Guide to Cultivating Complex Analysis

§1.2 polar multiplication; Example 10.2.6, PDF p.226.. Uses the local-to-global root-continuation issue. The signed route cards, parity classification and no-jump alternative-choice proof are derived for the unit circle. Evidence: Inspected the analytic-continuation example online on 2026-09-25. The classroom model does not claim to prove the source's general continuation theory; exact half-angle routes are checked in `ga1-checks.py`.

<https://www.jirka.org/ca/ca.pdf>

Can a network hide a high point?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for average-ready learners. The shortcut makes the network choice consequential; uniqueness is a separate proof continuation. Signed differences and fractions need adult support for some learners.

A solved shortcut network and a proof that any finite averaging network with a fixed boundary in each connected component cannot hide a new maximum.

Prerequisites and preparation

Core: Averages of two or three numbers, including halves, and reasoning from a largest value. Solving the shortcut network can use simultaneous equations or balancing; the core proof needs all averaging rules to hold at once.

Extensions: Subtracting two complete assignments for uniqueness and boundary sensitivity. The continuum harmonic-function connection requires analysis and is not proved here.

Reading and representation: Rules may be read aloud. Square nodes are fixed boundary values; circles must equal the average of all neighbors. Edges express constraints, not travel or pipes with lengths.

Materials: Printed five-node path and shortcut network, reusable number counters or pencil, blank graph space. Label edges equally; physical edge length has no weight. Preparation: about 5 minutes.

5-10 minutes freely balance one average; 15 solve the path and predict a shortcut; 5 movement/reset; 15 invent attempted hidden peaks and explain failure; 15-20 optional uniqueness and sensitivity. Distribute later pages when ready.

Launch and facilitation

Keep the square endpoints 0 and 12. Choose the three circle values so each is the average of its neighbors. Then add the printed shortcut and predict which circles must rise or fall before recalculating.

Student page 1: Change one connection

Core: averages and halves

At every circle, its number equals the average of all adjacent numbers. Squares are fixed and do not obey this rule. All edges count once with equal weight, regardless of drawing length. The original network is 0-a-b-c-12.

Can a network hide a high point?

1 / prompt and solution

Choose numbers for a, b, c that make every circle obey the rule at the same time. Check your whole assignment. Can equal steps along the path help?

Solution: The assignment is $a=3, b=6, c=9$. Checks: $3=(0+6)/2$, $6=(3+9)/2$, $9=(6+12)/2$. The equations force successive differences equal: $a-0=b-a=c-b=12-c$, so all four differences are 3. Any correct method is acceptable; updating one circle once is not a simultaneous solution.

1 / hints to offer only after exploration

1. If a is halfway between 0 and b , compare those two gaps.

2 / prompt and solution

Now connect a directly to c , keeping every old edge. Before solving, predict which of a, b, c will rise, fall or stay. Find a legal new assignment and explain your prediction.

Solution: The shortcut gives $3a=b+c$, $2b=a+c$, $3c=a+b+12$. Solving gives $a=9/2$, $b=6$, $c=15/2$. Thus a rises, c falls, and b stays. Reflection of the graph interchanges 0 and 12 and a and c , so the new values remain symmetric about 6; direct substitution verifies them. An initial intuitive prediction about stronger averaging across the shortcut is welcome, but the final simultaneous check is needed.

2 / hints to offer only after exploration

1. The middle circle b still has exactly two neighbors.
2. Try keeping values symmetric about 6.

Student page 2: Build a hiding place

Core proof: the greatest-value argument

Draw a finite network of at most six nodes. Choose at least one fixed square in every connected component; use fixed values between 0 and 12. Circles must obey the same averaging rule. Your challenge is to make some circle strictly above 12.

3 / prompt and solution

Try to design a network and legal values that hide a circle above 12. Keep a record of one attempt. If an average fails, mark the failed circle instead of silently changing the rule.

Solution: No legal design satisfying these conditions can succeed. For a useful failed example, draw fixed $0-a-b-12$ and try $a=b=13$; a 's average is 6.5 and b 's is 12.5, so both fail. More elaborate proposals should be checked node by node. The impossibility follows in prompt 4; an unsuccessful search alone is not a proof.

Can a network hide a high point?

3 / hints to offer only after exploration

1. Try making several neighboring circles equally high.

4 / prompt and solution

Assume a legal network has a largest value M greater than every fixed value. At a circle with value M , what must all its neighbors equal? Follow a path to a square and finish the argument. Also explain the lower bound.

Solution: Every neighbor is at most M . Their average equals M only if every neighbor equals M ; one smaller neighbor would make the average smaller. Repeat along a path to a fixed square, which exists in that component. That square would equal M , contradicting M being larger than every fixed value. Apply the same argument to the smallest value, or negate all values, to show no circle is below the minimum fixed value. Finiteness gives an attained maximum; the boundary-in-each-component condition provides the path.

4 / hints to offer only after exploration

1. An average of numbers no bigger than M equals M only in one case.

5 / prompt and solution

Break exactly the boundary condition: add two circles joined to each other, disconnected from every square. Can this pair carry 100 while the old network stays legal? What claim fails?

Solution: Yes. Give both new circles 100; each has one neighbor of value 100, so each averages correctly. Leave the old network solution unchanged. The added component has no fixed boundary, so the global boundary bound and uniqueness fail: any constant can be used on that pair. Disconnectedness alone is not fatal when each component has its own fixed boundary.

5 / hints to offer only after exploration

1. A one-neighbor average is that neighbor's value.

Student page 3: Could two answers both work?

Proof continuation: subtraction of assignments

Use any finite graph with a nonempty fixed boundary in each component. A “solution” means all circles obey the rule simultaneously. You may reuse the maximum and minimum result from the previous page.

Can a network hide a high point?

6 / prompt and solution

Suppose two different assignments solve exactly the same network and fixed values. At each node subtract the second number from the first. What rule do these differences obey? Decide whether both assignments can exist.

Solution: Differences are zero at every fixed square. At a circle, subtraction commutes with averaging, so the difference is the average of neighboring differences. The maximum principle bounds every difference above by 0 and below by 0. Thus all differences are zero and the assignments coincide. This proves at most one solution; it does not by itself prove existence for every graph.

6 / hints to offer only after exploration

1. Apply the earlier argument to differences, which may be negative.

7 / prompt and solution

On the shortcut network, raise only the right square from 12 to 16. Before solving, bound how far any circle can move. Then find the three new values and test your bound.

Solution: Subtract the old solution from the new one. Boundary differences are 0 and 4, so every interior difference lies in $[0,4]$. Scaling the original shortcut solution by $16/12$ gives $a=6, b=8, c=10$, checked by the same averaging equations. Changes are $3/2, 2, 5/2$, all between 0 and 4. The general argument proves increasing one boundary by $\delta \geq 0$ cannot lower an interior value or raise it by more than δ , whenever the two solutions exist.

7 / hints to offer only after exploration

1. The difference network has simpler fixed values than either original network.

Optional facilitator extension 1

Gate: Connected components.

Does the maximum principle require the whole graph to be connected?

Solution: No. It suffices that every connected component has at least one fixed boundary node. Apply the argument separately in each component. A component containing only fixed nodes is harmless. An isolated interior node has no defined neighbor average and is excluded by the rule.

Can a network hide a high point?

Optional facilitator extension 2

Gate: Difference maximum principle; distinguish graph connectivity from interior connectivity.

Can every circle increase strictly when one fixed value rises?

Solution: Not necessarily. A circle in a different component does not change. Even in a connected whole graph, another fixed square can separate its interior component from the changed boundary. Example path fixed 4-a-fixed 0-b-fixed 0: raising the left fixed value to 8 raises a from 2 to 4 but leaves b=0. Strict influence needs an interior path to the changed boundary that does not pass through other fixed squares.

Mathematics and prior use

This is a discrete Dirichlet problem. The maximum propagates along neighbors, yielding both a maximum principle and uniqueness. It shares the mechanism of harmonic potential theory without asserting convergence to a continuum model.

Prior use: Week 10 supplies network familiarity. The accepted AP-01 trial already develops an averaging chance skyline on a path, equal-gap uniqueness, and a ferry preserving the skyline. Here the path is only a reset: adding an undirected a-c edge changes the interior values, learner-designed networks lead to maximum propagation, and subtraction proves graph-wide uniqueness and boundary sensitivity. Skip the path calculation for learners returning from AP-01. Actual classroom use remains unknown.

Source: John Hunter, Notes on Partial Differential Equations

§2.3, Theorems 2.15, 2.17 and 2.18, printed pp.26-28 / PDF pp.32-34.. The source gives continuous maximum and uniqueness principles. All graphs and the finite maximum-propagation proof are independently supplied; the graph is a counterpart rather than a discretization theorem. Evidence: Inspected the maximum/minimum and difference-of-solutions arguments online on 2026-09-25. Rational Gaussian elimination verifies the new path, shortcut and sensitivity instances.

https://math.ucdavis.edu/~hunter/pdes/pde_notes.pdf

What does a real picture miss?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced/specialist. The algebra choices are genuine and the real-slice surprise is substantial. A four-real-dimensional object cannot be faithfully supplied as an ordinary crossing drawing; the paired diagrams are deliberately honest.

All solutions are parameterized and distinguished from their real slice; a later change of coordinates exposes why the zero equation has two branches while the constant-one equation does not split that way.

Prerequisites and preparation

Core: Complex arithmetic including $i^2 = -1$, factorization, and two complex coordinates. The core payoff is an exhaustive solution set with two branches and a proof that a real slice can conceal almost all solutions. This is not an elementary-age activity.

Extensions: Continuity for the no-branch-switch proof; a supplied necessary local-disc condition for the optional complex-curve singularity proof. No calculus is needed for the explicit $uv=c$ parameterizations.

Reading and representation: Symbol-heavy. Read z and w as two separate complex numbers, hence four real coordinates in total. Paired Argand diagrams show each coordinate, not an embedding of the entire solution set in a plane.

Materials: Two paired Argand grids per investigation, colored/shaped markers for linked z, w choices, pencils. A final blank u, v pair supports the change of coordinates. Preparation: about 5 minutes.

5-10 minutes choose complex inputs and multiply by i ; 20 solve and classify branches; 5 reset; 15 inspect real slices and continuous paths; 15-25 compare the constant-one equation. The manifold extension needs a separate discussion.

Launch and facilitation

Choose a complex number z and find every complex w satisfying $z^2 + w^2 = 0$. Draw the paired coordinates. Give a partner a different z , then ask what a picture restricted to real z and w would show.

Student page 1: Choose the coordinates

Core: complex algebra

Solve $z^2 + w^2 = 0$ over the complex numbers. A point in this problem is a pair (z, w) , not a single point on one ordinary graph. The two Argand grids record its two coordinates.

What does a real picture miss?

1 / prompt and solution

Choose $z=1+i$ or $z=2-i$, then find every possible w . Plot each linked pair on the two grids. Challenge a partner with a third z of your own.

Solution: For $z=1+i$, $w=-1+i$ or $1-i$. For $z=2-i$, $w=1+2i$ or $-1-2i$. For any chosen $z=a+bi$, the complete possibilities are $w=-b+ai$ and $w=b-ai$, coinciding only for $z=0$. Each gives $w^2=-z^2$. Completeness is justified by factorization in prompt 2.

1 / hints to offer only after exploration

1. First try multiplying your z by i .
2. Try the negative of a solution as well.

2 / prompt and solution

Find a formula describing every solution pair. Prove that no others are missed, and determine exactly where your two formulas give the same pair.

Solution: $z^2+w^2=(w-iz)(w+iz)$. Since complex numbers have no zero divisors, the product is zero exactly when $w=iz$ or $w=-iz$. Thus the branches are (z,iz) and $(z,-iz)$ for arbitrary $z\in\mathbb{C}$. They meet when $iz=-iz$, hence $2iz=0$ and $z=w=0$. Both directions of the factorization argument establish an exhaustive classification.

2 / hints to offer only after exploration

1. Factor as a difference of two squares using $(iz)^2=-z^2$.

Student page 2: Restrict the view

Core continuation: real slices; path question uses continuity

The two branches each have a freely chosen complex parameter. A restriction to real values of some coordinates is a slice. It is not a picture of every complex solution.

3 / prompt and solution

Find all solutions when both z and w must be real. Then allow z to stay real but w to be complex. Describe both slices and explain why the first picture could be misleading.

Solution: If both are real, $z^2, w^2 \geq 0$, so only $(0,0)$ solves the equation. If $z=x\in\mathbb{R}$ and w may be complex, solutions are (x,ix) and $(x,-ix)$ for any real x . Thus the all-real slice is one point, but every complex neighborhood of $(0,0)$ contains nonzero solutions on both branches. An isolated point in a real slice does not establish an isolated complex zero.

What does a real picture miss?

3 / hints to offer only after exploration

1. Use nonnegativity only when the coordinates really are real.

4 / prompt and solution

Begin at $(1, i)$ and try to move continuously through solutions to $(1, -i)$, without ever passing through $(0, 0)$. Can a route exist? If you allow the origin, give an exact route.

Solution: No origin-avoiding route exists. Away from the origin, $z \neq 0$ and the continuous ratio w/z is always $+i$ or $-i$. It cannot change between these separated values on a connected time interval. Allowing the origin, for $0 \leq s \leq 1/2$ take $(z, w) = (1-2s, i(1-2s))$; for $1/2 \leq s \leq 1$ take $(2s-1, -i(2s-1))$. The two pieces meet at $(0, 0)$ and remain on the respective branches.

4 / hints to offer only after exploration

1. Can w/z change continuously between its two allowed values?

Student page 3: Change the equation

Advanced algebra continuation: an invertible complex change of coordinates

Set $u = w - iz$ and $v = w + iz$. Both new coordinates are still complex. Investigate $z^2 + w^2 = 1$ with this new notation; the equation no longer asks for a product to be zero.

5 / prompt and solution

Compute uv , then recover z and w from u and v . Use this to give every solution of $z^2 + w^2 = 1$ with one freely chosen parameter. State any forbidden parameter value.

Solution: $uv = z^2 + w^2$. Adding and subtracting give $w = (u+v)/2$ and $z = (v-u)/(2i)$, so the change is invertible. For $uv = 1$ choose any $u \in \mathbb{C} \setminus \{0\}$, set $v = 1/u$, and use those inverse formulas. Conversely every solution has $u \neq 0$ and $v = 1/u$, so none are missed. Unlike $uv = 0$, neither coordinate may vanish.

5 / hints to offer only after exploration

1. Add the definitions; then subtract them.

What does a real picture miss?

6 / prompt and solution

Choose $u=2$ or $u=i$ and construct the corresponding (z,w) . Can you give a continuous route through constant-one solutions from $u=1$ to $u=-1$ without meeting a branch junction?

Solution: For $u=2$, $v=1/2$ and $(z,w)=(3i/4, 5/4)$, giving $-9/16+25/16=1$. For $u=i$, $v=-i$ and $(z,w)=(-1,0)$. A route $u=e^{i\theta}$, $v=e^{-i\theta}$, $0 \leq \theta \leq \pi$, gives $z=-\sin \theta$ and $w=\cos \theta$, a real semicircle from $(0,1)$ to $(0,-1)$. The parameter never reaches zero and the inverse coordinate map is continuous; every nonzero u has a unique paired v , so there is no meeting of two branches in this parameterization. This explicit route does not rely on a planar drawing of C^2 .

6 / hints to offer only after exploration

1. Move u halfway around the unit circle and set $v=1/u$.

Optional facilitator extension 1

Gate: Topology of a punctured disc and the supplied local-disc necessary condition for a smooth complex curve.

Why is the origin of $z^2+w^2=0$ not a smooth complex curve point? Use this supplied necessary condition: a smooth complex curve has a neighborhood homeomorphic to an open disc in C .

Solution: Every sufficiently small relative neighborhood of the origin contains nonzero points from both branches. Removing the origin separates it into the two nonempty sets $w=iz$ and $w=-iz$; the ratio $w/z=\pm i$ separates them. If a disc neighborhood existed, take a smaller disc inside its coordinate chart. Removing its center would leave a connected punctured disc, while its image has the two separated branch pieces, a contradiction. Away from the origin only one branch is locally present and z is a complex coordinate. This proves the singularity using the explicit local structure; a vanishing gradient alone would not suffice for an arbitrary defining equation.

Optional facilitator extension 2

Gate: Complex rational functions and the meaning of a holomorphic coordinate.

For a nonzero complex constant c , parameterize all solutions of $z^2+w^2=c$.

Solution: Use $u \in C \setminus \{0\}$, $v=c/u$, $w=(u+c/u)/2$ and $z=(c/u-u)/(2i)$. The formulas and inverse $u=w-iz$ give a global complex parameterization by C^* . In particular this is a smooth complex curve; the explicit holomorphic charts certify that statement without relying on a gradient test.

What does a real picture miss?

Mathematics and prior use

The polynomial zero set is a complex analytic variety. An invertible linear change gives $uv=0$, while $uv=1$ has a single $\mathbb{C}^*$ parameterization. The optional singularity argument uses local topology, not merely a vanishing gradient.

Prior use: No existing week supplies this complex-variable content; this deliberately keeps a substantial algebra prerequisite.

Source: Jiří Lebl, Tasty Bits of Several Complex Variables

§6.5, Definition 6.5.5 and Example 6.5.6, PDF pp.184-185.. Adapts the source's intersecting complex lines from a minus-sign to a plus-sign equation by a complex linear coordinate change. The $uv=c$ comparison and chosen numerical examples are derived here. Evidence: Inspected the local manifold/regular-point definition and explicit intersecting-branch example on 2026-09-25. Exact Gaussian-rational checks verify all printed examples and the inverse coordinate formulas.

<https://www.jirka.org/scv/scv.pdf>

Build an angle machine

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with specific algebra/trigonometry gates. The choice of informative versus redundant tests and the false-candidate challenge are substantive; circle measuring alone is not the core investigation.

An invented triple-angle polynomial with a proof for every real angle, followed by a composition rule that connects angle multiplication to polynomial substitution.

Prerequisites and preparation

Core: Cosine as a unit-circle horizontal coordinate, signed fractions, and polynomial algebra. The satisfying proof uses cosine addition identities or their supplied symmetric form; finite numerical checks alone do not prove the machine works.

Extensions: Polynomial composition and the fact that a nonzero polynomial has finitely many roots for the all-real composition identity. No calculus or approximation-theory theorem is assigned.

Reading and representation: Symbolic text. The machine input is $x = \cos \theta$, not the angle θ . Start with a circle projection before using the variable x .

Materials: Unit-circle diagram with selected angle rays, two-column blank input/output workspace, graph paper for an optional graph. No electronic calculator is necessary. Preparation: about 5 minutes.

10 minutes choose angle tests on the circle; 15 build a candidate cubic; 5 reset; 15-20 replace finite evidence with a recurrence proof; 15-20 compose machines. The last page is a natural separate continuation.

Launch and facilitation

An angle machine sees only the horizontal coordinate $x = \cos \theta$. You want it to output $\cos(3\theta)$. Choose angle tests that might determine an odd cubic $ax^3 + bx$, then try to break your candidate.

Student page 1: Choose useful tests

Core: unit-circle cosine and algebra

The input x is the horizontal coordinate of an angle arrow: $x = \cos \theta$. The desired output is $\cos(3\theta)$. Start by searching among odd cubics $P(x) = ax^3 + bx$. The printed rays include $0^\circ, 60^\circ, 90^\circ, 120^\circ, 180^\circ$; select tests deliberately.

Build an angle machine

1 / prompt and solution

Choose two of the printed angles and use them to determine a and b , if you can. If your tests do not determine them, explain why and choose a more useful pair.

Solution: The 0° and 180° tests both give $a+b=1$. The 60° and 120° tests both give $a+4b=-8$ (after multiplying and accounting for sign). The 90° test gives $0=0$ and no information. Thus one endpoint test and one of $60^\circ/120^\circ$ determine $a=4, b=-3$. Any pair within one class, or involving 90° with only one informative class, is insufficient. This explains all pairs among the five offered angles.

1 / hints to offer only after exploration

1. At 90° , any odd cubic outputs zero.
2. Do your two tests really give two different equations?

2 / prompt and solution

Choose another angle to challenge your candidate. Also compare θ and $-\theta$: do equal inputs ask the machine for conflicting outputs? State what these checks do and do not prove.

Solution: For example $\theta=45^\circ$ gives $x=\sqrt{2}/2$ and $4x^3-3x=-\sqrt{2}/2=\cos 135^\circ$. θ and $-\theta$ have the same input and equal desired outputs because cosine is even. In general angles sharing a cosine are congruent to $\pm\theta$ modulo 360° , so their triple-angle cosines agree. The checks support a candidate but do not establish the identity for every angle.

2 / hints to offer only after exploration

1. Try an angle whose input is not 0, $\pm 1/2$ or ± 1 .

Student page 2: From evidence to an identity

Core proof: cosine addition and polynomial recurrence

You may use $\cos(A+B)+\cos(A-B)=2\cos A \cos B$. Define $T_0(x)=1$, $T_1(x)=x$, and $T_{n+1}(x)=2xT_n(x)-T_{n-1}(x)$ for $n \geq 1$.

3 / prompt and solution

Use the recurrence to build T_2 and T_3 . Then explain why $T_n(\cos \theta)=\cos(n\theta)$ for every integer $n \geq 0$ and every real θ .

Solution: $T_2=2x^2-1$ and $T_3=4x^3-3x$. The identity holds for $n=0$ and $n=1$. If it holds for n and $n-1$, substitute $x=\cos \theta$ into the recurrence and use the supplied cosine identity with $A=n\theta$ and $B=\theta$ to obtain $\cos((n+1)\theta)$. Induction proves every n . Each recurrence step uses addition and multiplication of polynomials, so every T_n is a polynomial.

Build an angle machine

3 / hints to offer only after exploration

1. Make the supplied addition identity look like “next = $2 \times \text{input} \times \text{current} - \text{previous}$.”

4 / prompt and solution

A rival adds $x(x^2-1)(x^2-1/4)$ to your cubic. Show why the new polynomial passes all five original angle tests but still fails somewhere. What distinguishes a fitted guess from your proof?

Solution: The added term vanishes at $x=0, \pm 1, \pm 1/2$, which are exactly the five distinct original inputs. At $x=\sqrt{2}/2$ ($\theta=45^\circ$), the added term is $(\sqrt{2}/2)(-1/2)(1/4)=-\sqrt{2}/16$, nonzero; thus it fails the desired output there. The recurrence proof covers every angle; agreement at finitely many inputs does not certify an unrestricted polynomial identity unless a sufficient degree bound is also established. The rival polynomial has degree five, outside the original cubic search class.

4 / hints to offer only after exploration

1. Factor values make the five successful checks immediate.

Student page 3: Feed one machine into another

Advanced continuation: composition and polynomial identities

A doubling machine has polynomial T_2 ; a tripling machine has T_3 . Feeding one machine's output into the next means substituting one polynomial for x in the other.

5 / prompt and solution

Predict what $T_2(T_3(x))$ and $T_3(T_2(x))$ do when $x=\cos \theta$. Choose one order to expand, and compare with a partner who expands the other.

Solution: Both give $\cos(6\theta)$ on $x \in [-1, 1]$. Direct expansion gives $T_2(T_3(x))=2(4x^3-3x)^2-1=32x^6-48x^4+18x^2-1$. Also $T_3(T_2(x))=4(2x^2-1)^3-3(2x^2-1)$, the same polynomial. Order does not matter for these particular multiplier machines.

5 / hints to offer only after exploration

1. Think about the angle before expanding the algebra.

Build an angle machine

6 / prompt and solution

State and prove a rule for $T_m(T_n(x))$. First work on $-1 \leq x \leq 1$. Supplied algebra fact: a nonzero polynomial of degree d has at most d distinct real roots. Can your rule extend outside the circle-input interval?

Solution: For $x = \cos \theta$, $T_m(T_n(x)) = T_m(\cos(n\theta)) = \cos(mn\theta) = T_{mn}(x)$. Every $x \in [-1, 1]$ is a cosine. The difference of the two polynomials has infinitely many roots in that interval, so by the supplied algebra fact it is identically zero. Thus $T_m \circ T_n = T_{mn}$ for all real x (indeed complex x), with integers $m, n \geq 0$. For x outside $[-1, 1]$, the polynomial identity remains true though there is no real θ with that cosine.

6 / hints to offer only after exploration

1. Subtract the two sides after proving agreement on the whole interval.

Optional facilitator extension 1

Gate: Factoring and square roots; radians optional.

Find the solutions of $T_3(x) = 0$ in $[-1, 1]$ using either angles or algebra.

Solution: $4x^3 - 3x = x(4x^2 - 3)$, so $x = 0, \pm\sqrt{3}/2$. In angle form $0 \leq \theta \leq \pi$ and $\cos(3\theta) = 0$ give $\theta = \pi/6, \pi/2, 5\pi/6$, yielding the same three roots. These exhaust the interval and, by factorization, all real roots.

Optional facilitator extension 2

Gate: Function input notation.

Why is $T_3(x)$ not $\cos(3x)$?

Solution: The variable x is a coordinate, not an angle. At $x = 0$, $T_3(0) = 0$ whereas $\cos(3 \cdot 0) = 1$, in either radians or degrees. The correct identity is $T_3(\cos \theta) = \cos(3\theta)$.

Mathematics and prior use

The machine is the Chebyshev polynomial T_3 . A trigonometric recurrence constructs all T_n ; composition turns multiplication of indices into substitution of polynomials.

Prior use: Week 3 repeated actions is a conceptual comparison; polynomial functions and projections create new substance.

Source: NIST Digital Library of Mathematical Functions

DLMF §18.5(i), equation 18.5.1; recurrence relations §18.9(i).. The source supplies the identity $T_n(\cos \theta) = \cos(n\theta)$. Test selection, fake-fit polynomial, derived recurrence and chosen composition examples are developed here. Evidence: Inspected equation 18.5.1 and the recurrence section on 2026-09-25. Exact polynomial coefficient arithmetic checks T_2, T_3, T_6 and the deceptive higher-degree fit.

<https://dlmf.nist.gov/18.5>

Can a rate law choose a future?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced/specialist, explicitly calculus-gated in accordance with user feedback. The freedom to choose a waiting time is a real solution choice, and the join check is indispensable. Classification strengthens the counterexample without disguising its prerequisites.

At least two fully checked different solutions with identical initial data, including a chosen delayed departure and its differentiability at the join. The advanced continuation classifies every nonnegative solution.

Prerequisites and preparation

Core: Calculus is essential: derivatives of powers, one-sided difference quotients at a join, square roots, and the meaning of a differential equation at every time. A sketch or slope game alone is not the satisfying investigation.

Extensions: Chain and product rules plus monotonicity for the complete classification and comparison equation. Local Lipschitz continuity is defined on the final page; its uniqueness theorem is discussed as a supplied result, not proved here.

Reading and representation: Symbolic piecewise functions. $y(t)$ is nonnegative height, $t \geq 0$ is time, and $y'(t)$ is the instantaneous derivative. At $t=0$ use the right derivative; at positive join times use both sides.

Materials: Three blank time-height axes, pencil, optional graphing tool. Calculus-capable learner and facilitator; no manipulative substitution is claimed to lower the prerequisite. Preparation: about 5 minutes.

10-15 minutes invent and test power-law candidates; 20 design and verify a delayed departure; 5 reset; 20 classify every nonnegative future. The comparison and Lipschitz page may take another 20-30 minutes.

Launch and facilitation

For every $t \geq 0$ obey $y' = 2\sqrt{y}$, with $y \geq 0$ and $y(0)=0$. Can two different differentiable motions obey all these instructions? Choose a candidate and let a partner attack its weakest point.

Student page 1: Invent a motion

Calculus core: derivatives of powers

A solution must be defined for all $t \geq 0$, be nonnegative, have $y(0)=0$, and satisfy $y'(t)=2\sqrt{y(t)}$ at every time. Use the right derivative at $t=0$. The equation specifies an instantaneous rate, not an update step.

Can a rate law choose a future?

1 / prompt and solution

Search among $y(t)=at^2$ and $y(t)=at^4$ with $a \geq 0$. Which choices of a obey the rate law for every time? Use them to find different futures from the same start.

Solution: For at^2 , $y'=2at$ and $2\sqrt{y}=2\sqrt{a}\cdot t$ for $t \geq 0$. Equality for every $t > 0$ requires $a=\sqrt{a}$, hence $a=0$ or 1 . Both obey the initial condition and the right-derivative condition at 0 . For at^4 , $y'=4at^3$ and $2\sqrt{y}=2\sqrt{a}\cdot t^2$; if $a > 0$ equality would force $t=1/(2\sqrt{a})$ for every $t > 0$, impossible. Thus only $a=0$ works in that family. We already have $y=0$ and $y=t^2$, which differ for every $t > 0$.

1 / hints to offer only after exploration

1. Check all positive t , not one convenient instant.
2. Remember $\sqrt{(t^2)}=t$ only because $t \geq 0$ here.

2 / prompt and solution

Both of your valid futures have slope 0 at the start. Explain why that fact does not force the graph to remain flat. What would count as a complete check of another proposed future?

Solution: A zero derivative at one time controls the tangent there, not later values. $y=t^2$ has right derivative 0 at 0 but is positive afterward and satisfies the equation everywhere. A complete check verifies the domain, nonnegativity, initial value, differentiability everywhere (including any joins), and equality of derivative and $2\sqrt{y}$ at every time. Matching a plotted slope field at selected points is evidence, not that check.

2 / hints to offer only after exploration

1. The derivative is an instantaneous statement.

Student page 2: Choose when to leave

Calculus core: piecewise functions and the join

Choose a waiting time $c > 0$, for example $3/2$ or 2 , and require $y(t)=0$ until $t=c$. Your job is to invent the rest of the function and defend the exact moment at which it starts moving.

3 / prompt and solution

Find a formula for a motion that waits until your c , then becomes positive. Verify the rate law on both open pieces $t < c$ and $t > c$.

Solution: Use $y_c(t)=0$ for $0 \leq t \leq c$ and $y_c(t)=(t-c)^2$ for $t \geq c$. On $t < c$ the derivative and right side are both 0 . On $t > c$, $y'=2(t-c)$ and $2\sqrt{y}=2|t-c|=2(t-c)$. The pieces agree in value at c . This verifies all times except the differentiability and equation check at the join, completed next. Any proposed alternative needs the same exact checks.

Can a rate law choose a future?

3 / hints to offer only after exploration

1. Translate a valid nonconstant future horizontally.
2. Use $t-c$ for the time since departure.

4 / prompt and solution

At $t=c$, calculate the difference quotient $[y(c+h)-y(c)]/h$ separately for $h<0$ and $h>0$. Is the joined motion differentiable, and does it obey the rate law at the join?

Solution: For $h<0$ sufficiently small to stay in the domain, the quotient is 0. For $h>0$ it is $h^2/h=h$. Both limits are 0, so $y'(c)=0=2\sqrt{y(c)}$. Thus the full piecewise motion is differentiable and satisfies the IVP. At $c=0$ the same construction is t^2 on its domain and the right derivative is 0. Waiting forever gives the separate solution $y\equiv 0$.

4 / hints to offer only after exploration

1. Agreement of the function values is not yet agreement of the derivative limits.

5 / prompt and solution

A partner tries the single formula $y=(t-c)^2$ for all $t\geq 0$, with $c>0$, avoiding a piecewise definition. Identify every obstruction to calling this a solution of our initial-value problem.

Solution: First $y(0)=c^2\neq 0$. Also for $0\leq t<c$, its derivative $2(t-c)$ is negative while the required rate $2\sqrt{y}=2|t-c|$ is positive (except at c). At $t=c$ the equation is true, and for $t>c$ it is true, but those partial successes do not repair the initial condition or earlier times.

5 / hints to offer only after exploration

1. Do not simplify $\sqrt{((t-c)^2)}$ to $t-c$ when $t<c$.

Student page 3: Are there stranger futures?

Advanced calculus continuation: chain rule and monotonicity

Consider any nonnegative differentiable solution on all $t\geq 0$. You may use: a function with nonnegative derivative is nondecreasing, and a function with derivative 1 on an interval equals t plus a constant.

Can a rate law choose a future?

6 / prompt and solution

Could a solution leave zero, return to zero, and leave again? Explain the possible shape of its set of zero times.

Solution: No: $y' = 2\sqrt{y} \geq 0$, so y is nondecreasing. Once positive, it stays positive. Since y is continuous, the zero times form either all $[0, \infty)$ or an initial closed interval $[0, c]$ for some finite $c \geq 0$; $c=0$ includes immediate departure. For the latter case, if c is the supremum of the zero times, continuity gives $y(c)=0$ and every $t > c$ has $y(t) > 0$.

6 / hints to offer only after exploration

1. Use the sign of the derivative before manipulating square roots.

7 / prompt and solution

On any interval where $y > 0$, set $v(t) = \sqrt{y(t)}$. Find v' . Use continuity at the last zero c to classify every nonnegative solution of the original problem.

Solution: For $t > c$, the chain rule gives $v' = y'/(2\sqrt{y}) = 1$. Thus $v = t + K$ on (c, ∞) . As t decreases to c , continuity gives $v \rightarrow 0$, so $K = -c$. Hence $y = (t - c)^2$ after c and $y = 0$ before it. Along with $y \equiv 0$, these are all nonnegative differentiable solutions on $[0, \infty)$. The calculation divides by $\sqrt{y}$ only on the strictly positive interval; it does not divide at zero. Prompt 4 proves every classified candidate really is a solution.

7 / hints to offer only after exploration

1. First restrict to positive y so division is legal.

Student page 4: What changes the answer?

Advanced calculus: product rule and a stated uniqueness hypothesis

For a scalar rate function f , local Lipschitz at 0 would require constants $L, \delta > 0$ with $|f(a) - f(b)| \leq L|a - b|$ for all permitted a, b within δ of 0. A standard uniqueness theorem uses local Lipschitz continuity (plus continuity in time). We inspect the hypothesis here, not prove that theorem.

8 / prompt and solution

Change the rate law to $y' = y$ with $y(0) = 0$. Differentiate $e^{-(t)}y(t)$ and determine every solution. Does a delayed departure remain possible?

Solution: The product rule gives $(e^{-(t)}y)' = e^{-(t)}(y' - y) = 0$. Therefore $e^{-(t)}y$ is constant, and its value at 0 is 0; thus $y \equiv 0$. This argument proves uniqueness directly for differentiable solutions, so no positive delayed departure is possible. The rate $f(y) = y$ also satisfies a Lipschitz bound with $L = 1$.

Can a rate law choose a future?

8 / hints to offer only after exploration

1. A product with zero derivative is constant.

9 / prompt and solution

For $f(y)=2\sqrt{y}$ on $y\geq 0$, compare $f(h)-f(0)$ with h as $h>0$ tends to 0. Then test the claim “every non-Lipschitz rate gives several nonnegative forward futures” using $y'=-2\sqrt{y}$, $y(0)=0$.

Solution: The ratio $|f(h)-f(0)|/h=2/\sqrt{h}$ grows without bound, so no local L works even on the nonnegative half-line. Failure of this sufficient uniqueness hypothesis alone does not force nonuniqueness. For $y'=-2\sqrt{y}$ with nonnegative y and $t\geq 0$, $y'\leq 0$ makes y nonincreasing; from initial value 0 and nonnegativity it follows $y\equiv 0$. The negative-sign rate has the same unbounded absolute Lipschitz ratio but a unique nonnegative forward future. The domain and forward-time restriction are part of that counterexample.

9 / hints to offer only after exploration

1. To disprove a universal claim, one carefully stated rate law suffices.

Optional facilitator extension 1

Gate: Derivatives of real powers and chain rule.

For $0<\alpha<1$, construct delayed solutions of $y'=y^\alpha$, $y(0)=0$, $y\geq 0$.

Solution: For $c\geq 0$ use $y=0$ up to c and $y=[(1-\alpha)(t-c)]^{1/(1-\alpha)}$ afterward. The exponent $\beta=1/(1-\alpha)>1$ makes the right difference quotient at c tend to 0. On $t>c$, direct differentiation yields $y'=y^\alpha$. The coefficient is essential. Waiting forever also works. The same monotonicity and positive-region transformation $y^{1-\alpha}$ classify all nonnegative forward solutions.

Optional facilitator extension 2

Gate: Separation or direct differentiation of a rational function; one-sided blow-up.

Does a unique local future necessarily exist for all time?

Solution: No. For $y'=y^2$ with $y(0)=1$ the solution $y=1/(1-t)$ exists only for $t<1$ and grows without bound as $t\rightarrow 1$ from below. Uniqueness and global existence are distinct; this connects to GA-34 and is not an additional conclusion about the square-root example.

Can a rate law choose a future?

Mathematics and prior use

The initial-value problem fails uniqueness. The square-root right side is continuous but not locally Lipschitz at zero; the example demonstrates why an existence/uniqueness theorem has substantive hypotheses. Non-Lipschitz does not itself imply nonuniqueness.

Prior use: No existing worksheet covers rate-law uniqueness; this is intentionally a calculus-gated extension upward.

Source: Gerald Teschl, Ordinary Differential Equations and Dynamical Systems

§2.2, Theorem 2.2 (Picard-Lindelöf), PDF p.49; Problem 2.6(ii), PDF p.50.. Uses the local Lipschitz uniqueness hypothesis and the square-root diagnostic. Every waiting solution, the join check, the full nonnegative classification and the negative-sign comparison are independently derived. The source theorem is stated on an open domain. Our half-line example is checked directly; extending the rate to $2\sqrt{\max(y,0)}$ on $\mathbb{R}$ gives the same failure of the Lipschitz hypothesis if an open-domain comparison is desired. Evidence: Inspected the uniform local Lipschitz hypothesis and square-root problem online on 2026-09-25. Algebraic identities and one-sided quotient forms are checked in ga1-checks.py; written calculus arguments carry the universal proof.

<https://www.mat.univie.ac.at/~gerald/ftp/book-ode/ode.pdf>

Choose a shape, then choose its future

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced/specialist. Strong for learners already comfortable with derivatives; moving a graph alone is not a satisfying substitute for the mathematical core. Unpiloted.

Two distinct fixed-end wave futures with the same initial shape, plus the derivative calculation explaining what initial velocity adds.

Prerequisites and preparation

Core: Calculus is the core: partial derivatives, sine/cosine and exponentials. Graph sliding is an entry, not a proof of a PDE.

Extensions: Energy comparison uses definite integrals and integration by parts.

Reading and representation: Formula reading in two variables; distinguish holding x fixed from holding t fixed. Signed amplitudes and integer spatial frequencies.

Materials: Printed sine profiles and axes, tracing paper or a transparent sleeve, pencil; calculator optional for $e^{(-t)}$. No physical heat apparatus. Preparation: about 8 minutes.

Allow 10 minutes to move and compare profiles, 15-25 to diagnose the equations, a short movement reset, then 20-30 to design fixed-end evolutions. Mode superpositions and energy can be a later meeting.

Launch and facilitation

Choose a sliding speed and a fading rate. Where will the peak be later, and what calculation would catch an incorrect evolution law?

Student page 1: Slide it or fade it

Core: partial derivatives and trigonometric functions

For all real x and $t \geq 0$, try $U_c(x,t) = \sin(x-ct)$ and $V_a(x,t) = e^{(-at)} \sin x$. Choose c from $-2, -1, 0, 1, 2$ and a from $0, 1, 4$. The two candidate laws are $W_{tt} = W_{xx}$ (wave) and $W_t = W_{xx}$ (heat). A subscript is a partial derivative; hold the other variable fixed. Angles use radians.

1 / prompt and solution

Choose c and a . Draw the profiles at $t=0$ and $t=\pi/2$. Predict what happens to the peak position and height. Choose a space-time point where you would test a disputed prediction.

Solution: For U_c , peaks occur at $x = \pi/2 + ct + 2\pi j$ and their height stays 1; at $t=\pi/2$ the displacement is $c\pi/2$, with the sign setting the direction. For V_a , peaks remain at $\pi/2 + 2\pi j$ and have height $e^{(-at)}$, hence $e^{(-a\pi/2)}$ at the chosen later time. The $a=0$ and $c=0$ choices are stationary. Any point is a legitimate test; one should choose a point at which the competing predicted values differ, rather than only a common zero.

Choose a shape, then choose its future

1 / hints to offer only after exploration

1. Track a marked peak, and distinguish its x-coordinate from its height.

2 / prompt and solution

Classify all the allowed c and a : which U_c and V_a obey each law everywhere? Prove each success and exhibit a failed equality for each excluded type.

Solution: $U_t = -c \cos(x-ct)$, $U_{tt} = -c^2 \sin(x-ct)$, $U_{xx} = -\sin(x-ct)$, so U_c obeys the wave law exactly when $c = \pm 1$. It never obeys the heat law: if $c \neq 0$, use $x=t=0$; if $c=0$, use $x=\pi/2, t=0$. $V_t = -aV$, $V_{tt} = a^2V$, $V_{xx} = -V$, so V_a obeys heat exactly for $a=1$ and obeys wave for no real a , since $a^2+1 > 0$ and V is not identically zero. Evaluating the rejected wave identities at a point with sine equal to 1 gives explicit failures.

2 / hints to offer only after exploration

1. An identity must hold at every point. One convenient point can disprove it, but cannot by itself prove it.

Student page 2: One initial shape, several futures

Core continuation: initial and boundary data

Now use only $0 \leq x \leq \pi$ and require $W(0,t)=W(\pi,t)=0$ for every $t \geq 0$. The initial shape is $\sin x$. Try $W_b(x,t) = \sin x(\cos t + b \sin t)$, with a real number b of your choice. The heat candidate is $H(x,t) = e^{-t} \sin x$.

3 / prompt and solution

Choose two different values of b . Verify that both W_b satisfy the wave law, the endpoint rule and the same initial shape. Show that their later futures differ. Explain why simply sliding $\sin(x-t)$ would not meet this endpoint rule.

Solution: For every b , $W_{tt} = -W = W_{xx}$, the sine factor vanishes at both endpoints, and $W(x,0) = \sin x$. Its initial velocity is $W_t(x,0) = b \sin x$. At $t = \pi/2$, $W_b = b \sin x$, so distinct b give different futures, already distinguished at $x = \pi/2$. The traveling wave $\sin(x-t)$ gives $-\sin t$ at $x=0$, which is not always zero; using the same initial graph does not repair its boundary failure.

3 / hints to offer only after exploration

1. At $t = \pi/2$ the cosine term disappears. Check an endpoint for all times, not just at the start.

Choose a shape, then choose its future

4 / prompt and solution

Choose b so the wave W_b and heat H have both the same initial shape and the same initial velocity. Find an exact later observation that separates them. What additional choice remains essential when specifying a future?

Solution: $H_t(x,0)=-\sin x$, so $b=-1$. At $x=\pi/2, t=\pi/2$, $W_{-1}=-1$ but $H=e^{(-\pi/2)}>0$. Their second time derivatives at zero are already different: $W_{tt}(x,0)=-\sin x$, while $H_{tt}(x,0)=\sin x$. Even matching two initial records does not make solutions of different laws coincide: the evolution equation itself is essential. Within the displayed wave family, specifying initial velocity fixes b ; no general PDE uniqueness theorem is being inferred.

4 / hints to offer only after exploration

1. Match the derivative at $t=0$ first; then choose a time that simplifies sine and cosine.

Student page 3: Design with more than one mode

Calculus continuation: separated modes and linearity

A mode has spatial shape $\sin(kx)$, where k is a positive integer. Test heat candidates $e^{(-\alpha t)}\sin(kx)$ with $\alpha \geq 0$ and wave candidates $\cos(\omega t)\sin(kx)$ with $\omega > 0$, keeping the fixed endpoints.

5 / prompt and solution

Choose k . Determine every α and ω for which the corresponding candidate obeys its law. Explain how changing the number of spatial oscillations changes decay and oscillation in time.

Solution: Two x -derivatives multiply $\sin(kx)$ by $-k^2$. The heat derivative multiplies by $-\alpha$, so $\alpha=k^2$. Two wave time derivatives multiply by $-\omega^2$, so $\omega^2=k^2$ and $\omega=k$ because $\omega > 0$. These are necessary because $\sin(kx)$ is nonzero somewhere and sufficient by substitution. Integer k ensures both endpoints vanish. Heat decay rate grows like k^2 ; wave angular frequency grows like k .

5 / hints to offer only after exploration

1. Keep the factor k each time you differentiate in x .

Choose a shape, then choose its future

6 / prompt and solution

Choose a nonzero real d and initial shape $\sin x + d \sin(2x)$. Build a heat evolution and, separately, a wave evolution with zero initial velocity from the modes. Can the heat evolution at any positive time be just one scalar multiple of the whole initial shape? Prove your answer.

Solution: Linearity gives $H_d = e^{-t} \sin x + d e^{-4t} \sin(2x)$ and $W_d = \cos t \sin x + d \cos(2t) \sin(2x)$. Differentiating each term verifies the respective PDE; both endpoints and the initial shape match, and W_d has initial velocity zero. If $H_d(x, t) = C(\sin x + d \sin(2x))$ for all x , evaluation at $x = \pi/2$ forces $C = e^{-t}$. Evaluation at $x = \pi/4$ then forces $e^{-4t} = e^{-t}$, since $d \neq 0$. This occurs only at $t = 0$. Thus for every positive time the heat shape changes, not merely its overall height.

6 / hints to offer only after exploration

1. Use one position where $\sin(2x) = 0$, then one where it is not zero.

Optional facilitator extension 1

Gate: Partial derivatives, definite integrals and integration by parts.

For W_b and H on the fixed interval, compare wave energy $E = (1/2) \int_0^\pi \pi (W_t^2 + W_x^2) dx$ with heat size $Q = \int_0^\pi \pi H^2 dx$. Derive their time behavior, and explain why wave amplitude at one point is not the same as wave energy.

Solution: Write $A = \cos t + b \sin t$ and $A' = -\sin t + b \cos t$. Since $\int \sin^2 x = \int \cos^2 x = \pi/2$, $E = \pi(A'^2 + A^2)/4 = \pi(1 + b^2)/4$, constant. $Q = \pi e^{-2t}/2$ decreases. More generally for sufficiently smooth fixed-end wave solutions, differentiating E and integrating by parts gives $E' = [W_t W_x]_0^\pi + \int W_t (W_{tt} - W_{xx}) dx = 0$; fixed endpoint values give $W_t = 0$ there. A standing wave can be zero everywhere at an instant while its velocity stores the energy. This is an exact model identity, not an experimental claim about a real lossy string.

Mathematics and prior use

Exact separated modes of the wave and heat equations; initial data, boundary data and superposition. No claim of general PDE existence or completeness follows from this selected family.

Prior use: Week 9 follows individual billiard trajectories; this follows a function across all positions. GA-33 studies one-variable rate equations; this family makes spatial derivatives and boundary conditions essential.

Choose a shape, then choose its future

Source: John K. Hunter, Notes on Partial Differential Equations

§5.1, printed p.127, heat initial-value problem; §7.1, pp.211-212, wave equation and two initial conditions.. The source provides the continuum equations and initial-data context. The parameter-selection and same-initial-velocity discrimination tasks are new exact calculations; no token model is substituted for the PDE. Evidence: The equations, fixed-boundary wave formulation and energy calculation were inspected. Every mode, diagnostic and chosen superposition is differentiated independently. Inspected 2026-09-25.

https://math.ucdavis.edu/~hunter/pdes/pde_notes.pdf

Program the folded interval

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with specific gates. Backward composition can be a substantial algebra step. Exact orbit construction is satisfying; a few irregular dots are not evidence of chaos. Unpiloted.

An engineered orbit and a proof of its least period; the later complete four-step inventory is a stronger destination.

Prerequisites and preparation

Core: Fractions, a two-piece function, affine composition and solving a linear equation. The complete period argument needs symbolic reasoning.

Extensions: General word counts use divisors and a finite recurrence; no theorem about chaos is assumed.

Reading and representation: Follow current-state L/R labels in time order. Exact rational arithmetic avoids decimal orbits manufactured by rounding.

Materials: Printed tent graph and $[0,1]$ number lines, L/R inverse cards handed out on page 2, pencil and optional fraction counters. Preparation: about 6 minutes.

10 minutes for forward trials; 15-25 for backward design; a movement reset; 15-25 for least-period comparisons. Save the general word theorem and complete inventory for another session if useful.

Launch and facilitation

Write a four-letter L/R program. Can you find an exact starting fraction whose successive positions repeat that program forever?

Student page 1: Write a program before guessing a start

Core: exact fractions and a folded rule

$T(x)=2x$ when $0 \leq x \leq 1/2$, and $T(x)=2-2x$ when $1/2 \leq x \leq 1$. The formulas agree at $1/2$. Record L for a current state in $[0, 1/2)$, and R for a current state in $[1/2, 1]$. A program lists current-state labels, before each update.

1 / prompt and solution

Choose a four-letter word using L and R. Try exact starting fractions to make those four labels repeat. Keep a successful orbit or explain precisely where a trial fails.

Solution: Any four-letter choice is legal. A trial must check four current-state half labels and return exactly to its initial value after four updates. For a facilitator example, LRRL is realized by $4/15 \rightarrow 8/15 \rightarrow 14/15 \rightarrow 2/15 \rightarrow 4/15$. A finite decimal resemblance is insufficient. An unsuccessful trial is useful if the first failed half or return equality is identified.

Program the folded interval

1 / hints to offer only after exploration

1. A repeated picture is a conjecture; an exact return equality gives something you can check.

2 / prompt and solution

Test starts $1/2$, 1 and 0 . Can a genuinely periodic orbit ever pass through $1/2$ or 1 ? Explain what this means for the L/R boundary convention.

Solution: The chain is $1/2 \rightarrow 1 \rightarrow 0 \rightarrow 0$. A periodic orbit reaching $1/2$ or 1 would then remain at zero and could not return to the nonzero point, so neither can lie on a periodic orbit. Zero is a fixed point and is labeled L. Thus periodic orbits have no midpoint ambiguity, although general forward trials can hit it and use the stated R convention.

2 / hints to offer only after exploration

1. Once zero is reached, what can the next value be?

Student page 2: Undo the machine

Core continuation: inverse branches and composition

For a target $y \in [0, 1]$, its two possible previous values are $L(y) = y/2$ and $R(y) = 1 - y/2$. Here L and R name inverse cards as well as the corresponding half. For word $s_0 s_1 s_2 s_3$, a return candidate solves $x = B_{s_0}(B_{s_1}(B_{s_2}(B_{s_3}(x))))$: undo the latest step first.

3 / prompt and solution

Use inverse cards to solve your own four-letter program. Then independently check every forward value, half label and the return. If your first word was too easy, choose another.

Solution: Compose the chosen affine cards to get $B(x) = ax + b$ with $a = \pm 1/16$. The only algebraic candidate is $x = b/(1-a)$. For LRRL, $B(x) = 1/4 + x/16$, so $x = 4/15$ and the forward orbit is $4/15, 8/15, 14/15, 2/15$. For LLRR the start is $2/15$ and it is the same cycle viewed at another point. All choices are handled by this method; page 4 proves validity rather than assuming branch inequalities.

3 / hints to offer only after exploration

1. Write the outermost inverse first but evaluate the innermost one first.

Program the folded interval

4 / prompt and solution

Compare LRRL, LRLR and LLLL. Find their exact starting points and least positive return times. Why is “the word has four letters” not enough to prove least period four?

Solution: LRRL starts at $4/15$ with four distinct states and least period 4. LRLR has inverse composition $3/8 \rightarrow x/16$, giving $2/5 \rightarrow 4/5 \rightarrow 2/5$ and least period 2. LLLL gives $x=x/16$, hence zero with least period 1. A four-step return permits earlier returns; inspect the full orbit or the repeated structure of the word.

4 / hints to offer only after exploration

1. Check whether the entire program is made by repeating a shorter block.

Student page 3: How many different cycles?

Continuation: classification and cyclic starting points

Count geometric cycles, not ways of choosing their starting point. A cycle of least period four contains four distinct states. You may organize words however you wish; no list of answers is supplied.

5 / prompt and solution

Find every cycle of least period four. Give exact fractions for its states and a way to check that you have not counted the same cycle twice.

Solution: There are three: $(2/17, 4/17, 8/17, 16/17)$, $(2/15, 4/15, 8/15, 14/15)$, and $(6/17, 12/17, 10/17, 14/17)$, with each arrow given by T and the last returning to the first. Representative words are LLLR, LLRR and LRRR respectively. Cyclic rotations of a word move the starting point on one cycle; these three cycles have disjoint state sets. The general word count on page 4 certifies completeness.

5 / hints to offer only after exploration

1. Organize cyclic rotations together; exact fractions can identify a shared cycle.

6 / prompt and solution

A partner says that all sixteen four-letter words must produce sixteen four-cycles. Refute the claim, and give a corrected count that separates fixed points, two-cycles and four-cycles.

Solution: Sixteen words give sixteen distinct points fixed by T^4 , not sixteen cycles. LLLL and RRRR give fixed points 0 and $2/3$. LRLR and RLRL give the two points $2/5$ and $4/5$ of one two-cycle. The remaining twelve points form three four-cycles. Thus there are two fixed-point cycles, one two-cycle and three four-cycles. Completeness of the sixteen-point assertion is proved on the next page.

6 / hints to offer only after exploration

1. First distinguish a point from its full cycle; then remove words repeating a shorter block.

Program the folded interval

Student page 4: Why backward programs always work

Proof continuation: affine fixed points and word periods

For any nonempty word w of length n , let B_w be its inverse-card composition in the same time-order convention. A word has least word period d if it is repeats of a length- d block and no shorter block, where d divides n .

7 / prompt and solution

Prove that B_w has exactly one fixed point in $[0,1]$, that this point really follows the repeated word, and that different length- n words give different points. Include the midpoint and endpoint cases.

Solution: $B_w(x)=ax+b$ maps $[0,1]$ into itself and has $a=\pm 2^{-(n)}$. Since $b=B_w(0)\geq 0$ and $a+b=B_w(1)\leq 1$, the candidate $x=b/(1-a)$ lies in $[0,1]$ and is unique. $T\circ L=T\circ R$ is the identity, including endpoints, so successively cancelling inverse cards gives $T^n x=x$ and puts each intermediate state in its prescribed closed half. By prompt 2, a periodic orbit cannot hit $1/2$ or 1 ; thus the closed-half placements give exactly the stated L/R labels. Zero is labeled L and realizes only the all-L word. A point has a unique current-half itinerary, so distinct length- n words cannot produce the same point.

7 / hints to offer only after exploration

1. An affine self-map with slope of magnitude below 1 can have its fixed point located using only its endpoint inequalities.

8 / prompt and solution

Prove that the least orbit period equals the least word period. Use this to justify the completeness of your four-cycle inventory.

Solution: Let the least word period be d , with $w=v$ repeated n/d times. The unique fixed point y of B_v is fixed by $B_w=(B_v)^{(n/d)}$, hence equals x by uniqueness; therefore $T^d x=x$. If the least orbit period is r , then r divides d : divide d by r and a smaller positive remainder return would contradict minimality. The length- n itinerary is consequently repeats of its first r symbols, so minimality of d gives $d\leq r$. Hence $r=d$. For $n=4$ there are $2^4=16$ words/points. Exactly two words have period 1 and two more have period 2; twelve have period 4 and yield three cycles. The listed three verified cycles therefore exhaust all possibilities.

8 / hints to offer only after exploration

1. A fixed point of the shorter inverse composition is also fixed by its repeated composition.

Program the folded interval

Optional facilitator extension 1

Gate: Divisors, finite sums and the proved all-word theorem.

Let P_n be the number of tent-map points with least period n . Derive a recurrence for P_n and use it to count least-period-six points and cycles.

Solution: There are exactly 2^n points fixed by T^n , one for each length- n word. Any least period of such a point divides n by division with remainder. These disjoint classes give $2^n = \sum_{d|n} P_d$, so $P_n = 2^n - \sum_{d|n, d < n} P_d$. Thus $P_1=2$, $P_2=2$, $P_3=6$, and $P_6=64-2-2-6=54$. Each least-period-six cycle contains exactly six points, giving nine cycles. This counts exact periodic behavior; it is not a proof of any unspecified meaning of chaos.

Mathematics and prior use

Periodic symbolic coding of the tent map at slope 2. Every finite word determines one fixed point of its inverse composition; shorter word periods determine shorter orbit periods.

Prior use: Week 4 modular orbits and AP-26 delayed control have finite orbit/state mechanisms. This is the actual continuous tent map, with folding, inverse branches and symbolic programs; it does not reuse the atlas LRR example as the main instance.

Source: Gerald Teschl, Ordinary Differential Equations and Dynamical Systems

§11.3, equation (11.13), printed p.297, slope-two tent map; §11.4, pp.299-301, inverse intervals and symbolic coding for slopes greater than two.. The actual tent map is retained. The chosen four-letter orbits, all-word fixed-point proof and exact least-period inventory are new derivations, not an invocation of the $\mu > 2$ conjugacy theorem. Evidence: The slope-two definition and the neighboring coding construction were inspected. The source coding theorem uses $\mu > 2$; our $\mu = 2$ periodic-word theorem and boundary handling are proved directly. Inspected 2026-09-25.

<https://www.mat.univie.ac.at/~gerald/ftp/book-ode/ode.pdf>

Can tiny payments break a budget?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for fraction-ready learners, with the infinite proof as the real gate. Avoid turning the launch into lengthy common-denominator arithmetic. Unpiloted.

A block construction that beats an arbitrary budget, contrasted with an exact geometric remainder.

Prerequisites and preparation

Core: Fractions, powers of two and comparing groups of terms. The core payoff requires reasoning about every proposed finite target.

Extensions: Comparing ordinary convergence with averages of partial sums needs sequence limits.

Reading and representation: Read $1/n$ and distinguish a payment from the total paid so far. Exact addition of long fraction lists is deliberately unnecessary.

Materials: Printed payment cards 1 through $1/16$, geometric strip, blank target cards and pencil. Cards show labels, not physical lengths proportional to all reciprocals. Preparation: about 6 minutes.

10 minutes to play with early payments, 15 to propose budgets and group cards, a short movement reset, then 20-30 for target guarantees and tail challenges. Infinite quantifiers may need a later discussion.

Launch and facilitation

Choose a total you want to beat. Which machine can be guaranteed to pass it, and how could you prove that without adding every payment?

Student page 1: Two payment machines

Core: fractions and chosen targets

Machine G pays $1, 1/2, 1/4, 1/8, \dots$. Machine H pays $1, 1/2, 1/3, 1/4, \dots$. The total after n payments is G_n or H_n , with $n \geq 1$. A budget is broken only when a total is strictly greater than it.

1 / prompt and solution

Choose two target budgets, one below 2 and one above 2. Try to break each using each machine. Record what your trials prove and what they leave uncertain.

Solution: G can break any budget $B < 2$ because $G_n = 2 - 2^{-(n-1)}$ tends to 2 from below; it cannot break $B \geq 2$. H eventually breaks every finite target. At this stage learners may have only verified particular finite successes; failure to cross in a finite trial proves no permanent upper bound. For example $B = 3/2$ is broken by both machines at $n = 3$ ($G_3 = 7/4, H_3 = 11/6$), while $H_{16} > 3$ supplies a later certificate for $B = 3$. The learner can choose other targets.

Can tiny payments break a budget?

1 / hints to offer only after exploration

1. A failed finite attempt is not yet a proof that a machine can never win.

2 / prompt and solution

Find a formula for G_n and its exact unpaid distance from 2. For a tolerance ε of your choice with $0 < \varepsilon < 1$, give a rule selecting n with $2 - G_n < \varepsilon$. Test $\varepsilon = 1/100$.

Solution: $G_n = 2 - 2^{-(n-1)}$, proved by doubling/subtracting the finite sum or induction. Its gap is exactly $2^{-(n-1)}$. Choose n with $2^{-(n-1)} > 1/\varepsilon$, which exists because repeated doubling is unbounded. For $\varepsilon = 1/100$ the least n is 8: the gap is $1/128$, while $n=7$ gives $1/64$. This proves convergence to 2 and a strict accuracy threshold, without an infinite subtraction trick.

2 / hints to offer only after exploration

1. Compare one finite sum with half of itself, or describe the remaining strip.

Student page 2: Build a certificate from groups

Core continuation: lower bounds, not exact totals

You may group consecutive H payments any way you like. Replacing every payment in a group by a smaller amount gives a lower bound for that group.

3 / prompt and solution

Use the cards through $1/16$ to prove $H_{16} > 3$ without finding their exact sum. Choose your own grouping and explain why your inequality is strict.

Solution: One grouping is 1; then $1/2$; then $1/3 + 1/4$; then terms $1/5$ through $1/8$; then terms $1/9$ through $1/16$. Each of the four later blocks is at least $1/2$, and the block $1/3 + 1/4$ is strictly larger than $1/2$. Thus $H_{16} > 1 + 4/2 = 3$. Other rigorous groupings are accepted; an equality-only estimate $H_{16} \geq 3$ does not by itself break the strict budget.

3 / hints to offer only after exploration

1. Can twice as many smaller terms still guarantee the same contribution?

Can tiny payments break a budget?

4 / prompt and solution

Let a partner choose any finite positive budget B . Give a finite stopping index guaranteed to break it, and prove your rule works for every B . Give one explicit guarantee for $B=10$; do not claim it is the first crossing.

Solution: For each $j \geq 1$, indices $2^{j-1}+1$ through 2^j contribute at least $1/2$. Hence $H_{2^k} \geq 1+k/2$. Choose any integer $k > 2(B-1)$, with $k \geq 1$; then $N=2^k$ gives $H_N > B$. For $B=10$, $k=19$ and $N=524288$ suffice. The construction works for every finite B , proving unboundedness. This deliberately loose bound is not a calculation of the earliest crossing.

4 / hints to offer only after exploration

1. Keep the number of blocks separate from the number of individual payments.

Student page 3: Can skipping payments save the budget?

Continuation: tails and changed rules

A proposed repair ignores the first N payments, with N a positive integer, but keeps every later payment. A different repair keeps only selected indices. These are different operations.

5 / prompt and solution

After any chosen N , show that the remaining H payments can still break any extra budget. Contrast this with the total of all G payments after its first N payments.

Solution: The H block $N+1, \dots, 2N$ has N terms at least $1/(2N)$, contributing at least $1/2$. The next block $2N+1, \dots, 4N$ does the same, and so on. After k such blocks the extra total is at least $k/2$, so choosing $k > 2B$ breaks any extra budget B . For G , the complete remaining tail has sum $2^{-(1-N)}$, by scaling the geometric sum; every finite remaining total is smaller. Removing a finite prefix does not repair harmonic divergence.

5 / hints to offer only after exploration

1. Double the ending index starting from N , rather than starting over at 1.

6 / prompt and solution

Choose infinitely many H payment indices so that their total does stay within a finite budget. Prove your choice works. Explain why this does not contradict the previous tail argument or the fact that both machines have payments tending to zero.

Solution: Keep indices $1, 2, 4, 8, \dots$, obtaining payments $1, 1/2, 1/4, \dots$ with total 2. More generally starting at index 2^r makes the total 2^{1-r} . The choice deletes infinitely many payments; it is not merely removal of a finite prefix. Both original payment sizes tend to zero, but the harmonic blocks keep a substantial cumulative contribution. Thus termwise smallness alone cannot certify convergence.

Can tiny payments break a budget?

6 / hints to offer only after exploration

1. Could one of the original machines hide inside a sparse selection of the other?

Optional facilitator extension 1

Gate: Signed numbers, limits and the distinction between two sequences.

Consider payments $1, -1, 1, -1, \dots$. Find the ordinary partial sums and the averages of the first n partial sums. Explain exactly what converges and what does not.

Solution: Partial sums S_n equal 1 for odd n and 0 for even n , so they do not have an ordinary limit. Their averages $A_n = (S_1 + \dots + S_n)/n$ are $1/2$ for even n and $(n+1)/(2n)$ for odd n , both tending to $1/2$. Saying this series has Cesàro value $1/2$ names a different summation convention; it does not make its ordinary partial sums converge. Pairing all terms as $(1-1)$ or starting a grouping differently cannot override the stated definition.

Mathematics and prior use

Convergence versus unbounded partial sums; term size alone is insufficient. Finite grouping proves arbitrary target guarantees without evaluating the harmonic constant or a logarithm.

Prior use: GA-03 uses geometric budgets to cover points and GA-29 uses Cantor-stage lengths; neither already asks why vanishing harmonic payments can defeat every budget. GA-34 later uses finite sums of shrinking time gaps for blow-up, a different conclusion.

Source: Jiří Lebl, Basic Analysis I

§2.5, Example 2.5.4 and Proposition 2.5.5, printed pp.88-89; Example 2.5.11, pp.90-91, harmonic grouping.. The classical series contrast is retained, but learners choose targets and repair a proposed tail-saving plan. The sparse-subsequence and partial-sum-average questions explicitly distinguish changing the payment rule from ordinary convergence. Evidence: The geometric remainder and harmonic block proof were inspected. Index conventions, strict inequalities, tail blocks and chosen guarantees are checked independently. Inspected 2026-09-25.

<https://www.jirka.org/ra/realanal.pdf>

How narrow can the error band be?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with specific algebra gates. The movable-line search is concrete, but an attractive graph is not a proof. Unpiloted.

A global lower bound met by one line, with a uniqueness proof.

Prerequisites and preparation

Core: Graphs, signed fractions, affine formulas and inequalities. No calculus is needed for the complete minimax proof.

Extensions: The general interval change uses algebra; best constants for continuous piecewise-linear targets use maxima and minima.

Reading and representation: Read $y=ax+b$ and absolute vertical error; distinguish an error bound from total band thickness and from average or perpendicular error.

Materials: Printed x^2 graphs, ruler, tracing paper or a transparent sleeve, pencil. A graphing device is optional, not a proof tool. Preparation: about 6 minutes.

10-15 minutes moving candidate lines; 15-25 for challenges and an optimality certificate; movement reset; 15-25 for uniqueness or a moved interval. The last page can be separate.

Launch and facilitation

Place a straight line where its worst vertical miss from the parabola is as small as possible. Your partner may challenge it at any x in the interval.

Student page 1: A line and a challenger

Core: graphs and vertical error

The target is $f(x)=x^2$ for $-1 \leq x \leq 1$. You may choose any line $p(x)=ax+b$. An error budget $E \geq 0$ is valid if $|x^2 - p(x)| \leq E$ for every x in the interval. The band is between $p-E$ and $p+E$, so its total vertical thickness is $2E$.

1 / prompt and solution

Choose a line and an error budget. Let a partner challenge them at any x in $[-1, 1]$. Revise your proposal and try to make the budget smaller. Mark the actual vertical gaps you compared.

Solution: Valid proposals vary. For example $p=0$ needs $E=1$, $p=1$ needs $E=1$, and $p=1/2$ works with $E=1/2$ because $0 \leq x^2 \leq 1$. Tilted proposals may fail at one endpoint or at an interior point. A finite set of successful tests supports a conjecture only; the validity condition concerns every x . Do not announce the optimum before learners have experimented.

How narrow can the error band be?

1 / hints to offer only after exploration

1. A tilted line can reduce one gap while increasing another.

2 / prompt and solution

A line passes exactly through the two endpoint values of the parabola. Find that line and its worst error on the whole interval. Can endpoint agreement alone certify a small uniform error?

Solution: The two endpoints are $(-1,1)$ and $(1,1)$, giving $p=1$. Its residual x^2-1 lies in $[-1,0]$, and the worst absolute error is 1 at $x=0$. Thus endpoint agreement alone says nothing about a small bound between them. It is legitimate to use the supplied continuous graph to conjecture this, then use $0 \leq x^2 \leq 1$ to prove it.

2 / hints to offer only after exploration

1. Challenge the line halfway between the endpoints.

Student page 2: A certificate against every line

Core proof: combine inequalities

For a general candidate $p=ax+b$, keep E as an unknown valid error bound. You may test particular x -values to force a lower bound, but must also verify your winning line everywhere.

3 / prompt and solution

Use $x=-1,0,1$ to prove that no line can have $E < 1/2$. Find a line attaining the bound, and verify its error on all of $[-1,1]$.

Solution: At zero, $|b| \leq E$. At the endpoints, $|1-b-a| \leq E$ and $|1-b+a| \leq E$. Averaging the two signed endpoint residuals gives $|1-b| \leq E$. Therefore $1 \leq |b| + |1-b| \leq 2E$, so $E \geq 1/2$. The constant $p=1/2$ has $|x^2-1/2| \leq 1/2$ for all $|x| \leq 1$, attaining the lower bound. Its signed residuals at $-1,0,1$ are $+1/2, -1/2, +1/2$.

3 / hints to offer only after exploration

1. The average of the two endpoint residuals removes the slope.

4 / prompt and solution

Prove that there is only one line attaining this best budget. Explain why verifying a candidate and proving it is the unique best candidate are separate tasks.

Solution: If $E=1/2$ is valid, $|b| \leq 1/2$ and $|1-b| \leq 1/2$ imply $b=1/2$. Then $|1/2-a| \leq 1/2$ requires $0 \leq a \leq 1$, while $|1/2+a| \leq 1/2$ requires $-1 \leq a \leq 0$. Hence $a=0$. Candidate verification proves attainment; these forced equalities exclude every other line with the same error. The proof does not assume that an optimal line must be horizontal merely from the symmetry of a picture.

How narrow can the error band be?

4 / hints to offer only after exploration

1. Use the two intervals allowed for b , then do the same for a .

Student page 3: Move and stretch the playing field

Algebra continuation: transfer a certificate

Choose a center h and radius $r > 0$. Now approximate x^2 on $[h-r, h+r]$. You may use $z = (x-h)/r$, so $-1 \leq z \leq 1$. Error still means vertical error.

5 / prompt and solution

Find the best affine line and exact error for your chosen interval. Prove optimality and uniqueness by transferring the earlier argument. Test the interval $[1, 3]$.

Solution: Write $x = h + rz$. Then $x^2 = h^2 + 2hrz + r^2z^2$. The affine part can be matched exactly; applying the established unique optimum to z^2 gives $p(x) = 2hx - h^2 + r^2/2$ and $E = r^2/2$. More formally, subtract $h^2 + 2hrz$ from any competing affine approximation and divide by r^2 ; this bijects competitors to lines on $[-1, 1]$ and scales their uniform errors by $1/r^2$. Thus both optimality and uniqueness transfer. For $[1, 3]$, $h = 2, r = 1$, giving $p = 4x - 7/2$ and $E = 1/2$.

5 / hints to offer only after exploration

1. Remove the part that is already linear before deciding how to approximate the remaining square.

6 / prompt and solution

Compare the optimal line with the line through both endpoint values on your chosen interval. Find the latter line and its exact worst error. Explain how the best line improves it.

Solution: The endpoint secant is $s(x) = 2hx - h^2 + r^2$. Since $x^2 - s(x) = (x-h)^2 - r^2$, its residual ranges from $-r^2$ at the center to zero at the endpoints; its worst error is r^2 . Lowering that secant by $r^2/2$ balances the positive endpoint residuals with the negative center residual, halving the error to $r^2/2$. This comparison follows from the full range, not just three measurements.

6 / hints to offer only after exploration

1. Complete the square in the residual and compare its largest and smallest values.

How narrow can the error band be?

Optional facilitator extension 1

Gate: Ordering real values and absolute-value inequalities.

For the continuous polygonal graph with vertices $(-2,1), (-1,4), (0,-2), (1,1), (2,0)$, find the unique best constant under maximum vertical error. Generalize to a continuous target whose minimum m and maximum M are attained.

Solution: The polygonal graph stays between its extreme vertex values -2 and 4 , so $c=1$ has worst error 3 . Any constant c has error at least $\max(|M-c|, |m-c|) \geq (M-m)/2$. The midpoint $c=(M+m)/2$ attains this for every target value between m and M . The simultaneous inequalities at m and M force that same c when the error is optimal, including $m=M$. Thus the best constant is unique; no claim is made that it is also the best tilted line for the polygonal graph.

Mathematics and prior use

Best uniform affine approximation and a three-point alternation certificate. General approximation theorems are contextual; this entire chosen problem is solved directly.

Prior use: Week 1 packing also joins a construction to a lower bound; this is a different continuous approximation problem. GA-07 uses Chebyshev identities, while this family discovers an error certificate without assuming a general alternation theorem.

Source: NIST Digital Library of Mathematical Functions

§3.11(i), minimax polynomial approximation, uniqueness and alternating extreme errors.. The classical x^2 target remains, with learner-chosen lines and adversarial test points. The full proof is supplied without invoking the general equioscillation theorem or substituting least squares. Evidence: The minimax definition and alternation context were inspected. The lower bound, uniqueness and translated-interval formulas are independently proved by elementary inequalities. Inspected 2026-09-25.

<https://dlmf.nist.gov/3.11>

Choose a photograph that reveals the motion

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with specific gates. Arrow play is accessible; complete classifications require modular and trigonometric reasoning. Exact-time results do not claim robustness under real measurement noise. Unpiloted.

A chosen extra time that distinguishes two candidate motions, with an exact statement of what ambiguity remains.

Prerequisites and preparation

Core: Quarter turns, fractions and trigonometric periodicity. Classifying aliases uses integer remainders; the final obstruction uses elementary polynomials.

Extensions: An ideal irrational-time observation uses irrational numbers and exact full-arrow positions.

Reading and representation: Read time versus angle carefully. All arrows start pointing right and turn counterclockwise at a constant nonnegative integer number of turns per second. Heights may be signed.

Materials: Two printed unit-circle dials, paper arrows or pencils, time-strip cards and blank signal axes. No real camera or high-speed measurement is required. Preparation: about 6 minutes.

10 minutes turning arrows and making snapshot records; 15-25 designing an extra observation; movement reset; 15-25 classifying missed frequencies or proving the limit of finite observations.

Launch and facilitation

Two motions agree at every scheduled photograph. Choose one new photograph to try to tell them apart, then ask which other motions could still remain hidden.

Student page 1: Make a shared record

Core: unit-circle turns and times

A unit arrow starts pointing right at time 0 and turns counterclockwise. Frequency k means k complete turns each second. Its tip is $(\cos(2\pi kt), \sin(2\pi kt))$; a height-only photograph records $s_k(t) = \sin(2\pi kt)$. Compare $k=1$ and $k=5$.

Choose a photograph that reveals the motion

1 / prompt and solution

Turn or draw both arrows at $t=0, 1/4, 1/2, 3/4$. Record their heights and full tip positions. Choose another scheduled time $n/4$ and test whether it separates the motions.

Solution: Both give tip positions $(1,0), (0,1), (-1,0), (0,-1)$ at the four listed times, with heights $0, 1, 0, -1$. At every additional integer n , their angles differ by $2\pi n$ at $t=n/4$, so neither heights nor full positions separate them. Their motion between observations is still different; the $k=5$ arrow completes one extra full turn between adjacent scheduled times.

1 / hints to offer only after exploration

1. Compare the number of complete extra turns between photographs.

2 / prompt and solution

Prove agreement at every scheduled time $t=n/4$, not only the ones you drew. Could the entire infinite regular record decide between these two frequencies?

Solution: The phase difference is $2\pi(5-1)n/4=2\pi n$, an integer number of turns. Sine and cosine both have period 2π , so agreement holds for every integer n . Even the whole infinite scheduled record is identical for the two candidates and cannot identify which one produced it. This is exact ambiguity, not a claim that the motions themselves coincide.

2 / hints to offer only after exploration

1. Use an arbitrary integer n in the angle difference.

Student page 2: Design one more observation

Core continuation: distinguish a stated class

You may choose one extra observation time between 0 and $1/4$. First distinguish only frequencies 1 and 5. Then broaden the candidate class to all nonnegative integer frequencies with the same starting arrow.

3 / prompt and solution

Choose an extra time that distinguishes the two heights and justify your choice exactly. Give one tempting extra time that would fail, and explain why.

Solution: For example $t=1/8$ gives $s_1=\sin(\pi/4)=\sqrt{2}/2$ and $s_5=\sin(5\pi/4)=-\sqrt{2}/2$, so it succeeds. A failing interior choice is $t=1/12$: $s_1=\sin(\pi/6)=1/2$ and $s_5=\sin(5\pi/6)=1/2$. The full arrows differ at that time, but a height-only measurement does not. Other exact times are accepted if their two sine values are proved different. The choice matters even between regular samples.

Choose a photograph that reveals the motion

3 / hints to offer only after exploration

1. Equal heights can come from two different angles on the circle.

4 / prompt and solution

Find all nonnegative integer k whose heights agree with frequency 1 at every $t=n/4$. Now also require agreement at $t=1/8$. Does that additional photograph uniquely identify frequency 1 in this enlarged class?

Solution: At $t=1/4$, agreement requires $\sin(\pi k/2)=1$, which for integer k means $k \equiv 1 \pmod{4}$. This condition is also sufficient for all n because the phase difference is then a whole number of turns. Such k are 1 or 5 modulo 8; at $t=1/8$ the respective sine values are $+\sqrt{2}/2$ and $-\sqrt{2}/2$. Therefore the surviving class is exactly $k \equiv 1 \pmod{8}$, namely 1, 9, 17, The extra photo separates 1 from 5 but leaves infinitely many integer alternatives.

4 / hints to offer only after exploration

1. First use one necessary photograph, then prove your remaining residue class works at all photographs.

Student page 3: What did the camera forget?

Algebra continuation: sensor choice and finite data

For the next task change the schedule to $t=n/2$, still with nonnegative integer frequencies and arrows initially pointing right. For the final task allow arbitrary continuous real-valued signals, not just rotating-arrow heights.

5 / prompt and solution

At the new schedule, classify every frequency that looks like $k=1$ when only height is recorded. Then classify them when the full arrow is recorded. Prove whether the richer record removes all ambiguity.

Solution: For any integer k , $\sin(\pi kn)=0$ at every integer n , so all nonnegative integers match the height record of frequency 1. The full arrow is $((-1)^{kn}, 0)$. At $n=1$ it matches frequency 1 exactly when k is odd, and odd k then work for every n . Thus full positions discard the even frequencies but still leave all odd frequencies. Richer observations need not completely determine the motion.

5 / hints to offer only after exploration

1. A zero height does not tell you whether an arrow points left or right.

Choose a photograph that reveals the motion

6 / prompt and solution

An observer makes any finite list of measurements at distinct times $t_1, \dots, t_m$. Starting with a continuous signal f , construct a different continuous signal agreeing at every measured time. What claim about unrestricted reconstruction does your construction rule out?

Solution: Take $g(t) = f(t) + \varepsilon \prod_{j=1}^m (t - t_j)$, with any nonzero real ε . The polynomial perturbation vanishes at every measured time and is nonzero at every other time, so g is continuous, distinct from f and gives the same finite record. For an empty record any nonzero constant perturbation works. Thus finitely many point values cannot uniquely identify an arbitrary continuous function. These alternatives need not be sinusoids; that is exactly why the enlarged candidate class must be stated.

6 / hints to offer only after exploration

1. Build a nonzero function with a zero at each prescribed measurement time.

Optional facilitator extension 1

Gate: Irrationality and equality of angles modulo a complete turn.

Keep only constant nonnegative integer frequencies with initial arrow pointing right. In an ideal exact model, show that one full-arrow observation at $t = \sqrt{2}$ distinguishes frequency 1 from every other such frequency. Does this contradict the finite-record obstruction?

Solution: Equal full-arrow positions require $2\pi(k-1)\sqrt{2}$ to be an integer multiple of 2π , so $(k-1)\sqrt{2} = m$ for some integer m . If $k \neq 1$ this would make $\sqrt{2} = m/(k-1)$ rational, impossible. Hence $k=1$ is unique within the specified integer-frequency family. There is no contradiction: the finite-record obstruction allowed arbitrary continuous functions, a much larger class. This argument assumes an exactly specified irrational time and exact angular observation; it is not a finite-precision recovery guarantee.

Mathematics and prior use

Aliasing as restriction of periodic functions to a sampling grid. The candidate class is always explicit; no general reconstruction theorem is inferred from finitely many observations.

Prior use: Week 4 clock periodicity is a useful prerequisite, but here the object is a continuous signal observed only at selected times. GA-10 programs actual iterations; this family studies information lost by observation, not a new update rule.

Choose a photograph that reveals the motion

Source: Daniel W. Stroock, Topics in Fourier Analysis

§1, printed p.2, periodic exponentials $e_m(x) = \exp(2\pi i m x)$ and periodic extension.. The source gives Fourier background rather than this camera puzzle or a sampling theorem. The worksheet uses exact rotating arrows and explicitly restricted frequency families. Evidence: The periodic-frequency framework was inspected. All sampling congruences, extra-time tests and finite-record counterexamples are proved independently. Inspected 2026-09-25.

https://ocw.mit.edu/courses/res-18-015-topics-in-fourier-analysis-spring-2024/mitres_18_015_s24_full_lec.pdf

Decode a picture from four sign patterns

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for learners ready for signed quantities and halves; the full arbitrary-vector proof is a real later gate. Counters should support cancellation rather than imply that all coefficients must be nonnegative. Unpiloted.

A unique reconstruction of an arbitrary four-square picture and a predicted symmetry in its coefficients.

Prerequisites and preparation

Core: Signed addition, fractions and four-entry arrays. The basis proof uses dot products or their equivalent weighted sums.

Extensions: Three binary coordinates and parity for the eight-pattern extension.

Reading and representation: Four square labels are always 00,01 on top and 10,11 below. Negative contributions are algebraic corrections; they are not negative physical light.

Materials: Four printed 2-by-2 sign cards, blank picture grids, two colors of counters and pencil. Counter colors distinguish positive from negative contributions. Preparation: about 8 minutes.

10-15 minutes combining cards; 15-25 decoding and inventing detectors; movement reset; 15-25 for reconstruction proof, swaps or missing measurements. The general basis explanation can be a later stop.

Launch and facilitation

Choose hidden amounts of the four sign cards and add them into a picture. Can your partner recover the amounts, and can you predict what swapping columns will do before decoding again?

Student page 1: Mix cards into a picture

Core: signed addition and four-square arrays

Read entries in order 00,01,10,11: top row first, then bottom. The cards are $C=(1,1,1,1)$, $R=(1,1,-1,-1)$, $S=(1,-1,1,-1)$, $D=(1,-1,-1,1)$. A weight multiplies every entry of its card. Add contributions square by square; any real weights are allowed.

Decode a picture from four sign patterns

1 / prompt and solution

Choose weights for the cards and make a picture. Give your partner only the resulting four entries and ask for the weights. You may use a large constant-card weight to make every final entry nonnegative, even if some other weights are negative.

Solution: Any chosen four real weights define a legal picture $c_C C + c_R R + c_S S + c_D D$. For example weights $(3, -1, 1/2, 1/2)$ give $(3, 1, 4, 4)$. Adding a sufficiently large C contribution raises every entry equally without changing the nonconstant coefficients. The partner may guess, solve equations or discover cancellation. Later tasks prove that decoding is always possible and unique; positivity of the final picture does not require positivity of each hidden weight.

1 / hints to offer only after exploration

1. A negative weight reverses every sign on that card.

2 / prompt and solution

Decode the supplied picture $(5, 0, 1, 2)$. Rebuild all four entries from your weights to verify the result. Would restricting every weight to be a nonnegative whole number change the problem?

Solution: The unique real weights are $(2, 1/2, 1, 3/2)$. The four reconstructed values are $2 + 1/2 + 1 + 3/2 = 5$; $2 + 1/2 - 1 - 3/2 = 0$; $2 - 1/2 + 1 - 3/2 = 1$; and $2 - 1/2 - 1 + 3/2 = 2$. Restricting weights to nonnegative integers would forbid this representation, since uniqueness later rules out a second integer one. Real coefficients, including fractions and negative values, are part of the stated model.

2 / hints to offer only after exploration

1. Try adding all entries, then adding after changing some signs.

Student page 2: Invent a detector for one card

Core proof: weighted sums and cancellation

A detector applies a sign to each picture entry, then adds the four signed values. For a sign card P , write $\langle v, P \rangle$ for that weighted sum. No normalization is built into this notation.

3 / prompt and solution

Choose one card and build a detector that ignores every other card. Explain why it works. Find the detector output on its own card and use it to justify your decoding scale.

Solution: Use the chosen card itself as the four detector signs. Each card has self-sum of squares 4. For distinct cards, their entrywise product has two $+1$ and two -1 entries, so its sum is zero. Thus $\langle \sum_Q Q, P \rangle = 4c_P$, and the weight is $\langle v, P \rangle / 4$. This verifies all four detectors, not only the selected one; the supplied target has detector totals 8, 2, 4, 6.

Decode a picture from four sign patterns

3 / hints to offer only after exploration

1. Compare matching and opposite signs when two cards are multiplied square by square.

4 / prompt and solution

Prove that every real four-entry picture is reconstructed by these four detector readings divided by four, and that no picture has two different weight lists.

Solution: For $v=(a,b,c,d)$, the weights are $(a+b+c+d)/4$, $(a+b-c-d)/4$, $(a-b+c-d)/4$ and $(a-b-c+d)/4$. In the first reconstructed entry all a contributions add to $4a$ and every other variable cancels; the other three entries similarly give b,c,d . Hence the formula reconstructs every picture. If two lists reconstructed the same picture, subtract the lists and apply each detector: 4 times each weight difference is zero. Thus all differences vanish. Counting four cards alone would not establish spanning or uniqueness.

4 / hints to offer only after exploration

1. Check an arbitrary entry with letters, then use a detector to rule out an invisible coefficient change.

Student page 3: Predict changes without decoding again

Continuation: symmetry and missing information

A column swap exchanges 00 with 01 and 10 with 11. A row swap exchanges 00 with 10 and 01 with 11. These are permutations of the picture squares, not new sign conventions.

5 / prompt and solution

Choose a row swap or column swap. Predict exactly which coefficients stay the same and which change sign, then test your rule on $(5,0,1,2)$. Explain using the cards themselves.

Solution: A column swap fixes C and R but negates S and D, so (c_C, c_R, c_S, c_D) becomes $(c_C, c_R, -c_S, -c_D)$. The target becomes $(0,5,2,1)$, with weights $(2, 1/2, -1, -3/2)$. A row swap fixes C and S but negates R and D; the target becomes $(1,2,5,0)$, with weights $(2, -1/2, 1, -3/2)$. Since permutations distribute over sums, checking their action on the four cards proves the rule for every picture.

5 / hints to offer only after exploration

1. Swap the entries on one sign card before attempting a whole new decoding.

Decode a picture from four sign patterns

6 / prompt and solution

An incomplete sensor reports only total brightness 8 and “top total minus bottom total” equal to 2. Find every nonnegative four-square picture consistent with it. Which further card readings make the picture unique?

Solution: Let top and bottom totals be T and B . $T+B=8$ and $T-B=2$ give $T=5, B=3$. Therefore all pictures are $(p, 5-p, q, 3-q)$, with $0 \leq p \leq 5$ and $0 \leq q \leq 3$. This is a complete family: every nonnegative consistent picture has this form, and every listed pair works. The missing S and D readings supply the remaining coefficients and hence a unique picture by prompt 4. One of those alone generally leaves one free parameter; special boundary data may happen to single out one point, but the pair guarantees uniqueness for all pictures.

6 / hints to offer only after exploration

1. First determine the two row totals. How freely can you distribute each total within its row?

Optional facilitator extension 1

Gate: Binary coordinates, parity and a finite cancellation proof.

For three binary coordinates, define $P_a(x) = (-1)^{(a_1x_1 + a_2x_2 + a_3x_3)}$, with a, x in $\{0, 1\}^3$. Prove the eight patterns reconstruct every real eight-entry array and find the scaling factor.

Solution: Each self dot product is 8. For $a \neq b$, choose a coordinate where they differ. Pair each x with the vector obtained by flipping that coordinate; the products $P_a(x)P_b(x)$ cancel in each pair. Hence distinct patterns are orthogonal. Alternatively summing $P_a(x)P_a(y)$ over a yields 8 if $x=y$ and 0 otherwise by the same pairing. Thus $c_a = (1/8) \sum_x v(x)P_a(x)$ and $\sum_a c_a P_a(y) = v(y)$, proving spanning directly. Applying a detector proves uniqueness. The division is by eight, not four, because there are eight coordinates.

Mathematics and prior use

The Walsh character basis of $(\mathbb{Z}/2\mathbb{Z})^2$ and diagonal action of translations. The four-square case is proved directly, including scaling and uniqueness; no infinite-group theory is claimed.

Prior use: Week 2 toggles uses binary changes and Week 6 coding uses message distinctions. Here values and weights are real, signed corrections add linearly, and orthogonal patterns decode a complete array. This is not another binary toggle worksheet.

Decode a picture from four sign patterns

Source: Rohan Dandavati, The Fourier Transform on Finite Groups: Theory and Computation

§3, Definition 3.1 and Theorem 3.3 on characters and orthogonality, printed pp.5-6; Theorem 3.8, character basis, p.7.. Four sign cards are actual characters in a fixed coordinate order. Chosen encoding, new target $(5,0,1,2)$, symmetry prediction and incomplete-measurement classification are independently constructed. Evidence: The character definition, orthogonality and basis argument were inspected. The worksheet uses unnormalized signed sums and derives the necessary division by four directly; it does not copy a normalization from the general reconstruction display. Inspected 2026-09-25.

<https://math.uchicago.edu/~may/REU2018/REUPapers/Dandavati.pdf>

Predict what the sliding window draws

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for the geometric core. A third convolution is an advanced calculus extension. The moving endpoint convention is the main practical risk. Unpiloted.

A full overlap formula explained by endpoint changes, with a successful inverse design from desired graph dimensions.

Prerequisites and preparation

Core: Length, subtraction and piecewise graphs. The convolution connection uses the supplied fact that an indicator integral equals length.

Extensions: A third window requires definite integration and continuity of derivatives.

Reading and representation: Keep the moving window's right endpoint t distinct from its center. Distinguish input indicator values at a boundary from the overlap length at that time.

Materials: Printed fixed strip of length 3 units, transparent moving strip of length 2 units, unit ruler and blank overlap graph. One unit may be one inch; compare strips on the same scale. Preparation: about 8 minutes.

10-15 minutes sliding and choosing observations, 15-25 to justify the full graph, a movement reset, then 15-25 to reverse roles or design widths. Integral notation may be introduced only after the geometric core.

Launch and facilitation

Choose where to place the short window. Predict the entire overlap graph before tracing many samples, then design windows for a graph you want.

Student page 1: Move the right endpoint

Core: lengths and prediction

The fixed window is $[0,3]$. The moving window is $[t-2,t]$, so t labels its right endpoint. Let $h(t)$ be the length covered by both windows. Touching at one point gives length zero. All lengths use the same unit.

1 / prompt and solution

Choose several positions t , including a first touch, a partly overlapping position and a fully nested position. Before making many measurements, sketch your prediction for the entire graph of $h(t)$. Mark where you think its rule changes.

Solution: Valid sample choices include $t=0$ giving 0, $t=1$ giving 1, $t=5/2$ giving 2, $t=4$ giving 1, and $t=5$ giving 0. The full graph has changes at 0,2,3,5, as proved in prompt 2. The learner may choose different samples and should retain an incorrect prediction for comparison. Measurements alone do not certify all intermediate values.

Predict what the sliding window draws

1 / hints to offer only after exploration

1. Notice which endpoint limits the overlap as you slide.

2 / prompt and solution

Derive $h(t)$ for every real t . Explain each breakpoint using the two windows, including endpoint contacts. Use the formula to check and revise your prediction.

Solution: The intersection starts at $\max(0, t-2)$ and ends at $\min(3, t)$, provided the latter is at least the former. Thus $h = \max(0, \min(3, t) - \max(0, t-2))$. It equals 0 for $t \leq 0$, t for $0 \leq t \leq 2$, 2 for $2 \leq t \leq 3$, $5-t$ for $3 \leq t \leq 5$, and 0 for $t \geq 5$. At shared case endpoints the formulas agree. The corners occur when a moving endpoint crosses a fixed endpoint; isolated contact points have zero length.

2 / hints to offer only after exploration

1. Find the later left endpoint and the earlier right endpoint before subtracting.

Student page 2: Build windows for a chosen graph

Algebra continuation: arbitrary positive widths

Now choose any lengths $a, b > 0$. The fixed interval is $[0, a]$ and the moving one is $[t-b, t]$. Its overlap function is $h_{(a,b)}$. You may temporarily assume $a \geq b$ when describing cases.

3 / prompt and solution

Find the full formula for your chosen lengths, then for arbitrary $a \geq b > 0$. Swap the roles of a and b and prove whether the whole graph changes.

Solution: For $a \geq b$, the cases are 0 when $t \leq 0$; t on $[0, b]$; b on $[b, a]$; $a+b-t$ on $[a, a+b]$; and 0 when $t \geq a+b$. When $a=b$ the plateau has zero duration, giving a triangle. Equivalently $h_{(a,b)}(t) = \max(0, \min(a, b, t, a+b-t))$, which is symmetric in a and b and proves the unchanged graph when roles swap. The general max/min interval-intersection formula gives the same conclusion.

3 / hints to offer only after exploration

1. The smaller width controls the peak, and the larger width determines how long it stays fully inside.

Predict what the sliding window draws

4 / prompt and solution

Choose a desired distance $S > 0$ between the first and last contact positions, and a peak height $H > 0$, for such unit-height windows. Decide exactly when your design is possible and find every ordered pair (a, b) that realizes it. Test $S=7, H=2$.

Solution: The closure of the positive-overlap interval is $[0, a+b]$, so $S=a+b$ and $H=\min(a, b)$. Hence a design exists exactly when $S \geq 2H > 0$. The widths are H and $S-H$ in either order; if $S=2H$ the two orders coincide and the graph is triangular. Conversely these widths produce the desired graph by prompt 3. For $S=7, H=2$ the ordered possibilities are $(5, 2)$ and $(2, 5)$, with plateau duration 3. The graph identifies the unordered pair but cannot identify which window was called fixed.

4 / hints to offer only after exploration

1. Use the sum and the smaller member to recover two positive lengths.

Student page 3: Why this is an integral operation

Continuation: a supplied integral interpretation

Define $f(x)=1$ on $[0, a]$ and 0 elsewhere, and $g(x)=1$ on $[0, b]$ and 0 elsewhere. Their convolution is $(f * g)(t) = \int_{-\infty}^{\infty} f(x)g(t-x)dx$. Supplied fact: the integral of an interval's indicator is its length; changing finitely many endpoint values does not change the integral.

5 / prompt and solution

Show exactly why this integral equals your overlap length. Explain what graph you would get if you instead slid $[t, t+b]$ while continuing to label its left endpoint t . Would making the input intervals open change the convolution?

Solution: The product is 1 exactly when $0 \leq x \leq a$ and $0 \leq t-x \leq b$, the latter equivalent to $t-b \leq x \leq t$. Its integral is therefore the length of $[0, a] \cap [t-b, t]$. Sliding $[t, t+b]$ gives $h_{(a,b)}(t+b)$, shifting the graph left by b ; for $(a,b)=(3,2)$ its support closure becomes $[-2, 3]$. Replacing closed inputs by open ones only changes membership at finitely many x for each t , so the integral and graph stay unchanged, although pointwise input values differ.

5 / hints to offer only after exploration

1. Solve both inequalities in $g(t-x)=1$ before drawing the moving interval.

Predict what the sliding window draws

6 / prompt and solution

Find the area under your 3-by-2 overlap graph. Then prove the area under $h_-(a,b)$ is ab for every positive a,b , using the graph's pieces. What does this area remember that the choice of fixed window does not?

Solution: For $(3,2)$, the two triangles each have area 2 and the central rectangle has area 2, totaling 6. For $a \geq b$, the two triangular areas total b^2 and the plateau rectangle contributes $b(a-b)$, giving ab . Swapping handles $a < b$; equality $a=b$ has no rectangle and still gives ab . The area is the product of the original window lengths, independent of which is fixed. This is a direct proof for the chosen indicators, not a proof of a general integral-exchange theorem.

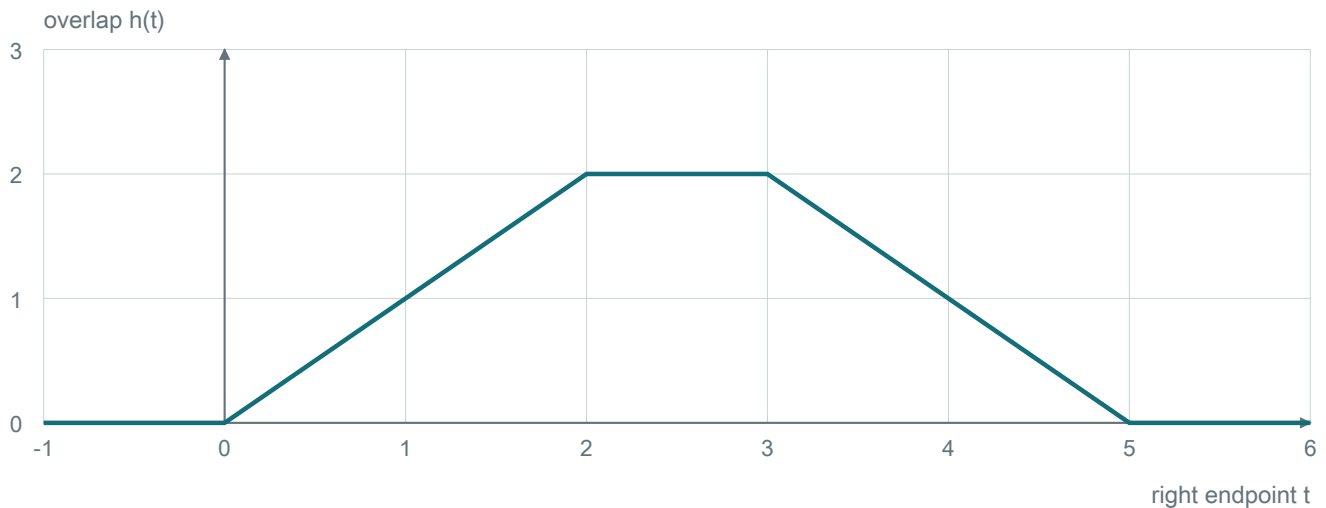
6 / hints to offer only after exploration

1. Keep the two triangular pieces and the plateau rectangle separate.

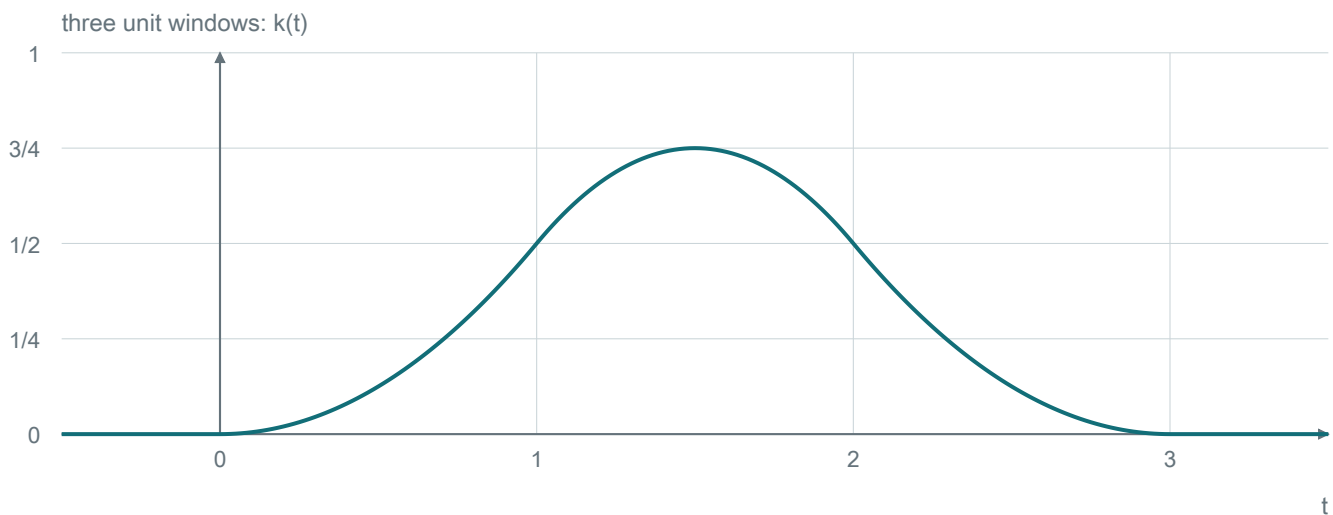
Worked overlap graphs

Predict what the sliding window draws

The 3-by-2 windows: the parameter t is the moving right endpoint. The graph below is a solution figure; keep it out of the student launch.



Corners at $t=0,2,3,5$; peak 2. Two triangular areas of 2 and a plateau area of 2 give total area 6.



The third-window graph joins with matching value and first derivative at $0,1,2,3$. Its second derivative jumps there. The peak is $3/4$ at $t=3/2$.

Worked overlap graphs

Optional facilitator extension 1

Gate: Definite integrals, piecewise polynomials and one-sided derivative comparisons.

Take three unit-width indicators. The first two convolve to $h(t)=t$ on $[0,1]$, $2-t$ on $[1,2]$, and zero elsewhere. Find the full third convolution $k(t)=\int_{-\infty}^{\infty} (t-1)^+ h(x)dx$ and determine whether k has a continuous first derivative.

Solution: Let $z_+=\max(z,0)$. Since $h(x)=x_+-2(x-1)_++(x-2)_+$, integrating over $[t-1,t]$ gives $k(t)=(t_+^2-3(t-1)_+^2+3(t-2)_+^2-(t-3)_+^2)/2$. Equivalently k is zero for $t\leq 0$; $t^2/2$ on $[0,1]$; $(-2t^2+6t-3)/2$ on $[1,2]$; $(3-t)^2/2$ on $[2,3]$; and zero for $t\geq 3$. The values and first derivatives match at $0,1,2,3$, so k is continuously differentiable. Its second derivative jumps at those points, so it is not twice continuously differentiable. The additional averaging has smoothed the corners by one derivative in this instance.

Mathematics and prior use

Convolution of interval indicators, commutativity, support and area. The exact continuum integral is represented by overlap length; no discretization or unjustified general transform theorem is used.

Prior use: No Week 1-10 task already computes this moving interval intersection. GA-14 discusses samples of a signal; this family derives a whole continuous graph and deliberately distinguishes that proof from sampled evidence.

Source: Stephen Boyd, EE102 Lecture 3: The Laplace Transform

Slides 3-29 and 3-30 (PDF pp.29-30), convolution definition, commutativity and transform calculation.. The source supplies the transform/convolution context. The 3-by-2 windows, choice of graph dimensions and full endpoint formulas are new; the core does not invoke a general Laplace inversion theorem. Evidence: The convolution definition and change-of-variable calculation were inspected. The worksheet extends causal indicators by zero to the real line and derives its interval overlaps, inverse design and third-window formula independently. Inspected 2026-09-25.

<https://web.stanford.edu/class/ee102k/laplace.pdf>

A graph feeds back its own mean

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced/specialist. The mean reduction is a strong surprise for integration-ready learners; graphs alone do not establish the compatibility condition. Unpiloted.

A complete unique/no-solution/family classification with necessity and sufficiency separately proved.

Prerequisites and preparation

Core: Definite integrals, functions and linear equations are core prerequisites. Necessary compatibility must be checked again by constructing the function.

Extensions: The changed rank-one kernel uses a weighted integral moment; no general compact-operator theorem is required.

Reading and representation: Read a function equation holding for every $x \in [0, 1]$; the integral uses a separate dummy variable. Signed functions are allowed.

Materials: Printed forcing graphs for x^2 and $2x-1$, feedback dial cards, pencil and blank graph space. No invented discrete averaging substitute. Preparation: about 5 minutes.

10 minutes trying chosen forcing/dial pairs; 15-25 deriving and verifying the mean condition; a movement reset; 15-25 for critical feedback and sensitivity. The changed kernel is a separate advanced continuation.

Launch and facilitation

Choose a forcing graph and a feedback dial. Look for a graph equal to that forcing plus the dial times its own mean. Could changing only the dial leave one, many, or no possible graphs?

Student page 1: Choose a forcing and a dial

Core: definite integrals and a function equation

All functions are continuous and real-valued on $[0, 1]$. Seek $u(x) = f(x) + \lambda m$ for every x , where $m = \int_0^1 u(t) dt$ is u 's own mean. Choose $f(x) = x^2$ or $f(x) = 2x-1$, and λ from $1/2, 1, 2$. Negative heights are allowed.

1 / prompt and solution

Choose a forcing and dial. Predict whether there is one solution, no solution, or more than one. Try a vertically shifted copy of the forcing graph and check whether its actual mean makes the feedback equation true.

Solution: For $f=x^2$, mean $F=1/3$: at $\lambda=1/2$, $m=2/3$ and $u=x^2+1/3$; at $\lambda=2$, $m=-1/3$ and $u=x^2-2/3$; at $\lambda=1$ there is no solution. For $f=2x-1$, $F=0$: at $\lambda=1/2$ or 2 the only solution is $u=2x-1$; at $\lambda=1$ every $u=2x-1+C$ works. A learner may choose any pair. Every displayed function can be checked by integrating it and substituting; the impossible case is justified by prompt 2 rather than a failed search.

A graph feeds back its own mean

1 / hints to offer only after exploration

1. A vertical shift by C adds C to the mean because the interval has length one.

2 / prompt and solution

Introduce the unknown mean m before trying to determine every value $u(x)$. Derive an equation that every solution must satisfy. Use it to justify one impossible case from the available choices.

Solution: Put $F = \int_0^1 f(t) dt$. Integrating the function equation gives $m = F + \lambda m$, hence $(1 - \lambda)m = F$. This is necessary for every solution because the interval length is 1. At $\lambda = 1, f = x^2$ it becomes $0 = 1/3$, a contradiction, so no solution exists. This reduction alone is not yet a sufficiency proof; a proposed m must reconstruct a function whose mean is indeed m .

2 / hints to offer only after exploration

1. The integral of the constant λm over $[0, 1]$ is λm .

Student page 2: Classify the whole equation

Core proof: necessity, construction and uniqueness

Keep the same equation but now let f be any continuous real function on $[0, 1]$ and let λ be any real number. Write $F = \int_0^1 f(t) dt$.

3 / prompt and solution

Handle $\lambda = 1$ completely. State exactly which f allow a solution and describe every solution when they exist. Prove both directions.

Solution: Necessity gives $F = 0$. If $F \neq 0$ no solution exists. If $F = 0$, every $u = f + C$, with arbitrary real C , has mean C and satisfies $u = f + m$. Conversely the original equation with $\lambda = 1$ forces $u - f = m$ to be constant, so every solution is on that list. There are infinitely many distinct solutions, including all real choices of C , whenever $F = 0$.

3 / hints to offer only after exploration

1. Do not divide by $1 - \lambda$ when it equals zero.

A graph feeds back its own mean

4 / prompt and solution

Handle every $\lambda \neq 1$. Give the solution, verify it directly and prove uniqueness. Put this case together with the critical case in a complete classification.

Solution: The only possible mean is $m = F/(1-\lambda)$, so the candidate is $u = f + \lambda F/(1-\lambda)$. Its integral is $F + \lambda F/(1-\lambda) = F/(1-\lambda)$, exactly m , proving sufficiency. Any other solution would have the same mean by the scalar equation and hence the same pointwise values by the original equation, proving uniqueness. This includes $\lambda = 0$, giving $u = f$. Together with prompt 3: $\lambda \neq 1$ gives one solution for every f ; $\lambda = 1, F \neq 0$ gives none; $\lambda = 1, F = 0$ gives exactly the one-parameter family.

4 / hints to offer only after exploration

1. Check the reconstructed mean rather than only repeating the scalar equation.

Student page 3: Repair the forcing, then perturb it

Continuation: exact sensitivity near the critical dial

You may adjust the forcing by a constant. First consider $f_a(x) = x^2 - a$ at $\lambda = 1$. Then move the dial to $\lambda = 1 - \delta$ with $0 < \delta \leq 1$ and use $f_\varepsilon(x) = x^2 - 1/3 + \varepsilon$.

5 / prompt and solution

Choose a so the critical equation has solutions. Describe all of them. If a is changed by any nonzero amount while λ stays exactly 1, what happens? Prove the claim.

Solution: The mean of f_a is $1/3 - a$, so the only successful adjustment is $a = 1/3$. Every solution is $u = x^2 - 1/3 + C$. Changing a by any nonzero amount makes the mean nonzero, so no solution exists at $\lambda = 1$. This is a compatibility failure, not a unique solution of very large magnitude. Exact zero mean is the necessary and sufficient condition at the critical dial.

5 / hints to offer only after exploration

1. A constant adjustment changes the forcing mean by exactly that amount.

6 / prompt and solution

For $\lambda = 1 - \delta$, compare the unique solution for ε with the one for $\varepsilon = 0$. Find their maximum absolute difference. Given an allowed solution error $\eta > 0$, how small must $|\varepsilon|$ be? Test $\varepsilon = 1/100$ at $\delta = 1/10$ and $\delta = 1/100$.

Solution: Here $F = \varepsilon$ and $m = \varepsilon/\delta$. Substitution gives $u_\varepsilon = x^2 - 1/3 + \varepsilon + (1-\delta)\varepsilon/\delta = x^2 - 1/3 + \varepsilon/\delta$. Thus $u_\varepsilon - u_0$ is the constant ε/δ and the maximum absolute difference is $|\varepsilon|/\delta$. Error at most η holds exactly when $|\varepsilon| \leq \eta\delta$. With $\varepsilon = 1/100$ the two specified δ values give errors $1/10$ and 1 . For every $\delta > 0$ there is still exactly one solution; growing sensitivity as δ approaches zero is distinct from the critical no-solution/many-solution alternatives.

A graph feeds back its own mean

6 / hints to offer only after exploration

1. Simplify the direct forcing shift together with the feedback shift before comparing solutions.

Optional facilitator extension 1

Gate: Definite integrals and weighted moments of functions.

Replace the feedback by $u(x)=f(x)+\lambda x \int_0^1 u(t)dt$. For continuous f , classify all solutions for every real λ . Exhibit one forcing with no solution at the critical dial and one with infinitely many.

Solution: Let $m=\int_0^1 u(t)dt$ and $M=\int_0^1 t f(t)dt$. Multiplying by x and integrating gives $m=M+(\lambda/3)m$ because $\int_0^1 t^2 dt=1/3$. If $\lambda \neq 3$, $m=M/(1-\lambda/3)$ and $u=f+[\lambda M/(1-\lambda/3)]x$; direct weighted integration verifies the mean and uniqueness follows. At $\lambda=3$ there is no solution if $M \neq 0$. If $M=0$, every $u=f+Cx$ has weighted moment $C/3$ and satisfies the equation, and the equation forces every solution to this form. For $f=x$, $M=1/3$ gives no solution at 3. For $f=x-2/3$, $M=1/3-1/3=0$ and the whole family works. The critical value moved because the kernel and moment normalization changed.

Mathematics and prior use

A rank-one Fredholm equation $u=f+\lambda Pu$, where P maps a function to its constant mean on the unit interval. The critical eigenvalue gives compatibility and nonuniqueness; the general Fredholm alternative is background only.

Prior use: GA-05 and delivered AP-01 use finite-network harmonic averages. Here the unknown is a continuous function, its continuum integral drives a rank-one equation, and a singular feedback parameter changes solvability. GA-33 solves differential equations; this is an integral equation.

Source: Michael O'Neil, Integral Equations and Fast Algorithms

Theorem 3, The Fredholm Alternative, printed p.54 (PDF p.55).. The original normalization and critical $\lambda=1$ are retained. New forcing choices, an adjustable zero-mean repair and a quantified near-critical perturbation deepen the investigation; no general compact-operator proof is claimed. Evidence: The compact-operator alternative and its compatibility statement were inspected. The selected constant and xt kernels are each solved completely by a scalar mean/moment calculation. Inspected 2026-09-25.

https://cims.nyu.edu/~oneil/courses/fa17-math2011/int_eq_notes_2017.pdf

Choose what best flat height means

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with fraction and algebra gates. The unequal widths make the choice of objective visible. Unpiloted; use the first two pages as a complete investigation before the parameter classification.

Prove that the same landscape has three different best heights under three precisely stated costs.

Prerequisites and preparation

Core: Fractions, squared and absolute errors, and signed algebra. Rectangle sums supply the integrals; completing the square and a full piecewise argument are the core.

Extensions: Finite weighted sums and expansion; Hilbert-space projection is context, not a prerequisite for the direct proof.

Reading and representation: Read a two-height graph, choose a loss rule and distinguish a unique optimizer from a whole interval of ties.

Materials: Printed landscape with width ratio 2:1, straight strip or ruler to move as a horizontal candidate, pencils and optional transparent sleeve. Preparation: about 5 minutes.

Explore candidate heights and rule cards for 10 minutes; allow 15 for the square certificate, a short break, then 20-30 for comparing objectives. Variable widths can be a later meeting.

Launch and facilitation

Place a flat line through this landscape. Choose how you will charge for errors, then let a partner propose a cheaper line.

Student page 1: Choose how errors cost

Core: weighted areas and squares

The landscape is $f(x)=0$ for $0 \leq x < 2/3$ and $f(x)=6$ for $2/3 \leq x \leq 1$. A candidate is any real constant height c . Rule cards: $Q(c) = \int_0^1 (f-c)^2 dx$; $A(c) = \int_0^1 |f-c| dx$; $W(c) = \max_{0 \leq x \leq 1} |f(x)-c|$. For these step functions, each integral is a sum of rectangle widths times their heights.

1 / prompt and solution

Choose one rule card and two candidate heights. Predict the winner before calculating. Challenge your partner to improve the winner, and record which rule your comparison uses.

Solution: Any two real choices are legal. For example $c=0,3$ give $Q=12,9$, $A=2,3$ and $W=6,3$. Thus the preferred candidate can depend on the chosen card. All comparisons use $Q=(2/3)c^2+(1/3)(6-c)^2$, $A=(2/3)|c|+(1/3)|6-c|$, $W=\max(|c|,|6-c|)$. A failed search is not yet an optimality proof.

Choose what best flat height means

1 / hints to offer only after exploration

1. Compute the two rectangle contributions separately.
2. Keep the horizontal widths even when the two heights look equally important.

2 / prompt and solution

Using rule Q, find the best height among ALL real c . Give a certificate that excludes every better competitor, and decide whether any different height ties.

Solution: $Q(c) = (2/3)c^2 + (1/3)(6-c)^2 = c^2 - 4c + 12 = (c-2)^2 + 8$. Hence the unique minimizer is $c=2$ and minimum cost 8. The square is nonnegative for every real c and is zero only at 2. The unweighted midpoint 3 costs 9 because the low region has twice the width.

2 / hints to offer only after exploration

1. Can you write the cost as a nonnegative square plus a constant?

Student page 2: One landscape, different demands

Core proof: absolute values and worst error

Keep the landscape unchanged. A rule measures what it says: area under squared error, area under absolute error, or the single worst vertical error.

3 / prompt and solution

Find every best height for rule A. Your proof must include candidates below 0 and above 6, not just the heights you tried.

Solution: For $c < 0$, $A = 2 - c$; for $0 \leq c \leq 6$, $A = 2 + c/3$; for $c > 6$, $A = c - 2$. Each outer branch decreases toward or increases away from $[0, 6]$, and the middle branch is strictly increasing. Thus $c=0$ is the unique minimum, with cost 2. Endpoint formulas agree.

3 / hints to offer only after exploration

1. Break the real line where the signs inside absolute values change.

Choose what best flat height means

4 / prompt and solution

Find every best height for rule W and prove the lower bound. Write a claim about “the best average height” that your three results refute, then repair the claim so it states a true mathematical question.

Solution: For any c , $6 = |6-0| \leq |6-c| + |c| \leq 2W(c)$, so $W \geq 3$. At $c=3$ both errors equal 3. If $W \leq 3$ then $|c| \leq 3$ and $|6-c| \leq 3$ force $c=3$, proving uniqueness. “One height is best regardless of how error is charged” is false: Q,A,W have respective optima 2,0,3. A repaired question names the landscape, allowed constants and exact objective.

4 / hints to offer only after exploration

1. A worst-error allowance must cover both heights at once.

Student page 3: Design the widths

Continuation: a parameter and complete cases

Now f_p is 0 on $[0,p)$ and 6 on $[p,1]$, where $0 < p < 1$. Both heights occupy a positive width. Keep all three cost definitions.

5 / prompt and solution

Classify all minimizing heights for Q, A and W as p varies. Identify any value of p where a unique answer becomes a whole set of answers.

Solution: $Q_p = p c^2 + (1-p)(6-c)^2 = (c-6(1-p))^2 + 36p(1-p)$, so its unique optimum is $6(1-p)$. For $0 \leq c \leq 6$, $A_p = 6(1-p) + (2p-1)c$; outside this interval moving toward it strictly reduces cost. Thus A has unique optimum 0 when $p > 1/2$, unique optimum 6 when $p < 1/2$, and every $c \in [0,6]$ when $p = 1/2$. W is independent of p in this open range and uniquely minimized at 3. Restricting $0 < p < 1$ avoids mixing absent regions or isolated endpoint values into the maximum convention.

5 / hints to offer only after exploration

1. For absolute loss, watch the sign of $2p-1$.
2. Both heights remain present, so the maximum rule still compares the same two errors.

6 / prompt and solution

Choose a desired squared-loss optimum h with $0 < h < 6$. Build the low-region width that produces it and prove it is the only such width. Can that same landscape have h as its unique absolute-loss optimum?

Solution: The required equation $6(1-p) = h$ gives the unique $p = 1 - h/6$, which lies in $(0,1)$. For example $h = 9/2$ needs $p = 1/4$. No interior h can be the unique A optimum: those unique optima are only 0 or 6; when $p = 1/2$, every interior height ties with every height in $[0,6]$. Even $h = 3$ is then a tie rather than a unique A optimum.

Choose what best flat height means

6 / hints to offer only after exploration

1. Use your parameter formula, then check which values the absolute rule can select uniquely.

Optional facilitator extension 1

Gate: Finite sums, expansion and positive weights.

Let heights $a_1, \dots, a_n$ occupy positive widths $w_1, \dots, w_n$ with $\sum w_i = 1$. Find the best constant for squared loss. Prove the residual has weighted sum zero and derive an identity certifying uniqueness.

Solution: Set $\mu = \sum w_i a_i$. Then $\sum w_i (a_i - \mu) = 0$. Expanding $a_i - c = (a_i - \mu) + (\mu - c)$ gives $\sum w_i (a_i - c)^2 = \sum w_i (a_i - \mu)^2 + (\mu - c)^2$; the cross term vanishes and the weights sum to one. Therefore μ uniquely minimizes the cost, even if all heights coincide. This is the orthogonality of the residual to constants in this finite step-function model.

Mathematics and prior use

Weighted mean, weighted median and midrange; direct projection onto constant functions for a step function in L^2 . No general Hilbert-space projection theorem is proved by a single example.

Prior use: GA-13 already minimizes maximum error for a parabola over all affine lines. This uses a step landscape, constants and three distinct losses, then changes the interval weights. GA-20 effort costs also compare choices of objective, but for motion under endpoint constraints rather than a fixed landscape.

Source: Sheldon Axler, Measure, Integration & Real Analysis

§8B, Definition 8.34 and Theorem 8.37, printed pp.227-228 (PDF pp.242-243).. Projection supplies context; the new landscape and every loss classification are derived directly. Evidence: Reopened the June 12, 2026 PDF on 2026-09-26; inspected nearest-point definition and residual orthogonality. No general projection existence theorem is assumed in the student proof.

<https://measure.axler.net/MIRA.pdf>

Defeat a claimed derivative bound

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced/specialist. Calculus and quantifiers are the satisfying core. Small-looking plots alone would miss the point; the exact endpoint derivative and norm ratio make the result decisive. Unpiloted.

A polynomial construction defeats every fixed bound on derivative size in terms of graph height alone.

Prerequisites and preparation

Core: Calculus is essential: polynomial derivatives, maximum norms and quantified counterexamples. Fixed-scale graph sketches are evidence; the all-bounds obstruction needs exact derivatives.

Extensions: Forward differences, bounded measurement errors and a worst-case triangle inequality. The integration comparison on page 3 needs definite integrals.

Reading and representation: Read statements with “for every function” and “for every proposed bound”; distinguish input size, derivative size and their ratio.

Materials: Fixed-scale axes on $[0, 1]$, polynomial trial cards, pencil; calculator optional for plotting but not needed for the proofs. Preparation: about 4 minutes.

Allow 10 minutes to choose an error budget and slope challenge, 15-20 for exact polynomial construction, and 20 for a universal obstruction. Stronger norms and integration belong to a later calculus discussion if needed.

Launch and facilitation

Make a graph stay inside a tiny height allowance while its slope at the right endpoint beats a chosen target. Keep the horizontal and vertical scales fixed.

Student page 1: Small height, chosen slope

Calculus core: powers and derivatives

All graphs use $0 \leq x \leq 1$ and fixed axis scales. For a continuous real function, write $\|f\|_\infty = \max_{0 \leq x \leq 1} |f(x)|$. A polynomial is allowed; no graph may change its horizontal domain to win.

1 / prompt and solution

Choose a height allowance $\varepsilon > 0$ and a slope target $M > 0$. Build a polynomial $a x^n$ with $\|a x^n\|_\infty \leq \varepsilon$ and derivative at $x=1$ at least M . Give a rule that works for every chosen pair. Also meet $\varepsilon=1/1000$ and $M=7$ exactly.

Solution: Take $a=\varepsilon$ and a positive integer $n \geq M/\varepsilon$. Since $0 \leq x^n \leq 1$, the norm is ε and the endpoint derivative is $\varepsilon n \geq M$. For the supplied target, $f=x^{7000}/1000$ has norm $1/1000$ and $f'(1)=7$. This is exact even when the narrow rise is hard to see on paper.

Defeat a claimed derivative bound

1 / hints to offer only after exploration

1. Where is x^n largest on this interval?
2. The endpoint derivative multiplies the coefficient by n .

2 / prompt and solution

For $f_n(x)=x^n/n$, $n=1,2,\dots$, find the exact norms of f_n and f_n' . Sketch two choices on the supplied common axes and explain which exact fact the sketch cannot certify on its own.

Solution: $f_n'=x^{n-1}$, with $f_1'=1$. Thus $\|f_n\|_\infty=1/n$ and $\|f_n'\|_\infty=1$, attained at $x=1$. For example $n=2$ and 8 both rise from 0 but reach $1/2$ and $1/8$; their endpoint slopes are both 1. A raster or hand drawing cannot certify the supremum or the universal n statement.

2 / hints to offer only after exploration

1. Find a bound valid on the whole interval and a point attaining it.

Student page 2: Challenge every bound

Calculus proof: quantifiers and uniform convergence

A proposed bound is a finite $C \geq 0$ such that $\|f'\|_\infty \leq C\|f\|_\infty$ for EVERY continuously differentiable real function f on $[0,1]$. Differentiation is called bounded under these norms if some such C exists.

3 / prompt and solution

Let a partner name C . Defeat the claim using a polynomial, then prove that your strategy defeats every finite C . Why is one unsuccessful candidate C not enough?

Solution: For f_n , the ratio $\|f_n'\|_\infty/\|f_n\|_\infty=n$. Choose an integer $n>C$; then $1>C/n$, contradicting the proposed inequality. The construction depends on arbitrary C and thus rules out all finite bounds. Refuting only $C=1$ would leave open a larger valid bound. Differentiation is linear, but linearity alone does not give boundedness in infinite-dimensional spaces.

3 / hints to offer only after exploration

1. Translate the proposed inequality into a statement about n .

Defeat a claimed derivative bound

4 / prompt and solution

A graph sequence converges uniformly to zero when its maximum absolute height tends to zero. Do f_n and their derivatives both do so? Would $g_n(x)=x^n$ establish the same small-input claim? Explain the endpoint role.

Solution: $\|f_n\|_\infty = 1/n \rightarrow 0$, but $\|f_n'\|_\infty = 1$ for every n , so the derivatives do not converge uniformly to zero. Their value at 1 stays 1. For each fixed $x < 1$, $f_n'(x) \rightarrow 0$, which is a different pointwise statement. The sequence g_n has $\|g_n\|_\infty = 1$ because $g_n(1)=1$; it does not give uniform convergence of inputs to zero. Its large derivative still gives unbounded ratios, but it does not establish the stated small-input convergence contrast.

4 / hints to offer only after exploration

1. Check $x=1$ before guessing a limit from the graph away from the edge.

Student page 3: Change the measurement

Advanced continuation: norms and integrals

Now measure a C^1 input by $\|f\|^* = \|f\|_\infty + \|f'\|_\infty$, while still measuring its derivative by $\|f'\|_\infty$. Separately, define $If(x) = \int_0^x f(t)dt$ for continuous real f , using supremum norms for both input and output.

5 / prompt and solution

Find the smallest constant C for $\|f'\|_\infty \leq C\|f\|^*$. Does any nonzero function attain equality for this smallest C ? Distinguish a sharp bound from an attained ratio.

Solution: $C=1$ works since $\|f'\|_\infty \leq \|f\|_\infty + \|f'\|_\infty$. For f_n the ratio is $1/(1+1/n) = n/(n+1) \rightarrow 1$, so no $C < 1$ works. If $f \neq 0$, then $\|f\|_\infty > 0$, making the ratio strictly below 1. Thus the operator norm is exactly 1, with no nonzero attaining input. The zero function trivially satisfies equality $0=0$ but has no defined output/input ratio.

5 / hints to offer only after exploration

1. Use the same witnesses, with their new input sizes.
2. A nonzero continuous function has positive maximum absolute value.

6 / prompt and solution

Find the smallest constant C for $\|If\|_\infty \leq C\|f\|_\infty$. Prove the bound for every continuous f , and find an input attaining it. Explain what changed between this experiment and task 5.

Solution: For $0 \leq x \leq 1$, $|If(x)| \leq \int_0^x |f(t)|dt \leq x\|f\|_\infty \leq \|f\|_\infty$. Thus $C=1$ works. For $f=1$, $If=x$ and both norms equal 1, so $C=1$ is sharp and attained. Task 5 kept differentiation and changed the input norm; this task changes the operator to integration and uses the original supremum norms.

Defeat a claimed derivative bound

6 / hints to offer only after exploration

1. Bound the integral by a rectangle of height $\|f\|_\infty$.

Optional facilitator extension 1

Gate: Signed bounded errors and finite differences; derivative approximation is a separate issue.

Suppose two measured values $z(x), z(x+h)$ each differ from true values $f(x), f(x+h)$ by at most ε , where $h > 0$ and $x, x+h \in [0, 1]$. Find the exact worst measurement contribution to the forward-difference error. Does this alone choose an optimal h for approximating f' ?

Solution: Writing the two errors as e_0, e_1 gives difference error $(e_1 - e_0)/h$, whose magnitude is at most $2\varepsilon/h$. Taking $e_1 = \varepsilon, e_0 = -\varepsilon$ attains it. This compares the measured difference with the true finite difference, not necessarily $f'(x)$. A truncation-error model involving regularity of f would be needed to balance that other error and justify an optimal h .

Mathematics and prior use

Unbounded differentiation $D: C^1[0, 1] \rightarrow C[0, 1]$ with both supremum norms, continuity under a stronger input norm, and bounded integration. Operator norm can be sharp without being attained by any nonzero input.

Prior use: GA-09 uses derivatives to test evolution laws; GA-17 shows amplification near a critical parameter. Here the derivative itself is the fixed operator and the input norm is the issue. The original x^n/n witness is retained but now supports learner-chosen thresholds, a complete no-bound argument and a norm repair.

Source: Sheldon Axler, Measure, Integration & Real Analysis

§6C, Definition 6.43 and Examples 6.44-6.45, printed pp.167-168 (PDF pp.182-183).. Use the norm and bounded-map definitions; all polynomial witnesses, changed-norm results and integration bounds are derived here. Evidence: Reopened on 2026-09-26. This precise §6C locator replaces the original broad Chapter 10 reference. The source examples are not falsely described as the worksheet derivative example.

<https://measure.axler.net/MIRA.pdf>

Choose a journey and its effort bill

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising for algebra-ready learners on finite schedules; advanced/specialist for the full variational claim. The cost is a mathematical rule, not an assertion about calories or fuel. Unpiloted.

An exact bill identity proves the unique cheapest schedule; adding a waypoint changes the winner in a controlled way.

Prerequisites and preparation

Core: Signed speeds, durations, squares and algebra suffice for finite schedules. The claim for every continuous piecewise- C^1 path has a separate calculus gate and uses the fundamental theorem.

Extensions: Definite integrals, endpoint constraints and equality in a nonnegative-square integral.

Reading and representation: Read a position-versus-time graph; distinguish signed displacement, total distance and a deliberately defined quadratic cost.

Materials: Position/time axes, speed-and-duration cards, pencil and optional counter moved along a number line. No physical energy measurement is required. Preparation: about 5 minutes.

Spend 10 minutes inventing two-speed trips, 15 comparing exact bills, then 15-20 generalizing schedules or adding a waypoint. Give the calculus page only to learners who know integrals, possibly at a later meeting.

Launch and facilitation

Get from position 0 to position 6 in three time units. You can rush, pause or go backward. What trip is cheapest when speed is squared before charging?

Student page 1: Invent a speed schedule

Core: signed displacement and weighted squares

Start at $y=0$ and finish at $y=6$ after time 3. First travel at constant signed speed v for time 1, then w for time 2. The bill is $E=v^2+2w^2$. This invented quadratic-effort rule is not a physical fuel law.

1 / prompt and solution

Choose two different legal schedules, one with a pause or backward motion. Draw their position graphs and compare their exact bills. What equation decides whether a proposed schedule finishes correctly?

Solution: Legality is $v+2w=6$. Examples: $(v,w)=(6,0)$ has bill 36 and a pause; $(0,3)$ has bill 18; $(-2,4)$ has bill 36 and initially goes backward; $(2,2)$ has bill 12. Draw straight graph segments through $(0,0),(1,v),(3,6)$. Any pair satisfying the endpoint equation is allowed, including negative speeds.

Choose a journey and its effort bill

1 / hints to offer only after exploration

1. A speed is charged for the length of time it is used.

2 / prompt and solution

Find the cheapest legal two-speed schedule and prove uniqueness among all real v, w . Try expressing a rival schedule as a change from your candidate.

Solution: Write $v=2+2a, w=2-a$; this describes every solution of $v+2w=6$. Then $E=(2+2a)^2+2(2-a)^2=12+6a^2$. The unique minimum is 12 at $a=0$, hence $v=w=2$. The certificate includes all signed speeds.

2 / hints to offer only after exploration

1. Use the endpoint equation to express both speeds with one free parameter.

Student page 2: Let a rival use more changes

Core proof: finite weighted sums

A rival may choose any finite list of durations $\Delta t_i > 0$ summing to 3 and real speeds v_i satisfying $\sum \Delta t_i v_i = 6$. The bill is $E = \sum \Delta t_i v_i^2$. Changes of speed occur only at the finitely many joins.

3 / prompt and solution

Can any such rival beat your cheapest schedule? Derive a formula for its excess cost above 12 and say exactly when it is zero.

Solution: Expand $\sum \Delta t_i (v_i - 2)^2 = E - 4 \sum \Delta t_i v_i + 4 \sum \Delta t_i = E - 24 + 12 = E - 12$. Therefore $E = 12 + \sum \Delta t_i (v_i - 2)^2 \geq 12$. Since every duration is positive, equality holds exactly when every $v_i = 2$. Different partitions can describe the same winning position path $y=2t$; they do not give different motions.

3 / hints to offer only after exploration

1. Use both the total-time and total-displacement constraints when expanding.

4 / prompt and solution

Now require the trip to pass through $y=4$ at time 1. Find and prove the new cheapest finite schedule, allowing any extra speed changes before or after that time. Does it refute your earlier theorem?

Solution: Split or refine the schedule at time 1. On $[0, 1]$, displacement 4 and duration 1 give $E_1 = 16 + \sum \Delta t_i (v_i - 4)^2 \geq 16$. On $[1, 3]$, displacement 2 and duration 2 give $E_2 = 2 + \sum \Delta t_i (v_i - 1)^2 \geq 2$. The unique winning motion uses speed 4 then 1, with bill 18. It does not refute the earlier minimum 12: the new constraint excludes $y=2t$, whose position at time 1 is only 2.

Choose a journey and its effort bill

4 / hints to offer only after exploration

1. Treat the two time intervals as separate endpoint problems.

Student page 3: Every admissible curve

Calculus continuation: FTC and equality in an integral

An admissible $y:[0,3]\rightarrow\mathbb{R}$ is continuous, is C^1 on each of finitely many closed subintervals, and satisfies $y(0)=0, y(3)=6$. One-sided derivatives at a join may differ; their point values do not affect $E=\int_0^3 (y')^2 dt$.

5 / prompt and solution

Prove the exact minimum and uniqueness for this whole class of curves, not only piecewise-linear schedules. Explain why the equality case determines the path at the joins as well.

Solution: The fundamental theorem on each piece telescopes to $\int_0^3 y' dt = 6$. Expanding gives $E = 12 + \int_0^3 (y' - 2)^2 dt \geq 12$. If equality holds, the continuous nonnegative integrand on every C^1 piece must be zero: a positive value would remain positive on a small interval and have positive integral. Thus $y' = 2$ within every piece. Continuity at joins and $y(0) = 0$ force $y(t) = 2t$ throughout, which attains 12. Derivative values assigned to individual join points are irrelevant.

5 / hints to offer only after exploration

1. Apply the fundamental theorem piece by piece before completing the square.

6 / prompt and solution

Test the family $y_a(t) = 2t + a t(3-t)$ in your effort formula. Then change the bill to total distance $L = \int_0^3 |y'| dt$. Find its minimum over all admissible paths and all values of a for which this family attains it. Does uniqueness survive?

Solution: $y_a' = 2 + a(3-2t)$, so $E = 12 + a^2 \int_0^3 (3-2t)^2 dt = 12 + 9a^2$. For distance, $L \geq \int_0^3 |y'| = 6$. Equality holds exactly for nondecreasing admissible paths: $L - 6 = \int_0^3 (|y'| - y')$, and continuity of y' on pieces rules out any negative derivative interval; conversely nonnegative derivatives imply monotonicity and $L = 6$. In this family the linear derivative is nonnegative throughout iff $2 - 3|a| \geq 0$, i.e. $|a| \leq 2/3$. Thus infinitely many distinct paths attain distance 6, while only $a = 0$ attains minimum quadratic effort.

6 / hints to offer only after exploration

1. For distance, compare $|y'|$ with y' point by point.
2. The minimum of a linear derivative on an interval occurs at an endpoint.

Choose a journey and its effort bill

Optional facilitator extension 1

Gate: Piecewise calculus and a global equality argument.

For prescribed endpoints $y(0)=A, y(T)=B$ with $T>0$, prove the exact quadratic minimum. Then prescribe finitely many time-position waypoints with strictly increasing times. State and prove the unique minimizing motion.

Solution: Set $v=(B-A)/T$. FTC and expansion give $\int_0^T (y')^2 = (B-A)^2/T + \int_0^T (y'-v)^2$. Equality forces $y=A+vt$ as above. For waypoints (t_i, b_i) , apply this identity on each interval: the minimum is $\sum (b_{i+1}-b_i)^2 / (t_{i+1}-t_i)$. Equality on every interval forces the unique piecewise-linear interpolation, which is admissible. Negative displacements and repeated positions are allowed; only repeated times were excluded.

Mathematics and prior use

A global variational minimum obtained by a square identity, including equality and waypoint constraints. An Euler-Lagrange stationary condition is not used as a sufficient minimum proof.

Prior use: Week 1 packing uses an attained lower bound; this changes the objects to speed schedules and paths. AP-12 minimizes travel time across media, while this fixes duration and minimizes a declared quadratic functional. GA-18 changes a loss rule for a landscape; the comparison here tests uniqueness of motion.

Source: UCL MATH0043, Calculus of Variations

§2, “Functionals and Extrema,” “The Statement of an Example Problem,” and Euler-Lagrange setup.. Use a functional on an explicitly admissible path class. The selected endpoint/waypoint instances and global square certificates are independently derived. Evidence: Reopened the functional setup on 2026-09-26. The source provides context rather than a claim that stationarity proves the worksheet global optimum.

https://www.homepages.ucl.ac.uk/~ucahmt0/latex_html/pandoc_chapter2.html

A mirror that is too short

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with coordinate and geometric proof gates. The first optimum is concrete; the shortened segment requires real abstraction and should not be settled by ruler precision. Unpiloted.

A global shortest route is certified by unfolding; when its contact becomes illegal, a separate argument proves the new endpoint winner.

Prerequisites and preparation

Core: Distances, equal-scale coordinates, reflection and the triangle inequality. The shortened-mirror proof uses a supplied strict-convexity tool with an explicit geometric justification; calculus is not required.

Extensions: Symbolic coordinates and a convex-combination argument covering all closed mirror segments.

Reading and representation: Read a legal-contact interval and distinguish an unattainable lower bound from a minimum actually achieved by a legal route.

Materials: Equal-scale coordinate mirror diagrams, ruler, string or two straight strips, pencil and tracing paper for reflection. Preparation: about 7 minutes.

Allow 10-15 minutes for string trials, 15 for unfolding, a short break, then 20 for the shortened mirror. The general segment theorem can be a later proof discussion.

Launch and facilitation

Touch the mirror once on the way from A to B. Choose the contact point and try to make the broken route shorter. Then lose the last part of the mirror.

Student page 1: Place the contact point

Core: a route, distances and a legal interval

$A=(-2,3)$, $B=(6,1)$. A legal route consists of the two straight segments $A \rightarrow M \rightarrow B$, where $M=(m,0)$ and $-3 \leq m \leq 5$. Minimize their total Euclidean length. Both endpoints lie above the mirror line.

1 / prompt and solution

Use string or drawn segments to choose and compare legal contact points. Try to improve your best route and record a conjectured winner. What would a proof need to exclude that your trials do not?

Solution: Any contacts in $[-3,5]$ are legal. For example $m=-2$ gives $3+\sqrt{65}$, $m=0$ gives $\sqrt{13}+\sqrt{37}$, and $m=4$ gives $\sqrt{45}+\sqrt{5}=4\sqrt{5}$. Measurements may suggest $m=4$, but cannot exclude the continuum of untried contacts or prove exact uniqueness. Later reflection supplies that certificate.

A mirror that is too short

1 / hints to offer only after exploration

1. Keep both legs straight and keep the contact on the allowed segment.

2 / prompt and solution

Write the exact length $F(m)$. Is the midpoint of the mirror always a shortest contact? Test that claim on this diagram with an exact comparison.

Solution: $F(m) = \sqrt{(m+2)^2 + 9} + \sqrt{(m-6)^2 + 1}$. The mirror midpoint is $m=1$, with $F(1) = \sqrt{18} + \sqrt{26}$. Its square is $44 + 12\sqrt{13} > 80$ because $\sqrt{13} > 3$, so $F(1) > 4\sqrt{5} = F(4)$. Thus the midpoint is not shortest. Both quantities are positive, so comparing squares is valid. A learner may use another exact comparison that disproves the midpoint claim.

2 / hints to offer only after exploration

1. The midpoint of $[-3, 5]$ is 1.
2. Use the Pythagorean theorem on each leg before comparing.

Student page 2: Unfold the route

Core proof: reflection and equality

Reflect a point across $y=0$ by changing the sign of its y -coordinate. You may use the triangle inequality: $|UV| + |VW| \geq |UW|$, with equality exactly when V lies on the straight segment from U to W .

3 / prompt and solution

Reflect B to B' . Construct the crossing of the straight segment AB' with $y=0$ and find its exact coordinate. Is it a legal contact?

Solution: $B' = (6, -1)$. $A + s(B' - A) = (-2 + 8s, 3 - 4s)$, so crossing $y=0$ requires $s = 3/4$ and gives $M = (4, 0)$. Since $4 \in [-3, 5]$, this contact is legal. The construction does not need a protractor: draw the reflected point and straight joining segment.

3 / hints to offer only after exploration

1. Solve the vertical coordinate first.

A mirror that is too short

4 / prompt and solution

Prove the exact shortest length and the uniqueness of its contact. State where reflection and the equality condition each enter the argument.

Solution: Reflection fixes every M on $y=0$ and preserves distance, so $|MB|=|MB'|$. Therefore $F(m)=|AM|+|MB'| \geq |AB'| = \sqrt{(8^2+4^2)} = 4\sqrt{5}$. The crossing $m=4$ attains the bound. Equality requires $M \in [AB']$, and this segment crosses the mirror line exactly once because its endpoint y -coordinates have opposite signs. Hence $m=4$ is the unique optimum.

4 / hints to offer only after exploration

1. A lower bound is a minimum only after a legal route attains it.

Student page 3: Remove part of the mirror

Proof continuation: a constrained minimum

Now only $-3 \leq m \leq 2$ is legal. Supplied tool: F is strictly convex, meaning $F((1-t)m+tn) < (1-t)F(m)+tF(n)$ for $m \neq n$ and $0 < t < 1$. Justification: apply the vector triangle inequality to each leg; vectors $(m+2, -3)$ and $(n+2, -3)$ cannot be positive parallel multiples unless $m=n$, and likewise for the other leg. Positive endpoint heights make the inequalities strict.

5 / prompt and solution

Choose the best contact on the shortened mirror. Compute its exact length. Explain why merely repeating the lower bound $4\sqrt{5}$ does not prove your new candidate is best.

Solution: The candidate is $m=2$, giving $F(2) = \sqrt{(4^2+3^2)} + \sqrt{((-4)^2+1)} = 5 + \sqrt{17}$. The previous lower bound remains true, but equality would require $m=4$, now illegal. Thus it gives only an unattained lower bound for this restricted problem. Trials suggest $m=2$ but the monotonicity argument in task 6 is the certificate.

5 / hints to offer only after exploration

1. Find where the unfolded equality contact lies relative to the remaining mirror.

6 / prompt and solution

Use the strict-convexity tool and the known unique minimum at 4 to prove that F strictly decreases to the left of 4 and strictly increases to its right. Finish the shortened-mirror proof, including uniqueness.

Solution: If $x < y < 4$, write $y = (1-t)x + t \cdot 4$ with $0 < t < 1$. Strict convexity gives $F(y) < (1-t)F(x) + tF(4) < F(x)$, because $F(4) < F(x)$. The endpoint case $y=4$ follows from uniqueness of the global minimum. Reversing the order similarly proves increase for $4 < x < y$: x is a strict convex combination of 4 and y , so $F(x) < F(y)$. Therefore every $m < 2$ in $[-3, 2]$ has $F(m) > F(2)$, proving the unique constrained minimum $5 + \sqrt{17}$.

A mirror that is too short

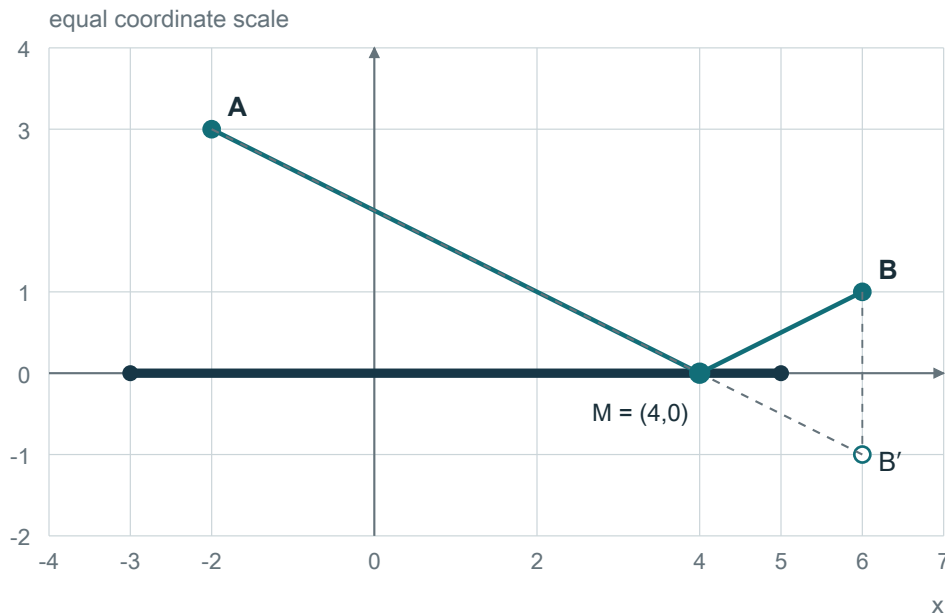
6 / hints to offer only after exploration

1. Express an intermediate contact as a weighted average of a farther contact and 4.

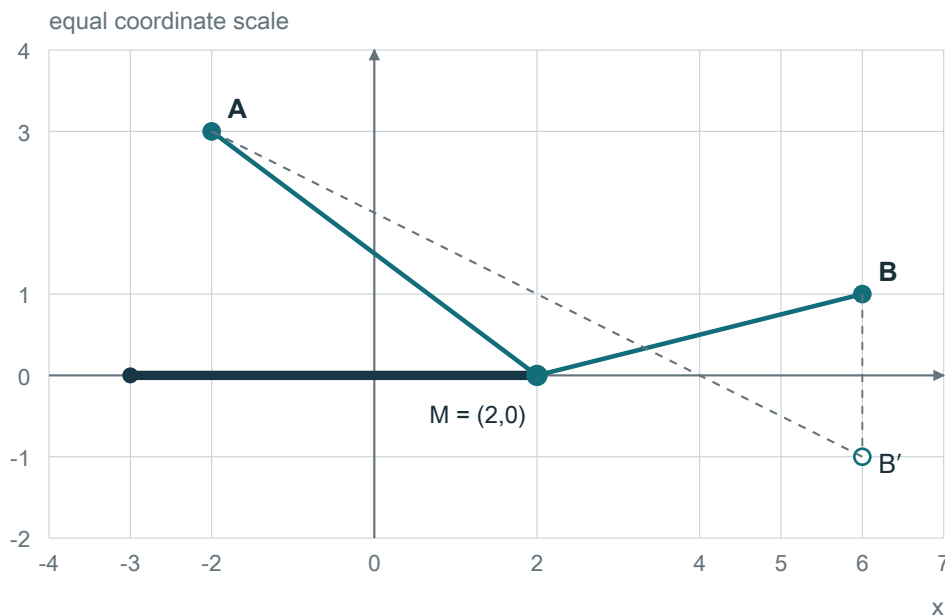
Worked mirror contacts

A mirror that is too short

Full mirror: the reflected straight segment crosses at $m=4$. Equality in the triangle inequality gives the unique minimum $4\sqrt{5}$.



Short mirror: the same reflected crossing is unavailable. Strict decrease to $m=4$ makes the legal endpoint $m=2$ uniquely best, with length $5+\sqrt{17}$.



Worked mirror contacts

Optional facilitator extension 1

Gate: Coordinate algebra, reflection and strict convexity.

Let $A=(a,h)$, $B=(b,k)$ with $h,k>0$. Find the shortest two-leg route touching any prescribed closed segment $[s,t]$ on $y=0$, where $s<t$. Give its unique contact for every relative position of the unconstrained crossing.

Solution: Reflect B to $(b,-k)$. The crossing occurs at fraction $h/(h+k)$ of the joining segment, hence $m^*=(ka+hb)/(h+k)$. The unrestricted length is $\sqrt{(b-a)^2+(h+k)^2}$, attained uniquely at m^* . Each distance term is strictly convex in m because its endpoint height is positive; the same argument as task 6 gives strict decrease before m^* and increase after. Thus the unique legal contact is s if $m^*<s$, m^* if $s\leq m^*\leq t$, and t if $m^*>t$. Evaluate $\sqrt{(m-a)^2+h^2}+\sqrt{(m-b)^2+k^2}$ there. Equality cases $m^*=s$ or t are included.

Mathematics and prior use

Reflection converts a broken-path minimum to an attained triangle-inequality bound. Strict convexity establishes the constrained endpoint rule; stationarity or visual symmetry alone is insufficient.

Prior use: Week 9 already uses billiard unfolding for trajectories and periods. This new instance centers continuous optimization, uniqueness and a finite-mirror obstruction; the old periodic path is not repeated. AP-12 changes speed across a boundary, whereas both legs here have the same distance cost.

Source: Anton Petrunin, Euclidean Plane and Its Relatives

§5D, Proposition 5.8 and Corollary 5.10, printed pp.39-40; §5F, Lemma 5.13, pp.41-42.. Use distance preservation by line reflection and the triangle-inequality framework. Both mirror intervals and the strict-convexity constraint proof are independently derived. Evidence: Reopened reflection and shortest-perpendicular proofs on 2026-09-26; no claim that the selected shortened-mirror instance is quoted from the source.

<https://arxiv.org/pdf/1302.1630>

Make a four-peg challenge

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for the planar core with oral rules and drawings; the exact certificate and general-dimensional theorem have separate gates. Unpiloted; give time to understand one-point hulls before challenging the theorem.

A strategy works for every four-point arrangement, including collinear and coincident cases, while three pegs can defeat it.

Prerequisites and preparation

Core: Convex hulls as stretched bands, planar cases and a complete argument. Exact certificates later require fractions and coordinate combinations; general dimension requires a separately supplied linear-algebra fact.

Extensions: Linear dependence in $\mathbb{R}^{d+1}$, signed coefficients and normalized convex combinations.

Reading and representation: Every label must belong to exactly one of two nonempty groups. Hulls include their boundaries, and the hull of one peg is just that point.

Materials: Printed coordinate peg boards, four labeled counters, two colored pencils or bands and blank arrangement grids. Avoid actual sharp pegs. Preparation: about 6 minutes.

Allow 10 minutes to invent and exchange arrangements, 15 for the two supplied boards, then 20-30 for cases and proof. Exact coordinate certificates are a later fraction-ready continuation.

Launch and facilitation

Place four pegs to make your partner's job hard. Your partner must split the labels into two groups whose stretched-band regions touch.

Student page 1: Challenge the splitter

Core: two nonempty groups and their hulls

Use four distinct labeled points. Assign every label to exactly one of two nonempty groups. A group's convex hull is the smallest filled convex region containing it: one point stays a point, two make a segment, and three noncollinear points make a filled triangle. Boundary contact counts as an intersection.

Make a four-peg challenge

1 / prompt and solution

Place four distinct pegs of your choice and trade the board with a partner. Find a legal split whose hulls meet, or try to explain why your arrangement defeats every split. Mark the actual common point when you succeed.

Solution: Every four-point arrangement permits a split, as proved later. For checking a chosen board, look for a point lying in the triangle of the other three; isolate it if present. Otherwise a noncollinear board is a convex quadrilateral and opposite pairs have intersecting diagonals. In a collinear board, isolate a nonextreme point. A convincing trial shows an actual shared point and assigns all labels once; merely overlapping bounding boxes is not enough.

1 / hints to offer only after exploration

1. A single peg can be one whole group.
2. Try separating alternate corners around the outside.

2 / prompt and solution

Solve both supplied boards: $A=(0,0), B=(6,0), C=(5,4), D=(0,3)$; and $A=(0,0), B=(6,0), C=(0,6), P=(2,1)$. Explain how the winning splits use different kinds of hulls.

Solution: For the first board, split $\{A,C\}$ and $\{B,D\}$; the two segments cross. For the second, split $\{P\}$ and $\{A,B,C\}$; P lies inside the triangle. The first witness is the intersection of two segments; the second is a one-point hull contained in a filled triangle. Exact common-point coefficients are derived later, so an accurate geometric construction is enough here.

2 / hints to offer only after exploration

1. For the second board, test whether P lies inside the outer triangle.

Student page 2: Can any arrangement defeat you?

Core proof: all cases, including degeneracy

You may use this elementary planar hull fact: four distinct points that are not all collinear either include a point in or on the triangle of the other three, or are the four corners of a convex quadrilateral. Diagonals of a convex quadrilateral intersect inside it.

Make a four-peg challenge

3 / prompt and solution

Prove that every four-distinct-point board has a legal split with meeting hulls. Include all-collinear points and three-collinear-plus-one-off-line points, and explain why your cases cover every board.

Solution: If one point lies in or on the triangle of the other three, isolate it. If no such point exists and the board is not collinear, the supplied hull fact gives four convex corners; split alternate corners to get the crossing diagonals. If all four are collinear, order them along the line and isolate either nonextreme point; the other three have a segment hull containing it. With exactly three collinear and one off-line, isolate the middle collinear point, which lies on an edge of the triangle of the other three. All groups are nonempty and all labels are used. These cases exhaust the supplied hull alternatives and their collinear boundary.

3 / hints to offer only after exploration

1. Do not silently discard a point sitting on a triangle edge.

4 / prompt and solution

Allow different labels to occupy the same location. Does the conclusion change? Then show that three points are not enough to guarantee success for every board.

Solution: If two labels coincide, put one alone and every other label in the second group; the matching label gives the same point in both hulls. If none coincide, task 3 applies. Thus the conclusion survives all coincidences. Three noncollinear points give a counterexample to a three-point guarantee: every partition has a singleton vertex versus the opposite segment, which does not contain that vertex. This says three can fail, not that every three-point arrangement fails.

4 / hints to offer only after exploration

1. Labels must be separated into groups even when their physical positions agree.

Student page 3: Certify contact with fractions

Continuation: convex combinations and coordinates

A convex combination is $\sum w_i P_i$ with $w_i \geq 0$ and $\sum w_i = 1$. It belongs to the hull of those points. Use the same two coordinate boards as before; no diagram measurement counts as an exact fraction.

5 / prompt and solution

Find the exact crossing of AC and BD in the first board. Express it as a convex combination from each group, and explain why the coefficients certify actual segment contact.

Solution: Write $A+r(C-A)=(5r,4r)$ and $B+s(D-B)=(6-6s,3s)$. From $4r=3s$ and $5r=6-6s$, $r=6/13, s=8/13$. The point is $(30/13, 24/13)=(7/13)A+(6/13)C=(5/13)B+(8/13)D$. Each pair of positive coefficients sums to 1, so the point lies inside both segments.

Make a four-peg challenge

5 / hints to offer only after exploration

1. Solve the two coordinate equations for the two segment parameters.

6 / prompt and solution

Express P from the second board as a convex combination of A, B, C . Move all terms to one side to obtain a signed relation with coefficient sum zero. Why do signed coefficients alone not describe a point inside one hull?

Solution: $P = (1/2)A + (1/3)B + (1/6)C$, since the B, C terms contribute $(2,0), (0,1)$, and weights sum to 1. Thus $(1/2)A + (1/3)B + (1/6)C - P = 0$ with coefficient sum 0. The relation compares two convex combinations, the second being $1 \cdot P$. A signed combination within a single group can leave its hull: for example $2B - A = (12,0)$ lies outside segment AB . Nonnegative weights and sum one are essential to a hull certificate.

6 / hints to offer only after exploration

1. Read the x - and y -coordinates first, then find the remaining weight at A .

Optional facilitator extension 1

Gate: Linear dependence and signed affine/convex combinations.

Supplied linear-algebra fact: any $d+2$ vectors in $\mathbb{R}^{d+1}$ are linearly dependent. Use the vectors $(p_i, 1)$ to prove that any $d+2$ labeled points in $\mathbb{R}^d$ split into two nonempty groups with intersecting hulls. Include zero coefficients and coincident points.

Solution: A nonzero relation $\sum \alpha_i (p_i, 1) = 0$ gives $\sum \alpha_i p_i = 0$ and $\sum \alpha_i = 0$. Since not all α_i vanish, at least one is positive and one negative. Set $S = \sum_{(\alpha_i > 0)} \alpha_i = \sum_{(\alpha_i < 0)} (-\alpha_i) > 0$. Dividing the point relation by S gives equal convex combinations of the positive and negative groups. Each is nonempty. Assign zero-coefficient labels arbitrarily to either group; adding them does not remove the existing convex-hull witness. Coincidences do not affect linear dependence or the argument. The supplied dimension fact is the higher-dimensional prerequisite.

Mathematics and prior use

Planar Radon partition, minimality of four in the worst case, and equal convex combinations extracted from a signed affine relation. The finite examples are not the proof for all arrangements.

Prior use: Week 1 tiling uses polygonal pieces but not intersecting hulls of a partition. The original four-peg idea is extended through adversarial arrangement choice, complete degeneracy cases and new exact convex-combination certificates. No previous trial uses these coordinate boards.

Make a four-peg challenge

Source: Daniel Hug and Wolfgang Weil, A Course on Convex Geometry

§1.2, Theorem 1.2.1 (Radon), printed p.22 (PDF p.22).. Use the affine-dependence partition mechanism; derive new planar examples and a hull-case proof before the linear-algebra extension. Evidence: Reopened the statement and complete positive/negative coefficient proof on 2026-09-26. Zero coefficients and nonempty groups are explicitly handled in this adaptation.

<https://users.fmf.uni-lj.si/lavric/hug%26weil.pdf>

Let the route turn a tangent arrow

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising only with spatial-vector readiness; the trigonometric continuation is advanced. A hand-carried arrow is approximate and the exact rule must be rehearsed before interpreting physical results. Unpiloted.

An exact calculation shows every initial tangent arrow rotates by a quarter-turn around the octant; reversing or retracing has a different effect.

Prerequisites and preparation

Core: Three-dimensional tangent vectors and a precisely supplied great-circle transport rule. Following a ball is entry; signed vector tracking is the proof. Varied longitude later needs sine/cosine components.

Extensions: Rotation composition, integer multiples of an angle and rational versus irrational periods.

Reading and representation: Distinguish the moving forward tangent from the transported arrow, and a leg's fixed plane normal from the sphere's outward normal. No compass rule is implied.

Materials: Ball marked with equator, north pole and two meridians; two differently colored paper arrows for forward tangent and carried vector; printed route and frame cards. A small flat card at a landmark helps show its tangent plane. Preparation: about 15 minutes.

Allow 10 minutes just to try tangent directions, then demonstrate the transport rule on one leg. Spend 15-20 on the octant route and a reset, then 20 on exact frame tracking. The longitude and area pages can be a separate trigonometry meeting.

Launch and facilitation

Carry an arrow around the marked spherical triangle. Keep it tangent by the stated rule, and do not turn the arrow when you turn the corner. Does coming home also restore its direction?

Student page 1: Carry an arrow around the corner

Core: spatial directions and an exact rule

Use the unit sphere with $A=(1,0,0)$, $B=(0,1,0)$, $C=(0,0,1)$. Route $A \rightarrow B \rightarrow C \rightarrow A$ follows the short great-circle arcs. On one leg let T be its forward unit tangent and N the fixed unit normal to that leg's oriented plane. Write the carried arrow as $aT+bN$ and keep a,b constant along that leg. At a corner keep the actual arrow fixed, then use the NEXT leg's T,N . Do not turn it with the walker.

Let the route turn a tangent arrow

1 / prompt and solution

Choose initial arrow $+y$, $+z$ or a mixture in A's tangent plane. Predict its final direction, then try the route on the ball. Record the arrow at each landmark and any disagreement between prediction and experiment.

Solution: For $+y$ the landmark sequence is $+y$ at A, $-x$ at B, $-x$ at C and $+z$ at A. For $+z$ it is $+z, +z, -y, -y$. A mixture $a \cdot y + b \cdot z$ follows the same linear combinations, ending $-b \cdot y + a \cdot z$. Physical errors are expected; the later frame cards supply the exact certificate. The arrow and forward tangent must be visibly separate, and no reset is allowed at corners.

1 / hints to offer only after exploration

1. Pause at a corner before changing the forward-tangent arrow.
2. Check that the carried arrow is still in the local tangent plane.

2 / prompt and solution

Starting from the same chosen arrow, compare $A \rightarrow B \rightarrow A$ with $A \rightarrow C \rightarrow B \rightarrow A$. Predict whether each restores the arrow. Explain why simply turning the arrow with the walker would be a different rule.

Solution: Retracing $A \rightarrow B \rightarrow A$ restores every initial arrow because reverse transport undoes the preserved frame coefficients. The reverse triangular route gives the inverse of the forward quarter-turn: $a \cdot y + b \cdot z$ maps to $b \cdot y - a \cdot z$, so a nonzero arrow is not restored. Rotating the arrow when the walker corners adds an extra rotation that the stated rule forbids; it can change or erase the effect being investigated. These predictions are proved by frame reversibility and task 4.

2 / hints to offer only after exploration

1. Reverse one leg while keeping the same physical arrow at the turning point.

Student page 2: Track the two basis arrows

Core proof: orthonormal frames and signed components

For $0 \leq t \leq \pi/2$ the supplied frames are: AB has $r = (\cos t, \sin t, 0)$, $T = (-\sin t, \cos t, 0)$, $N = +z$; BC has $r = (0, \cos t, \sin t)$, $T = (0, -\sin t, \cos t)$, $N = +x$; CA has $r = (\sin t, 0, \cos t)$, $T = (\cos t, 0, -\sin t)$, $N = +y$. In each case T, N are perpendicular unit vectors tangent at r .

Let the route turn a tangent arrow

3 / prompt and solution

Use the frame cards to compute the complete route first for initial $+y$ and then for initial $+z$. At each corner explicitly rewrite the arriving vector in the next frame.

Solution: Initial $+y$ equals T on AB and reaches B as $-x$. At B it equals $-N$ on BC , so remains $-x$ through C . At C it equals $-T$ on CA , so ends $+z$. Initial $+z$ equals N on AB , so stays $+z$ to B . At B it equals T on BC , reaching C as $-y$. At C it equals $-N$ on CA , so ends $-y$. These decompositions keep the physical vector fixed at corners.

3 / hints to offer only after exploration

1. Substitute $t=0$ and $t=\pi/2$ into each T before tracking the coefficients.

4 / prompt and solution

Find the final vector for every initial $a \cdot y + b \cdot z$. Prove the map preserves length and identify its signed angle in A 's tangent plane, where positive means from $+y$ toward $+z$. Justify the inverse and retracing claims.

Solution: Each leg preserves linear combinations of its two frame vectors, and changes of frame are linear. Task 3 therefore gives $(a,b) \mapsto (-b,a)$, or $a \cdot y + b \cdot z \mapsto -b \cdot y + a \cdot z$. Its squared length is $b^2 + a^2$, and it is the positive $\pi/2$ rotation in the stated orientation. Its inverse is $(a,b) \mapsto (b,-a)$. Traversing a leg backward uses the reversed forward tangent and plane normal; coefficients become $-a, -b$, and preserving them brings the same physical vector back. Therefore reversing a full route reverses the order of inverse leg maps, while an out-and-back cancels.

4 / hints to offer only after exploration

1. A linear map is determined by the images of $+y$ and $+z$.

Student page 3: Choose a different longitude

Trigonometric continuation: changing frames

Keep $A=(1,0,0)$, $C=(0,0,1)$, but choose $B=(\cos \alpha, \sin \alpha, 0)$, $0 < \alpha < \pi$. Travel the equator $A \rightarrow B$ and the short meridians $B \rightarrow C \rightarrow A$. Put $c=\cos \alpha, s=\sin \alpha$. Endpoint frames: AB has $T:+y \rightarrow (-s, c, 0)$, $N=+z$; BC has $T:+z \rightarrow (-c, -s, 0)$, $N=(s, -c, 0)$; CA has $T:+x \rightarrow -z$, $N=+y$.

Let the route turn a tangent arrow

5 / prompt and solution

Derive the returning vector for initial $+y$ and $+z$ using these frames. Then give the formula for an arbitrary initial $a \cdot y + b \cdot z$. Do not assume an angle-area theorem.

Solution: Let $u = (-s, c, 0)$. Initial $+y$ reaches B as $u = -N_{BC}$, remains u to C, then decomposes as $-s T_{CA} + c N_{CA}$. At A it becomes $c \cdot y + s \cdot z$. Initial $+z$ reaches B unchanged, follows T_{BC} to $-B = (-c, -s, 0)$ at C, then equals $-c T_{CA} - s N_{CA}$; it ends $-s \cdot y + c \cdot z$. Linearity gives $(a, b) \mapsto (ac - bs, as + bc)$, a positive rotation by α . The formula was obtained entirely from the supplied transport rule.

5 / hints to offer only after exploration

1. At C, read the x and y components as the CA frame coefficients.

6 / prompt and solution

Choose a longitude separation and an initial vector; compute the final vector and test its length. Include the case $\alpha = \pi/3$, initial $+y$. Can a nonzero vector be unchanged after one trip when $0 < \alpha < \pi$?

Solution: For $\alpha = \pi/3$ and $+y$, the result is $(1/2)y + (\sqrt{3}/2)z$, with length 1. Any chosen (a, b) follows task 5; expanding $(ac - bs)^2 + (as + bc)^2 = (a^2 + b^2)(c^2 + s^2) = a^2 + b^2$ proves exact length preservation. An unchanged vector solves $(c - 1)a - sb = 0$ and $sa + (c - 1)b = 0$. The determinant $(c - 1)^2 + s^2 = 2(1 - c) > 0$ for $0 < \alpha < \pi$, so only $a = b = 0$ works.

6 / hints to offer only after exploration

1. Use $c^2 + s^2 = 1$, and keep the zero vector separate from a direction arrow.

Student page 4: What did the route measure?

Continuation: area and limits of a model

The unit sphere has area 4π . A lune of longitude width α occupies fraction $\alpha/(2\pi)$ of the sphere; its northern half is the triangle $A \rightarrow B \rightarrow C \rightarrow A$. Angles are measured in radians.

7 / prompt and solution

Compute the triangle's area and compare it with the rotation you proved. Why is this a proved relation for this family of routes, but not a proof about all closed curves on every surface?

Solution: The full lune has area $(\alpha/(2\pi)) \cdot 4\pi = 2\alpha$, and its northern half has area α . The independently computed transport angle is also α . Both statements apply to this equator-and-two-meridians family on the unit sphere. We did not compute arbitrary spherical curves, prove a general area-holonomy theorem, or treat surfaces with varying curvature. The general theorem needs additional differential geometry.

Let the route turn a tangent arrow

7 / hints to offer only after exploration

1. Use rotational symmetry and the north/south division of the lune.

8 / prompt and solution

A partner proposes carrying an arrow fixed in surrounding three-dimensional space. Test a fixed +y arrow along AB. Give a closed route that does restore every arrow, and explain why curvature does not force every closed route to turn it.

Solution: On the equator $r(t)=(\cos t, \sin t, 0)$, the dot product of fixed +y with $r(t)$ is $\sin t$, nonzero in the interior. Therefore that arrow is not tangent there and violates the transport rule. $A \rightarrow B \rightarrow A$ restores all arrows because the reverse leg is the inverse map. Curvature allows route-dependent rotation; it does not say every possible closed itinerary has nonzero rotation.

8 / hints to offer only after exploration

1. A tangent vector at r on the unit sphere is perpendicular to r .

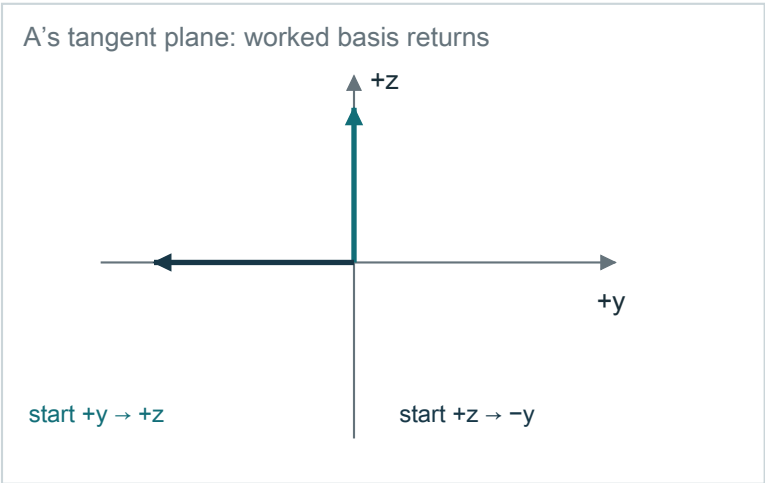
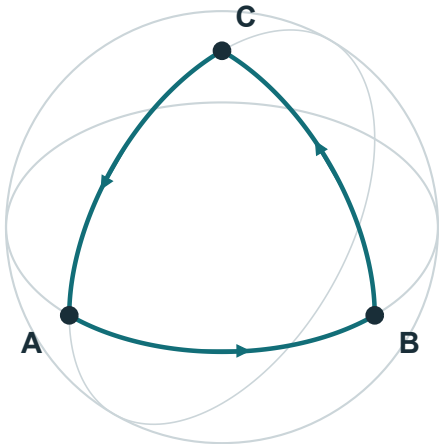
Worked transport ledger

Let the route turn a tangent arrow

This page contains return directions. Keep it out of the launch. Coordinate directions are ambient x,y,z ; a corner changes the frame, not the arriving vector.

start at A	after AB, at B	after BC, at C	after CA, at A
+y	$-x = -N_{BC}$	$-x = -T_{CA}$	+z
+z	$+z = T_{BC}$	$-y = -N_{CA}$	-y

Thus $a \cdot y + b \cdot z$ returns as $-b \cdot y + a \cdot z$: a $+\pi/2$ rotation in the tangent plane, preserving $a^2 + b^2$.



For longitude α , the same calculation gives $(a \cos \alpha - b \sin \alpha) \cdot y + (a \sin \alpha + b \cos \alpha) \cdot z$. The triangle area is α : half a lune of area 2α . This calculation proves the relation for this family, not the general Gauss-Bonnet theorem.

Worked transport ledger

Optional facilitator extension 1

Gate: Trigonometric rotation composition and rational/irrational multiples.

Repeat the longitude- α route k times. For a nonzero initial arrow, determine exactly when it returns. Find the first return for $\alpha=\pi/3$ and decide what happens if $\alpha/(2\pi)$ is irrational.

Solution: The proved matrix is a rotation by α ; composing k copies gives angle $k\alpha$. A nonzero vector returns exactly when $\cos(k\alpha)=1$ and $\sin(k\alpha)=0$, equivalently $k\alpha\in 2\pi\mathbb{Z}$. For $\alpha=\pi/3$ the least positive k is 6. If $\alpha/(2\pi)$ is irrational, no positive integer k can make its product an integer, so no exact finite return occurs. This is not a claim about approximate return or measurement precision.

Mathematics and prior use

Parallel transport along great circles and holonomy on a specific spherical triangle. For a unit-speed great circle, $T'=-r$ and a fixed N has $N'=0$, so the supplied $aT+bN$ rule has derivative normal to the sphere and agrees with parallel transport. The general Gauss-Bonnet theorem is context, not proved here.

Prior use: Week 9 billiards and GA-21 mirror unfolding concern a flat plane. GA-01 composes rigid box turns, whereas this follows vectors in changing tangent planes on a curved surface. The original octant example is retained as entry and grows into a proved rotation for every initial vector and variable longitude.

Source: Anton Petrunin and Sergio Zamora Barrera, What Is Differential Geometry? Curves and Surfaces

Chapter 16, pp.134-138: parallel fields, Proposition-Definition 16.2, Exercise 16.5 and Proposition 16.6; semisolution 16.5, p.202; Proposition 17.10, p.143.. Use the actual tangent-field/transport definition and octant example; derive exact frame calculations for all initial vectors and variable longitude independently. Evidence: Reopened on 2026-09-26. Inspected v' normal-to-tangent-plane definition, transport existence statement, octant semisolution and spherical-area relation. Our +angle convention is explicitly +y toward +z at A.

<https://arxiv.org/pdf/2012.11814>

Test a reversible reshaping

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for tracing and puncture choices; promising with explicit topology facts for the proof. Physical string is an approximation to a space, so the one-point convention needs careful demonstration. Unpiloted.

A point-deletion contradiction rules out a loop-to-segment homeomorphism, and an explicit ellipse map demonstrates what a successful reversible map requires.

Prerequisites and preparation

Core: Tracing connected pieces and reasoning with a point-deletion invariant. The formal proof uses the definition of homeomorphism and supplied interval/continuous-image facts; arithmetic is not needed.

Extensions: Planar polygonal paths and a supplied finite-union-of-lines fact for a disk.

Reading and representation: A mathematical puncture deletes one point, not a whole graph edge or a short physical strand. Joined crossings and disjoint curves must be distinguished.

Materials: String loop, open string, removable dot markers and printed loop/segment/Y/joined-eight/disjoint-loops diagrams. No cutting is needed to test imagined point removal. Preparation: about 5 minutes.

Allow 10 minutes of free reshaping and tracing, 15 for chosen punctures, then 15-25 for impossibility arguments. Formula-map tests and the disk extension are separate continuations.

Launch and facilitation

Bend a loop and an open string into many shapes. Invent a test that every truly reversible reshaping would have to pass.

Student page 1: Choose the point to remove

Core: trace pieces without jumping gaps

The pictures represent every point along the drawn strands, not just marked vertices. “Delete a point” means that single mathematical point is unavailable. A connected piece can be traced between any of its retained points in these examples. The figure-eight crossing and the Y center are joined; the two separate loops are not joined.

Test a reversible reshaping

1 / prompt and solution

Choose points to delete from the circle and the closed segment. Try different positions and record the number of pieces left. Can a segment endpoint trick a test based on just one careless choice?

Solution: A circle minus any single point is one open arc, hence one connected piece. Deleting a segment endpoint leaves one piece; deleting an interior point leaves two. Thus choosing only an endpoint would make both examples give one and fail to distinguish them. The useful contrast is the existence of a segment cut point versus the circle's every-point property.

1 / hints to offer only after exploration

1. Mark the missing point, then trace all retained strands without crossing it.

2 / prompt and solution

Choose useful punctures in the Y, the joined figure-eight and the two disjoint loops. Classify the possible numbers of remaining pieces for each shape. Explain the role of a junction instead of treating every drawn point alike.

Solution: Y: deleting its joined center gives three pieces; deleting an arm interior gives two; deleting an endpoint gives one. Joined figure-eight: deleting the joining point gives two open arcs; deleting any other point leaves the rest connected, since the broken loop remains attached to the intact loop through the join. Two disjoint loops: every one-point deletion leaves two pieces, one open arc and one intact circle. They already had two components before deletion. All statements concern one point, not removal of an entire edge.

2 / hints to offer only after exploration

1. An intact second loop may still connect both ends of a punctured strand through the join.

Student page 2: Turn the test into a proof

Topology proof: supplied facts and matching points

A homeomorphism is a continuous bijection with continuous inverse. Supplied facts: an interval is connected, and a continuous image of a connected space is connected. A homeomorphism restricts to a homeomorphism after deleting matching points, because the map and inverse remain continuous on their remaining subspaces.

Test a reversible reshaping

3 / prompt and solution

Prove that the circle cannot be homeomorphic to $[0,1]$. Choose a point strategically, and justify why every corresponding punctured circle is connected.

Solution: Suppose $h:[0,1] \rightarrow S^1$ is a homeomorphism. Remove $1/2$ and $h(1/2)$. The remaining interval has the two nonempty relatively open pieces $[0, 1/2)$ and $(1/2, 1]$, so is disconnected. A circle minus any point p is connected: rotate p to $(1,0)$; the remaining circle is the continuous image of the connected interval $(0, 2\pi)$ under $t \rightarrow (\cos t, \sin t)$. Rotation back preserves connectedness. The inverse of the restricted homeomorphism would send a connected punctured circle onto a disconnected punctured interval, contradiction.

3 / hints to offer only after exploration

1. Choose the interval point first, since you know how to make its deletion disconnect.

4 / prompt and solution

Use matching-point deletion to distinguish the Y from a segment and the joined figure-eight from a circle. Also distinguish the joined figure-eight from two disjoint circles.

Solution: Deleting the Y center leaves three components, whereas deleting any point of a closed segment leaves at most two. A restricted homeomorphism induces a bijection of connected components: a connected component maps into one, and the continuous inverse prevents a larger connected component from appearing. Thus Y and segment are not homeomorphic. Deleting the figure-eight join gives two components, but deleting any circle point leaves one, so these are not homeomorphic. The joined eight is connected before deletion and two disjoint circles are disconnected, already ruling out that homeomorphism.

4 / hints to offer only after exploration

1. When a puncture is not needed, use connectedness of the whole space.

Student page 3: Test actual maps

Continuation: bijections, inverses and coordinates

A continuous map onto a shape is not automatically a reversible reshaping. Test which of the definition's requirements each candidate satisfies.

Test a reversible reshaping

5 / prompt and solution

The map $(x,y) \mapsto x$ sends the unit circle onto $[-1,1]$. The map $t \mapsto (\cos 2\pi t, \sin 2\pi t)$ sends $[0,1]$ onto the circle. Why does neither contradict your impossibility proof? Exhibit the failure explicitly.

Solution: Both maps are continuous and onto, but neither is one-to-one. Projection identifies $(0,1)$ and $(0,-1)$, so it has no inverse function. The parametrization identifies $t=0$ and $t=1$, so it also has no inverse function on the stated domain. A picture covering the target does not meet the homeomorphism definition.

5 / hints to offer only after exploration

1. Find two different inputs sharing one output.

6 / prompt and solution

Build a homeomorphism from the unit circle to the ellipse $(X/2)^2 + Y^2 = 1$. Give its inverse, check both compositions and explain continuity. What geometric features did this successful map change?

Solution: Use $h(x,y) = (2x,y)$ and $g(X,Y) = (X/2,Y)$. The circle equation becomes $(X/2)^2 + Y^2 = x^2 + y^2 = 1$; conversely the ellipse maps into the circle. Both compositions are the identity, proving bijectivity and inverse formulas. Coordinate scaling is continuous, so both h and g are continuous. Lengths and most angles change; the connectedness and puncture properties used above do not.

6 / hints to offer only after exploration

1. Stretch one coordinate and undo that same stretch.

Optional facilitator extension 1

Gate: Planar paths, convexity, and the supplied finite-line fact.

A circle and a closed disk both remain connected after deleting any one point. Show this does not make them homeomorphic: compare deleting two distinct points. You may use the fact that finitely many lines cannot cover an open disk.

Solution: A circle minus two distinct points is the union of two disjoint open arcs, hence has two components. For a closed disk with two deleted points d_1, d_2 , take any retained p, q (if equal there is nothing to prove). Choose an interior z outside the finitely many lines through p or q and either deleted point, and outside the deleted points themselves. Such z exists by the supplied fact. Convexity keeps segments pz, zq in the disk, and their line choices prevent either from hitting a deleted point. Thus the punctured disk is path-connected. The same construction proves the one-point disk claim; a punctured circle is an open arc. A homeomorphism deleting corresponding pairs would preserve connectedness, contradiction.

Test a reversible reshaping

Mathematics and prior use

Connectedness preserved by continuous maps and homeomorphisms, restriction after deletion, cut points and limits of a one-point invariant. No complete classification of spaces follows from component counts.

Prior use: Week 1 has discrete tiling moves. GA-06 compares a crossing algebraic curve with disjoint components; GA-27 uses connected sublevel sets. This task focuses on homeomorphisms, matching punctures and explicit failed/successful maps rather than visual resemblance.

Source: Romyar Sharifi, Point-Set Topology

Chapter 3, §3.1, Definition 3.1.1 and Proposition 3.1.8.. Use connectedness and preservation under continuous images. The puncture comparisons and explicit path constructions are derived here. Evidence: On 2026-09-26 direct page opens failed; retrieved and read the indexed primary-page text of the definition and proposition with proof. No uninspected page content is claimed.

<https://www.math.ucla.edu/~sharifi/notes/topology-ch03.html>

Source: Romyar Sharifi, Point-Set Topology

Chapter 2, Definition 2.1.13 and Examples 2.1.16.. Use the continuous-bijection-with-continuous-inverse definition. The circle/ellipse and failed maps are independent examples. Evidence: Read the indexed primary-page definition and homeomorphism examples on 2026-09-26.

<https://www.math.ucla.edu/~sharifi/notes/topology-ch02.html>

Find the flood's joining height

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with coordinate-inequality gates; the Morse interpretation is advanced/specialist. Physical shading can hide a point connection, so equality at zero must be stated and used. Unpiloted.

Prove the exact joining threshold and all component counts in the actual square, then design the smallest water level for a forced crossing.

Prerequisites and preparation

Core: Signed squares, coordinate inequalities and explicit paths in a clipped square. Calculus is required only for the final gradient/Hessian interpretation, not for the exact component proof.

Extensions: Quadratic inequalities in x^2 , path components, gradient and Hessian for a new double-well model.

Reading and representation: Flooded means height $\leq$ water level, including equality. Count regions, not boundary curves; every path stays in the stated square.

Materials: Printed equal-scale grids of $[-1, 1]^2$, transparent sleeves or tracing paper, two colored pencils; clay saddle is optional and cannot replace the exact inequality. Preparation: about 8 minutes.

Explore point tests and chosen water levels for 10 minutes, sketch the common levels for 15, then allow 20 for safe paths and barriers. Give the calculus interpretation later only to learners with partial derivatives.

Launch and facilitation

Choose where to walk between the top and bottom of this landscape. How high must the water be before your entire route lies underwater?

Student page 1: Choose a water level and a route

Core: squares, inequalities and wet paths

Work only in the square $-1 \leq x \leq 1$, $-1 \leq y \leq 1$. Ground height is $f(x, y) = x^2 - y^2$. At level c , a point is wet exactly when $f \leq c$; equality is included. A wet route is a continuous path whose every point is wet.

Find the flood's joining height

1 / prompt and solution

Choose one negative and one positive water level and test points of your own. Then sketch the exact wet regions at $c=-1/4, 0, 1/4$ on the supplied grids, labeling their boundary inequalities.

Solution: Any chosen levels and points can be tested by $x^2 - y^2 \leq c$. At $-1/4$, the wet region is $y^2 \geq x^2 + 1/4$ inside the square: upper and lower lobes, with $|x| \leq \sqrt{3}/2$ at $y = \pm 1$. At 0 it is $|y| \geq |x|$, two closed wedges meeting at the origin. At $1/4$ it is $x^2 \leq y^2 + 1/4$ inside the square, containing the entire y -axis; for $|y| \geq \sqrt{3}/2$ every x in $[-1, 1]$ is wet. Boundaries belong to the wet region. Example tests: $(0, 0)$ has height 0; $(1, 0)$ has height 1; $(0, 1)$ has height -1 ; $(1/2, 0)$ has height $1/4$.

1 / hints to offer only after exploration

1. Rearrange the inequality before shading.
2. Do not omit points where the inequality is an equality.

2 / prompt and solution

Try to connect $U=(0, 1)$ to $D=(0, -1)$ by a wet route at each common level. Give an explicit successful route or an obstruction that rules out every route. At which level do the two sides first join?

Solution: For $c=-1/4$, every wet point has $y \neq 0$; any continuous U-to-D path must pass through $y=0$ by the intermediate value theorem, where $f=x^2 \geq 0$, contradiction. At $c=0$ and $c=1/4$ the vertical segment $x=0$ is wet since $f(0, y) = -y^2 \leq 0$. Thus the first possible joining level is 0, including the single connecting point $(0, 0)$. A small drawing gap at zero would be misleading.

2 / hints to offer only after exploration

1. Track the y -coordinate of a proposed path rather than only its drawing.

Student page 2: Classify every flood level

Core proof: bounds, connectedness and a barrier

The domain remains the closed square. A component is a maximal connected wet piece. You may use that any continuous route from positive y to negative y passes through $y=0$.

Find the flood's joining height

3 / prompt and solution

Classify the number of wet components for every real c , including the first nonempty level, the joining level and when the whole square is wet. Prove that each claimed piece really is connected within the square.

Solution: Since $-1 \leq f \leq 1$, $c < -1$ gives no wet points; $c = -1$ gives exactly $(0,1), (0,-1)$, two singleton components. For $-1 < c < 0$, no wet point has $y=0$, so upper and lower sets are separated by the relatively open signs of y . Each upper point can move vertically to $(x,1)$, decreasing f , then horizontally along the top to $(0,1)$; wet top points obey $|x| \leq \sqrt{1+c}$. The lower piece has the analogous route via $y=-1$. Thus exactly two components exist for $-1 \leq c < 0$. For $c \geq 0$, every wet point joins the origin by $r \mapsto r(x,y)$, $0 \leq r \leq 1$: $f(r(x,y)) = r^2 f(x,y) \leq c$, whether f is positive or negative. Hence there is one component. At $c \geq 1$ the entire square is wet; for $0 \leq c < 1$ it is connected but not the whole square because $(1,0)$ is dry. These boundary events were proved in the clipped domain.

3 / hints to offer only after exploration

1. For a negative level, move toward the top or bottom edge before joining sideways.
2. For nonnegative levels, test a straight radial path.

4 / prompt and solution

Choose a forced crossing point $(a,0)$, with $|a| \leq 1$. Among all U-to-D routes that pass through it, find the smallest water level making an entire route wet. Give both a lower bound and a route attaining it, including $a=0$ and $a=\pm 1$.

Solution: Every such route visits $(a,0)$, whose height is a^2 , so its maximum height and required level are at least a^2 . Go from U along $y=1$ to $(a,1)$, vertically to $(a,-1)$, then along $y=-1$ to D. The top/bottom portions have $f=x^2-1 \leq 0 \leq a^2$, and the vertical portion has $f=a^2-y^2 \leq a^2$, with equality at its crossing. Thus the exact optimum is a^2 . All segments stay in the square. At $a=0$ it reduces to the axis route with threshold 0; at $a=\pm 1$ the threshold is 1.

4 / hints to offer only after exploration

1. A mandatory visited point is already a lower-bound certificate.

Student page 3: What makes this a Morse saddle?

Calculus continuation: partial derivatives and Hessian

A critical point has zero gradient. The Hessian is the matrix of second partial derivatives. A critical point is nondegenerate if its Hessian is invertible; its index is the number of negative eigenvalues. These derivative tests concern the function near an interior point.

Find the flood's joining height

5 / prompt and solution

Find all critical points and the Hessian of $x^2 - y^2$. Compare with the bowl $x^2 + y^2$ near zero: what different component event occurs as c crosses 0?

Solution: For f , $\nabla f = (2x, -2y)$ vanishes only at $(0,0)$, and Hessian $\text{diag}(2, -2)$ is invertible with index 1. It is the nondegenerate saddle model. The bowl has gradient $(2x, 2y)$ and positive Hessian $\text{diag}(2, 2)$, index 0. Its square-clipped sublevel is empty for $c < 0$, the origin at $c = 0$, and a nonempty connected region for $c > 0$ (a disk for $0 < c \leq 1$, then clipped), so a component is born rather than two joining. A local Morse interpretation does not replace the earlier global path proof.

5 / hints to offer only after exploration

1. Compare the signs of the two second derivatives.

6 / prompt and solution

The landscape $g = x^4 - y^4$ has the same wet set as f at level 0. Verify that statement, then test whether g has a nondegenerate critical point at the origin. Why can a matching drawing not justify invoking the Morse lemma?

Solution: Because fourth powers preserve the order of $|x|$ and $|y|$, $g \leq 0$ iff $|x| \leq |y|$, the same wedges as $f \leq 0$. But $\nabla g = (4x^3, -4y^3)$ vanishes at the origin and its Hessian $\text{diag}(12x^2, -12y^2)$ is the zero matrix there. Thus the critical point is degenerate, and the nondegeneracy hypothesis of the Morse lemma fails. Equal zero-level sublevel pictures do not certify Hessian signature or the local smooth normal-form theorem.

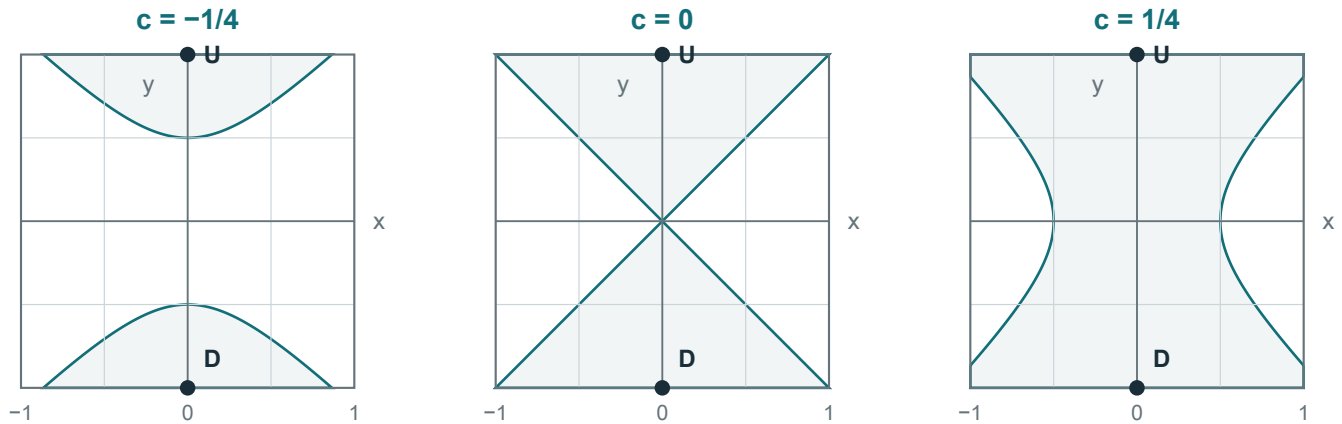
6 / hints to offer only after exploration

1. Compute the second derivatives at the point, not merely their signs away from it.

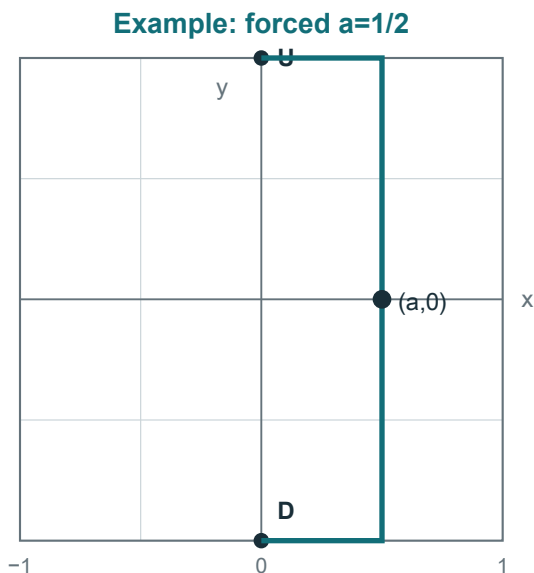
Worked wet regions and barrier route

Find the flood's joining height

Shading includes its boundary. All regions are clipped to the closed square. The touching point at zero is retained, so the two wedges form one connected set.



For $c < 0$, wet points of one sign of y connect vertically to the corresponding square edge and then along that edge. For $c \geq 0$, radial contraction to the origin stays wet because $f(rp) = r^2 f(p)$. At $c = -1$ only U, D remain; below -1 nothing is wet. At $c \geq 1$ the whole square is wet.



The crossing forces $c \geq a^2$. The top-edge / vertical / bottom-edge route attains $c = a^2$: edge heights are at most 0 and vertical heights are $a^2 - y^2 \leq a^2$. This remains legal at $a = 0$ and $a = \pm 1$.

Worked wet regions and barrier route

Optional facilitator extension 1

Gate: Polynomial inequalities, path components and partial derivatives.

On the whole plane, let $F(x,y)=(x^2-1)^2+y^2$. Classify the components of $\{F \leq c\}$ for every real c and find all critical points and Hessians. Give a direct path argument, rather than inferring the answer from a landscape sketch.

Solution: For $c < 0$ the set is empty. For $c \geq 0$ the allowed x obey $|x^2-1| \leq \sqrt{c}$, so $\max(0, 1-\sqrt{c}) \leq x^2 \leq 1+\sqrt{c}$. At $c=0$ only $(\pm 1, 0)$ occur. For $0 < c < 1$ the allowed x form two disjoint intervals of opposite sign. Every vertical fiber is an interval containing $(x, 0)$, and each allowed x interval on the axis is wet, proving exactly two components. For $c \geq 1$ the allowed x form one interval, so the same fiber-to-axis paths prove connectedness; at $c=1$ they first join through $(0, 0)$. $\nabla F = (4x(x^2-1), 2y)$ gives critical points $(\pm 1, 0), (0, 0)$. Hessian $\text{diag}(12x^2-4, 2)$ is $\text{diag}(8, 2)$ at the minima and $\text{diag}(-4, 2)$ at the saddle. Its saddle level is 1. This is a new explicitly solved model, not a general Morse theorem.

Mathematics and prior use

Sublevel-set component changes, a minimax path barrier and the local nondegenerate saddle model. Square-boundary events are proved directly and are not applications of a smooth compact-manifold slab theorem.

Prior use: GA-24 develops connectedness and point-deletion invariants. This changes the space with a numerical level and proves connections by paths and barriers. The original three saddle levels remain a common entry, but the new investigation classifies every real level and solves a learner-chosen constrained crossing.

Source: Patrick Gillespie, Morse Theory Notes

Lemma 1.12 (Morse lemma), p.3; Theorem 2.9 (Regular Interval Theorem), p.7; §3, Definition 3.1 and Theorem 3.3, p.9.. Use the local saddle/Hessian interpretation as context. The clipped-square and double-well sublevel classifications are independently proved by inequalities and paths. Evidence: Reopened the nondegeneracy hypothesis and regular-interval boundary conditions on 2026-09-26. The square has corners and does not satisfy the quoted smooth-boundary setup; no such theorem is applied to it.

https://web.math.utk.edu/~afreire/teaching/m663f21/Morse_Theory_Notes.pdf

How much can one question guarantee?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising first pilot for fraction-ready learners. The adversarial search creates a reason to invent bisection. Use page 3 only with calculus-ready learners.

A best possible worst-case search strategy, an exact root bracket, and an explanation of its guarantee.

Prerequisites and preparation

Core: Fractions, comparing squares, and interval reasoning. Calculus is needed only on page 3.

Extensions: Derivatives for Newton iteration; continuity and the intermediate value theorem for a root certificate.

Reading and representation: Short numeric records; a facilitator may read procedural instructions.

Materials: Pencil, printed interval strips and bracket record; calculator optional. A partner supplies left/right answers in the hidden-number game. Preparation: about 8 minutes.

5-10 minutes to handle the materials or choose examples; 15-20 explore page 1; 5 share conjectures; 15-20 develop page 2. Page 3 is a separate continuation when its tools are ready; allow another 30-45 minutes. Give one page at a time.

Launch and facilitation

Find a number whose square is two, using questions that answer only “too large” or “too small.” Decide what information to keep after every question.

Student page 1: Hide a number

Core: fractions and comparisons

An unknown real number r lies strictly between 1 and 2. Each test asks which side of your chosen number contains r ; an exact hit wins. Your partner may delay choosing r , but every reply must stay consistent with all earlier replies. Keep the entire interval still possible. Your strategy must handle unlucky replies too.

1 / prompt and solution

Choose a first test. Your partner gives the reply leaving the wider possible interval. Continue for four questions, always choosing your next test. Record the surviving interval after each reply.

Solution: The partner keeps the longer of the two subintervals; tied sides are either legal. For midpoint tests, four replies leave width $1/16$, with endpoints depending on replies. Any still-possible interior point realizes the finite reply history, so the adversary is consistent. Exact hits need not occur in a worst case.

How much can one question guarantee?

1 / hints to offer only after exploration

1. Draw both possible pieces before hearing the reply.

2 / prompt and solution

Find a strategy that guarantees width at most $1/16$ after four questions. Prove that no comparison strategy can guarantee a smaller width for every secret number.

Solution: Always choose the midpoint: each reply halves the width, giving $1/16$. For any test in an interval of width W , at least one side has width at least $W/2$. A partner can retain that side at each stage, so four questions leave width at least $1/16$. An interior secret in the final interval makes all replies true. This proves the worst-case lower bound.

2 / hints to offer only after exploration

1. What can the partner keep if your test is off-center?

Student page 2: A bracket for a root

Core continuation: squares; a supplied continuity theorem

Now the target is the positive solution of $x^2=2$. A bracket $[a,b]$ has $a^2 \leq 2 \leq b^2$ and $1 \leq a \leq b \leq 2$. Supplied theorem: a continuous function taking opposite signs at two endpoints has a zero between them. Polynomials are continuous.

3 / prompt and solution

Start with $[1,2]$. Make four midpoint tests, recording each exact square and the bracket you keep. Give one estimate and a proved bound on its error.

Solution: Tests $3/2, 5/4, 11/8, 23/16$ have squares $9/4, 25/16, 121/64, 529/256$. The remaining bracket is $[11/8, 23/16]$, width $1/16$. Its midpoint $45/32$ has error at most $1/32$. Signs and continuity retain a root; strict increase of x^2 on positive numbers shows it is the same unique positive root.

3 / hints to offer only after exploration

1. Keep the lower and upper witnesses, not only your latest guess.

4 / prompt and solution

Before doing more arithmetic, how many midpoint tests guarantee bracket width below $1/100$? Would opposite signs still certify a root for a function allowed to jump? Make a counterexample.

Solution: After n tests width is $2^{-(n+1)}$, so seven suffice and six do not give this width bound: $1/128 < 1/100 < 1/64$. The step function $f(x) = -1$ for $x < 0$ and $+1$ for $x \geq 0$ has opposite endpoint signs on $[-1, 1]$ but no zero. Continuity is essential.

How much can one question guarantee?

4 / hints to offer only after exploration

1. Compare powers of 2 with 100; then try a graph with a jump.

Student page 3: A tangent can make no progress

Calculus continuation: derivatives

Newton's rule is $x_{\text{next}} = x - f(x)/f'(x)$, whenever $f'(x) \neq 0$. Use $f(x) = x^3 - 2x + 2$. This page concerns the rule's guarantee, not a race to many decimal places.

5 / prompt and solution

Try Newton's rule from $x=0$ and from $x=1$. Can this rule continue forever without approaching any root? Explain exactly what happens.

Solution: $f'(x) = 3x^2 - 2$. From 0 the rule gives 1; from 1 it gives 0. Thus $0 \leftrightarrow 1$ repeats forever with nonzero denominators and nonzero function values. The sequence does not converge and neither point is a root.

5 / hints to offer only after exploration

1. Check whether your second answer is familiar.

6 / prompt and solution

Find a root bracket of width $1/4$ inside $[-2, -1]$, using two midpoint tests. Explain why this certificate remains valid even though Newton's rule cycled elsewhere.

Solution: $f(-2) = -2$ and $f(-1) = 3$. At $-3/2$, $f = 13/8 > 0$; at $-7/4$, $f = 9/64 > 0$. The retained bracket is $[-2, -7/4]$. Continuity and endpoint signs certify a root there. The independent Newton trajectory at 0 and 1 does not affect this invariant.

6 / hints to offer only after exploration

1. Keep a negative-value endpoint and a positive-value endpoint.

Optional facilitator extension 1

Gate: Fractions, powers and the longer-side argument.

Suppose queries must use the same fraction q of the current width from its left end, where $0 < q < 1$. Find the worst-case width after n tests and the best q .

Solution: Each reply retains at most $\max(q, 1-q)$ of the width, and the longer-side adversary attains that factor every time. Starting width 1 gives $\max(q, 1-q)^n$. This is minimized uniquely by $q = 1/2$, with value 2^{-n} .

How much can one question guarantee?

Mathematics and prior use

The root-containing interval is an invariant. The longer-side adversary gives an optimal information bound for comparisons; continuity transfers the interval method to an actual equation.

Prior use: AP-06 in this expansion studies a different Newton cycle for x^3-5x and a safeguarded hybrid. This family adds optimal comparison search and an explicit root-error certificate. GA-02 supplies a related continuity obstruction; skip repeated elementary bisection for experienced returners.

Source: MIT 2.086, Numerical Computation for Mechanical Engineers, Unit VI

§29.2.3, bisection discussion, printed p.432; §29.2.5, Newton pathologies.. The source supplies background definitions and theorem context. The worksheet instances, learner choices, elementary derivations and answer keys are independently developed. Evidence: Inspected the bisection paragraph and Newton iteration context. The hidden-number lower bound and both polynomial calculations are independently derived. Checked 2026-09-25.

https://ocw.mit.edu/courses/2-086-numerical-computation-for-mechanical-engineers-spring-2013/60af88c658ba3e83cdf7c43289aeb5b4_MIT2_086S13_Unit6_Textbook.pdf

Find every shortest route around a cylinder

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for learners who know or can use the Pythagorean theorem. Physical string exploration has a genuine proof destination. Younger learners can compare routes, but the complete certificate uses lengths.

Two different shortest surface routes and a lower bound excluding every hidden winding.

Prerequisites and preparation

Core: Spatial unrolling and the Pythagorean theorem. Integer shifts support the all-routes proof; no calculus.

Extensions: Absolute-value comparisons for a moving endpoint; distance formula for arbitrary circumference and height.

Reading and representation: Labels and short diagrams suffice; no calculus reading.

Materials: Printed 6-by-4-inch rectangle (spare copy to cut), scissors, tape, string, ruler, pencil and the repeated-strip map. Join the two 4-inch edges edge-to-edge without overlapping the paper; keep end caps open. Print at actual size. Preparation: about 10 minutes.

5-10 minutes to handle the materials or choose examples; 15-20 explore page 1; 5 share conjectures; 15-20 develop page 2. Page 3 is a separate continuation when its tools are ready; allow another 30-45 minutes. Give one page at a time.

Launch and facilitation

Join the short edges into a cylinder. Connect a bottom-rim point to the top point halfway around. Find two visibly different routes and decide whether one can be shortened.

Student page 1: A route the seam hides

Core: paper construction, length comparison and the Pythagorean theorem

Cut out the rectangle from a spare copy. Tape the two short edges of the 6-by-4 rectangle edge-to-edge, without overlapping the paper, to make a cylinder. A is on the bottom rim at the seam; B is on the top rim halfway around. Routes must stay on the curved side, including its rims. No end caps or shortcuts through the interior.

Find every shortest route around a cylinder

1 / prompt and solution

Use string to find two different routes from A to B. Choose a cut that makes one route easy to draw flat. Can a different cut reveal a route you first missed?

Solution: The two opposite winding directions give equally good candidates. Cutting at an inconvenient seam can split a route into pieces; moving the cut or using repeated rectangles shows it as one continuous path. Arbitrary longer string paths are valid experiments; ask what straightening can preserve.

1 / hints to offer only after exploration

1. Put a fresh seam where it will not split your chosen route.

2 / prompt and solution

Draw a route that becomes straight when unrolled. Calculate its length, then find a different route with the same length. Can you yet rule out every shorter route?

Solution: Straight displacements $(3,4)$ and $(-3,4)$ both have length 5 and roll to distinct routes. Their equal length alone does not rule out another path; page 2 supplies the complete lower bound.

2 / hints to offer only after exploration

1. Use a 3-by-4 right triangle, in either horizontal direction.

Student page 2: A map with many copies

Core continuation: integer shifts and a lower bound

The repeated strip has $A=(0,0)$. Copies of B occur at $(3+6k,4)$, one for every integer k . Any surface path starting at A can be followed continuously across adjacent copies of the rectangle without jumping.

3 / prompt and solution

Choose three copies of B, including one farther away. Draw a straight route to each, calculate its length, and roll or trace how the extra winding appears.

Solution: For $k=0$ and -1 lengths are 5; $k=1$ and -2 give $\sqrt{97}$. In general the straight length is $\sqrt{(3+6k)^2+16}$. All these segments stay between heights 0 and 4 and roll to allowed surface paths. A farther lift adds winding, not an illegal interior shortcut.

3 / hints to offer only after exploration

1. The vertical displacement stays 4 even after another wrap.

Find every shortest route around a cylinder

4 / prompt and solution

Prove that there are exactly two shortest routes. Your argument must include curved routes and arbitrarily many windings, not just the copies printed on this page.

Solution: A rectifiable path lifts to some endpoint $(3+6k, 4)$ with unchanged length, at least that endpoint's straight-line distance. Nonrectifiable paths have infinite length. For every integer k , $|3+6k| \geq 3$, with equality only at $k=0, -1$. Straight segments attain length 5 in those two classes. Equality in the plane requires a straight segment (up to reparametrization), so these are exactly two geometric shortest routes.

4 / hints to offer only after exploration

1. First fix the lifted endpoint; then compare all possible endpoints.

Student page 3: Move the destination

Continuation: piecewise formulas and comparison

Keep A at the bottom seam. Move B around the top rim: its horizontal offset is d with $0 \leq d < 6$, and its height stays 4. Its copies are $(d+6k, 4)$.

5 / prompt and solution

Choose offsets on both sides of $d=3$. Find the winning lift for each. Locate exactly where the winner changes and where there is a tie.

Solution: For $0 \leq d < 3$ the unique shortest lift is $k=0$. For $3 < d < 6$ it is $k=-1$. At $d=3$ they tie; no other lift ties. The distance is $\sqrt{(16+\min(d, 6-d)^2)}$. At $d=0$ the vertical route is uniquely shortest.

5 / hints to offer only after exploration

1. Compare only after explaining why every other lift is farther away.

6 / prompt and solution

A long helical route looks perfectly straight in every small unrolled patch. Does that make it globally shortest? Give a specific counterexample from your map.

Solution: The segment from $(0, 0)$ to $(9, 4)$ rolls to a locally straight helical route of length $\sqrt{97}$. The two routes with horizontal displacement ± 3 have length 5, so local straightness does not ensure global minimality. Small pieces can be shortest between nearby points while the whole route loses to a different winding.

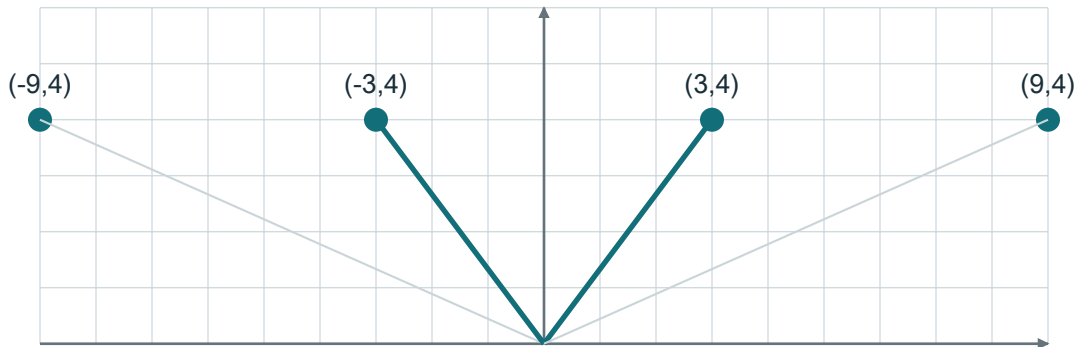
6 / hints to offer only after exploration

1. Compare routes with different lifted endpoints.

Find every shortest route around a cylinder

Worked diagrams / facilitator only

The two winning lifts



Displacements ± 3 give length 5. Displacements ± 9 already give $\sqrt{97}$. The integer bound $|3+6k| \geq 3$ excludes every other winding, including copies outside this drawing.

Optional facilitator extension 1

Gate: Pythagorean theorem and absolute-value inequalities.

For circumference $C > 0$, height $H > 0$ and $0 \leq d < C$, classify shortest lateral-surface paths.

Solution: Lift endpoints to $(d+kC, H)$. The minimum length is $\sqrt{H^2 + \min(d, C-d)^2}$. It is attained by one geometric path except at $d=C/2$, where two tie. At $d=0$ only the vertical path wins. The same straightening and integer-shift lower bound proves completeness.

Mathematics and prior use

Unrolling is a local length-preserving map. Different winding classes end at different lifted copies; minimizing those exact distances separates locally straight from globally shortest paths.

Prior use: New continuous geometry. GA-23 studies curvature on a sphere; this cylinder is locally flat while its global wrapping still matters.

Source: Anton Petrunin and Sergio Zamora Barrera, What Is Differential Geometry?

§18G, intrinsic isometries, printed p.160: wrapping the plane into a cylinder; Chapters 14-15 for path length.. The source supplies background definitions and theorem context. The worksheet instances, learner choices, elementary derivations and answer keys are independently developed. Evidence: Inspected the length-preserving cylinder map. The 6-by-4 lift minimization, completeness of the candidate list and moving-offset classification are independently derived. Checked 2026-09-25.

<https://arxiv.org/pdf/2012.11814>

Can every pair agree and everyone disagree?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate. The strongest moment is replacing many pair checks with two extremal witnesses. The plane continuation is a new gate, not compulsory.

A two-interval certificate for every failure on a line, followed by exact counterexamples when hypotheses change.

Prerequisites and preparation

Core: Ordering endpoints and drawing intervals. Signed coordinates and inequalities enter on page 3.

Extensions: Convex combinations and Radon's theorem only for a later planar Helly proof, not required here.

Reading and representation: Read short endpoint labels; symbols for inequalities are needed only in the planar extension.

Materials: Pencils or two colors; printed number-line strips and coordinate windows. Transparent strips are optional, not required. Preparation: about 10 minutes.

5-10 minutes to handle the materials or choose examples; 15-20 explore page 1; 5 share conjectures; 15-20 develop page 2. Page 3 is a separate continuation when its tools are ready; allow another 30-45 minutes. Give one page at a time.

Launch and facilitation

Each person marks an allowed interval for a meeting location. Every pair can agree somewhere. Can the entire group agree? Try hard to produce a failure before making a rule.

Student page 1: Try to defeat the group

Core: closed intervals on a line

Each person accepts every point in one closed interval $[left, right]$, including endpoints. A pair agrees if their intervals overlap. The whole group agrees only if one point belongs to every interval.

1 / prompt and solution

Find all common choices for $[0,4]$, $[2,6]$, $[3,5]$ and $[1,7]$. Then choose four new intervals: try to make every pair agree while the whole group disagrees.

Solution: The first common set is $[3,4]$. The requested four-interval counterexample cannot exist. Let L be the greatest left endpoint and U the least right endpoint: pairwise agreement of the two intervals attaining those extremes forces $L \leq U$, and every point of $[L, U]$ belongs to every interval. Let the failed construction precede this proof.

Can every pair agree and everyone disagree?

1 / hints to offer only after exploration

1. Keep the attempted counterexample visible; ask which two intervals would cause trouble.

2 / prompt and solution

A shortcut checks that everyone agrees with one favorite interval. Make three intervals that pass this shortcut but have no common choice.

Solution: Favorite $[0,6]$, with $[0,1]$ and $[5,6]$, passes the favorite check but the two short intervals are disjoint. The shortcut does not check all pairs. Any such explicit example is valid.

2 / hints to offer only after exploration

1. Let the favorite interval be generous.

Student page 2: The smallest certificate

Core continuation: an extremal argument

Use any finite list of nonempty closed bounded intervals. Endpoints need not be integers. You may point to a few decisive intervals instead of checking every pair.

3 / prompt and solution

Explain how to find the entire common interval from just two endpoint extremes. If the common part is empty, show that two of the original intervals already prove failure.

Solution: Set $L = \max$ left endpoints and $U = \min$ right endpoints. All common points must lie between L and U ; every point there is in every interval. Thus the intersection is $[L, U]$ if $L \leq U$ and empty otherwise. When $L > U$, an interval starting at L and one ending at U are disjoint. They are different because each original interval is nonempty.

3 / hints to offer only after exploration

1. Use the latest start and earliest finish.

4 / prompt and solution

Design a family whose only common point is 3. Then explain why sharing a single endpoint counts as agreement and why your proof works even if the two decisive extremes come from the same interval.

Solution: For example $[0,3]$, $[3,6]$, $[2,4]$ has intersection $\{3\}$. Closed intervals include endpoints, so $L=U$ is allowed. If both extremes come from one interval, its nonemptiness already gives $L \leq U$; otherwise pairwise intersection gives this inequality. Every interval contains all of $[L, U]$.

Can every pair agree and everyone disagree?

4 / hints to offer only after exploration

1. A narrow common set does not need positive length.

Student page 3: Change the allowed shapes

Continuation: signed coordinates and inequalities

In the square $-2 \leq x \leq 2$, $-2 \leq y \leq 2$, let A allow $x \geq 0$, B allow $y \geq 0$, and C allow $x+y \leq -1$. Each allowed region is convex: the segment between two allowed points stays allowed.

5 / prompt and solution

Shade the three regions. Mark a witness for each pair's agreement. Prove there is no point accepted by all three. Which feature of the earlier setting changed?

Solution: Pair witnesses: $A \cap B$ contains $(0,0)$; $A \cap C$ contains $(0,-1)$; $B \cap C$ contains $(-1,0)$. All lie in the square. A common point would satisfy $x+y \geq 0$ and $x+y \leq -1$, impossible. The sets remain convex, but the space changed from a line to a plane. Pairwise intersection is insufficient in dimension two.

5 / hints to offer only after exploration

1. Add the first two inequalities.

6 / prompt and solution

Return to a line but permit gaps: allowed sets are $\{0,1\}$, $\{1,2\}$ and $\{0,2\}$. Check every pair and the whole group. State exactly what your interval theorem does and does not promise.

Solution: Pair intersections are $\{1\}$, $\{0\}$ and $\{2\}$, respectively; the triple intersection is empty. These are not intervals, because points between allowed endpoints may be missing. The theorem concerns finite families of nonempty closed bounded intervals; its proof does not cover arbitrary sets or pairwise-intersecting planar convex regions.

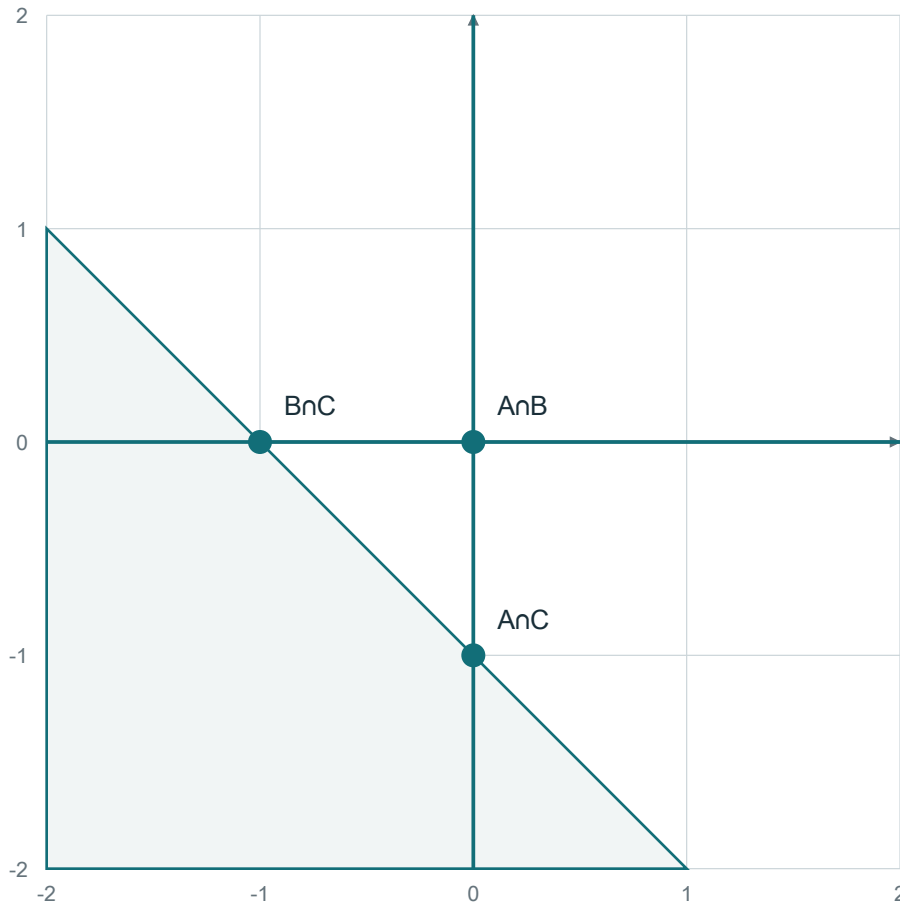
6 / hints to offer only after exploration

1. A different witness for each pair need not be one shared witness.

Can every pair agree and everyone disagree?

Worked diagrams / facilitator only

Three pair witnesses, no triple witness



A allows the right half; B allows the upper half. C is the shaded triangle $x+y \leq -1$ inside the square. Its upper-right boundary stays away from $A \cap B$.

Can every pair agree and everyone disagree?

Optional facilitator extension 1

Gate: Unbounded sets; compactness only for the positive infinite result.

Does the finite conclusion remain true for infinitely many closed intervals? Test $[n, \infty)$ for $n=1,2,\dots$, then contrast closed bounded intervals.

Solution: Every finite subfamily of $[n, \infty)$ intersects, but the whole intersection is empty: no real number is at least every positive integer. These intervals are unbounded. For a pairwise-intersecting family of nonempty closed bounded intervals, fix one interval I . Every finite subfamily together with I intersects by the finite result; compactness of I then gives a common point. This last step is a supplied compactness theorem, not established by drawing.

Mathematics and prior use

This is the one-dimensional finite Helly theorem, proved directly. The planar and nonconvex counterexamples identify why the line argument has restricted scope.

Prior use: New convex-feasibility family. GA-22 supplies a possible later proof tool; no prior worksheet is assumed.

Source: Daniel Hug and Wolfgang Weil, A Course on Convex Geometry

§1.2, Theorem 1.2.2, finite Helly theorem, printed p.22; Theorem 1.2.3, compact families, p.23.. The source supplies background definitions and theorem context. The worksheet instances, learner choices, elementary derivations and answer keys are independently developed. Evidence: Inspected both theorem statements and finite proof. The interval certificates and bounded planar counterexample are independently checked; compactness is explicitly supplied in the optional extension. Checked 2026-09-25.

<https://users.fmf.uni-lj.si/lavric/hug%26weil.pdf>

Predict the cut before using scissors

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate with time for construction and surprises. Insist on a prediction and an explanation after cutting. One impressive demonstration alone misses the mathematics.

A flat seam proof predicting connected pieces, boundary loops and the effect of a center cut.

Prerequisites and preparation

Core: Tracing joined edges and counting circuits. No arithmetic beyond small counts; adult reading can help.

Extensions: Reasoning about a repeated seam permutation for equal-width lanes. Knotting and linking are outside the classification.

Reading and representation: Colored arrows can replace written labels; oral reasoning is sufficient at entry.

Materials: Several long paper strips, tape, scissors, two marker colors and printed seam diagrams. Precut strips if useful. Make a plain band and one with a single half-turn. Preparation: about 12 minutes.

5-10 minutes to handle the materials or choose examples; 15-20 explore page 1; 5 share conjectures; 15-20 develop page 2. Page 3 is a separate continuation when its tools are ready; allow another 30-45 minutes. Give one page at a time.

Launch and facilitation

Build one band with matching ends and one with a half-turn. Trace an edge all the way back to your start. Predict what will happen if each band is cut exactly along its middle.

Student page 1: Two bands, two predictions

Core: construction and tracing

Band P joins the ends of a strip without a turn. Band M joins them after one half-turn. Draw the center line before taping. A boundary loop follows an actual edge until it first returns to its starting point.

1 / prompt and solution

Trace every boundary loop of each band. Then predict how many separate pieces and how many boundary loops each piece will have after a cut exactly along the center line. Record before cutting.

Solution: Before cutting, P is an annulus with two boundary loops; M is a Möbius band with one. Center-cut P gives two pieces, each an annulus with two boundary loops. Center-cut M gives one annulus with two boundary loops. Predictions may be wrong; retain them to explain the surprise.

Predict the cut before using scissors

1 / hints to offer only after exploration

1. Do not count visible edges from only one side of the object.

2 / prompt and solution

Cut, trace and compare. Can one connected piece have two boundary loops? Give evidence that distinguishes “pieces” from “boundary loops.”

Solution: Yes: the uncut plain band and the center-cut Möbius band are each one connected piece with two boundary loops. A boundary trace counts an edge circuit; connected pieces count components of the whole surface. Touching, nesting or linking two separate strips does not join their surfaces.

2 / hints to offer only after exploration

1. Keep a finger on one boundary while a partner tests whether the paper itself is connected.

Student page 2: Explain on a flat strip

Core continuation: follow a seam rule

Name the upper half U and lower half L before the center cut. Plain gluing matches upper to upper and lower to lower. Half-turn gluing matches upper to lower and lower to upper. The cut separates U and L along their long middle edge, but their end identifications remain.

3 / prompt and solution

Draw the seam journeys after cutting P and M. Use these journeys to prove the number of connected pieces without relying on the photograph or physical result.

Solution: P has separate journeys $U \rightarrow U$ and $L \rightarrow L$, hence two components. M has $U \rightarrow L \rightarrow U$, so both halves belong to one component. Every point in either rectangular half connects to its seam, so this accounts for all points. A crossing in the drawing is not an extra paper connection.

3 / hints to offer only after exploration

1. Cross a seam, then continue until the lane name repeats.

4 / prompt and solution

For the cut Möbius band, explain why the connected result has matching ends when viewed as one longer strip, and why it has two boundary loops. An ordinary band with matching ends is called an annulus.

Solution: Following the combined U then L strip crosses two reversing seams. Two reversals restore the transverse direction, so the long strip has matching ends: it is an annulus. Its two long edges therefore each return to themselves and form two boundary loops. This determines its surface type, without claiming it can be unlinked or untwisted in space.

Predict the cut before using scissors

4 / hints to offer only after exploration

1. Track a small arrow across both seam crossings.

Student page 3: A different cut to predict

Continuation: three lanes

Make a fresh half-turn band with two parallel cut lines dividing its width into three equal lanes: upper U, center C and lower L. Predict first, then cut both lines completely. At the seam, U joins L, C joins C and L joins U.

5 / prompt and solution

Predict the connected pieces. For each, decide whether the result is an ordinary band (annulus) or a Möbius band. Track lane names and whether a small arrow across the strip's width reverses when a journey closes.

Solution: There are two components: $U \rightarrow L \rightarrow U$ makes one longer annulus because two seam reversals cancel; $C \rightarrow C$ makes one Möbius band because one reversal remains. The annulus has two boundary loops and the Möbius band one, totaling three. The pieces may be linked in the physical model, but that does not make them one connected surface.

5 / hints to offer only after exploration

1. The center lane and the outer lanes have different journey lengths.

6 / prompt and solution

Invent a claim that confuses linked pieces, connected pieces or boundary loops. Give your partner a seam-based correction using one of the bands you made.

Solution: For example "linked strips are one piece" is false: the three-lane cut has two surface components even when linked. "One piece has one boundary" is false for an annulus. A valid correction names the exact count and follows the corresponding lane or boundary journey; surface adjacency comes only from intact paper or the specified seam.

6 / hints to offer only after exploration

1. Say what is being counted before giving a number.

Predict the cut before using scissors

Worked diagrams / facilitator only

Cut components are cycles of lane names

Cut	Seam cycles	Result
Plain band, center	$U \rightarrow U; L \rightarrow L$	Two annuli
Half-turn, center	$U \rightarrow L \rightarrow U$	One annulus
Half-turn, three lanes	$U \rightarrow L \rightarrow U; C \rightarrow C$	One annulus + one Möbius band

One reversing seam gives a Möbius band. Two reversing seams give an annulus. These classify surfaces; linking in space is a separate question.

Optional facilitator extension 1

Gate: A permutation of lane labels and parity.

Divide a half-turn band into m equal lanes and cut every lane boundary. How many components and which surface types result?

Solution: The seam sends lane i to $m+1-i$ and reverses its transverse direction. Each two-cycle forms an annulus (two reversals). If m is odd, the middle fixed lane forms a Möbius band (one reversal). Thus there are $\text{floor}(m/2)$ annuli plus one Möbius band when m is odd: $\text{ceil}(m/2)$ components and m boundary loops in total. This does not classify their linking.

Mathematics and prior use

The gluing rule defines a quotient surface. Its action on lanes and edge labels determines connectedness and boundary components. Intrinsic surface type is distinct from the object's linking in three-dimensional space.

Prior use: Week 1 used a ribbon to track a combinatorial structure. This investigation instead identifies the edges of a surface and predicts cuts from seam permutations. Week 5 transformations do not by themselves establish these quotient-surface claims.

Source: Allen Hatcher, Algebraic Topology

Chapter 0, printed p.2: Möbius band and core-circle deformation retraction.. The source supplies background definitions and theorem context. The worksheet instances, learner choices, elementary derivations and answer keys are independently developed. Evidence: Original atlas inspected the surface model. This expansion independently derives every cut and boundary count from the displayed seam permutation; it does not attribute these cutting puzzles to Hatcher. Checked 2026-09-25.

<https://pi.math.cornell.edu/~hatcher/AT/AT.pdf>

Where did the starting value go?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced, calculus-gated investigation. A meaningful debugging problem, not merely a transform-table exercise. Allow a second meeting for the integral justification.

An initial value located in the boundary term, two different recovered trajectories, and direct verification in the original equation.

Prerequisites and preparation

Core: Derivatives, integration by parts, improper integrals and elementary exponential algebra.

Extensions: Partial fractions and the integrating-factor argument for uniqueness. No elementary replacement is claimed.

Reading and representation: Symbolic mathematical reading; a shared table reduces memory demand.

Materials: Pencil, printed time graphs and a short transform table. Graphing software is optional; no electronics needed. Preparation: about 8 minutes.

5-10 minutes to handle the materials or choose examples; 15-20 explore page 1; 5 share conjectures; 15-20 develop page 2. Page 3 is a separate continuation when its tools are ready; allow another 30-45 minutes. Give one page at a time.

Launch and facilitation

A system moves toward level one according to $y'=1-y$. Predict its behavior when it starts empty and when it starts at level two. Can an algebraic shortcut preserve that difference?

Student page 1: Two starts, one rule

Core: derivatives and exponential functions

A system follows $y'=1-y$ for $t \geq 0$. Choose a starting value a from -1 , 0 or 2 . Compare with a partner who chose a different start. No transform rule has been assumed yet.

1 / prompt and solution

Predict the initial slope and which level the system moves toward. Choose a candidate of the form $y(t)=1+Ce^{(-t)}$. Determine C and verify the rate rule and starting value.

Solution: The initial slope is $1-a$, and $C=a-1$. Thus $y=1+(a-1)e^{(-t)}$, with $y'=-(a-1)e^{(-t)}=1-y$ and $y(0)=a$. Starts -1 and 0 increase toward 1 ; start 2 decreases toward 1 . No finite positive time reaches 1 unless $a=1$.

Where did the starting value go?

1 / hints to offer only after exploration

1. Use the starting value to choose C before differentiating.

2 / prompt and solution

A shortcut claims that every starting value gives $Y(s)=1/[s(s+1)]$ and hence $y(t)=1-e^{-t}$. Find an exact check that exposes the mistake. What information must an algebraic shortcut preserve?

Solution: At $t=0$ the proposed solution is 0, so it fails any chosen start $a \neq 0$, even though it satisfies the differential equation. The starting value must be retained. A rate equation alone does not select one trajectory.

2 / hints to offer only after exploration

1. Check the initial condition separately from the rate equation.

Student page 2: Recover the missing term

Core continuation: improper integrals and integration by parts

Define $Y(s)=\int_0^\infty e^{-st}y(t)dt$. Work with real $s>0$ for these examples. Supplied table: $L(1)=1/s$, $L(e^{-t})=1/(s+1)$, $L(te^{-t})=1/(s+1)^2$. Linearity follows from integration. Do not assume the derivative rule.

3 / prompt and solution

Integrate $e^{-st}y'(t)$ by parts first from 0 to T. Then let T increase. For your candidate from page 1, justify the boundary limit and derive $L(y')$. Where does a appear?

Solution: On $[0, T]$, the integral is $e^{-sT}y(T)-y(0)+s\int_0^T e^{-st}y(t)dt$. Our y is bounded, so for $s>0$ the first term tends to 0 and all needed integrals converge absolutely. Therefore $L(y')=sY-a$. The lower endpoint contributes the missing $-a$.

3 / hints to offer only after exploration

1. Keep both endpoint terms until the final step.

4 / prompt and solution

Transform $y'+y=1$ while keeping a visible. Recover $y(t)$, then prove that two differentiable solutions with the same initial value must agree.

Solution: The equation is $(s+1)Y-a=1/s$, hence $Y=1/[s(s+1)]+a/(s+1)=1/s+(a-1)/(s+1)$. The table returns $y=1+(a-1)e^{-t}$, already directly verified. For uniqueness, the difference z obeys $z'+z=0, z(0)=0$, so $(e^t z)'=0$ and $z=0$. This avoids relying on an unproved transform-injectivity theorem.

Where did the starting value go?

4 / hints to offer only after exploration

1. Subtract two possible solutions and multiply their difference by e^t .

Student page 3: Change the input

Continuation: transform algebra and a maximum

Now $y' + y = e^{-t}$, $y(0) = 0$. The same transform table applies. First sketch what a fading input might do to the output, then calculate.

5 / prompt and solution

Find the transformed equation and a candidate $y(t)$. Verify both the rate equation and the initial value directly. Does this output keep increasing forever?

Solution: $(s+1)Y = 1/(s+1)$, so $Y = 1/(s+1)^2$ and $y = t e^{-t}$. Its derivative is $(1-t)e^{-t}$, giving $y' + y = e^{-t}$ and $y(0) = 0$. It increases for $0 \leq t < 1$, reaches $1/e$ at $t = 1$ and then decreases toward 0. The difference argument from page 2 proves uniqueness.

5 / hints to offer only after exploration

1. Match the denominator with the supplied table, then differentiate.

6 / prompt and solution

At the peak, compare the input e^{-t} with the loss $y(t)$. Explain the peak using the original rate rule, and show why a transform table alone would not explain the missing starting-value term.

Solution: At $t = 1$ both input and loss equal e^{-1} , so $y' = \text{input} - \text{loss} = 0$. Before that time input exceeds loss; after it loss exceeds input. The table can suggest the candidate, but the derivative identity's boundary calculation is what explains how initial data enters. Initial-value verification remains essential.

6 / hints to offer only after exploration

1. Rewrite the differential equation as input minus loss.

Optional facilitator extension 1

Gate: Calculus and exponential algebra.

Solve $y' + ky = c$, $y(0) = a$, for $k > 0$ and constant c . Identify the steady level and prove uniqueness.

Solution: The solution is $y = c/k + (a - c/k)e^{-kt}$, verified by differentiation and the initial value. Its limit is c/k . The transform equation $(s+k)Y = a + c/s$ gives the same expression for $s > 0$. If two solutions exist, $z' + kz = 0, z(0) = 0$; $(e^{kt}z)' = 0$, hence $z = 0$.

Where did the starting value go?

Mathematics and prior use

The one-sided Laplace transform changes differentiation into algebra only after accounting for its boundary term. Direct time-domain verification and a difference equation establish the complete initial-value solution.

Prior use: New transform family. GA-16 develops convolution geometrically; this offers a later route to transformed differential equations and system response.

Source: Stephen Boyd, EE102 Laplace Transform

Slides 3-19 through 3-21, derivative rule and integration-by-parts derivation; slide 3-32, basic transform table.. The source supplies background definitions and theorem context. The worksheet instances, learner choices, elementary derivations and answer keys are independently developed. Evidence: Inspected the derivative boundary-term calculation and table. All worksheet equations, convergence conditions and uniqueness arguments are independently derived and directly verified. Checked 2026-09-25.

<https://web.stanford.edu/class/ee102k/laplace.pdf>

Infinitely many doublings before the deadline

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced, calculus-gated. The shrinking time gaps offer a vivid launch; the real investigation distinguishes uniqueness from duration of existence.

A verified trajectory with an exact finite blow-up time, and a proof that the other side of an asymptote is not its continuation.

Prerequisites and preparation

Core: Calculus is the core gate: derivatives, rational functions and limits. Fraction work alone is a warm-up.

Extensions: Separation or a reciprocal substitution; local uniqueness is stated explicitly when used.

Reading and representation: Symbolic fractions and rate notation; discussion can precede formal theorem reading.

Materials: Pencil, time strips, value cards and blank graphs. A calculator can check predictions but is not the proof. Preparation: about 8 minutes.

5-10 minutes to handle the materials or choose examples; 15-20 explore page 1; 5 share conjectures; 15-20 develop page 2. Page 3 is a separate continuation when its tools are ready; allow another 30-45 minutes. Give one page at a time.

Launch and facilitation

A quantity grows at a rate equal to its square. Starting at one, could it double over and over before time one? Predict the times before looking at the candidate formula.

Student page 1: Choose a target height

Core: rational functions, derivatives and limits

A quantity obeys $y'=y^2$ and starts at $y(0)=1$. A proposed trajectory is $y(t)=1/(1-t)$, initially for $0 \leq t < 1$. The formula is a candidate to check, not a graph-based guess.

1 / prompt and solution

Choose three target heights greater than 1. Find their arrival times from the candidate formula. Include a very large target, and decide whether any finite target reaches time 1.

Solution: For a chosen height $H > 1$, the arrival time is $t=1-1/H$, strictly between 0 and 1. Examples $H=2, 4, 100$ give $1/2, 3/4, 99/100$. Every finite H arrives before 1; times tend to 1 as H increases without bound. Different targets should be chosen by learners.

Infinitely many doublings before the deadline

1 / hints to offer only after exploration

1. Solve $H=1/(1-t)$ for t .

2 / prompt and solution

Verify the rate rule and starting value. Find the times for heights $2, 4, 8, \dots, 2^n$ and the gaps between consecutive doublings. What do their total times approach?

Solution: $y' = 1/(1-t)^2 = y^2$ and $y(0)=1$. Height 2^n arrives at $t_n = 1 - 2^{-n}$, with $t_0=0$. The n th gap is $t_n - t_{n-1} = 2^{-n}$. Their partial sums are $1 - 2^{-n}$, tending to 1. Infinitely many finite target heights can occur before this finite limiting time.

2 / hints to offer only after exploration

1. Subtract neighboring arrival times rather than adding a long list.

Student page 2: Is the far branch a rescue?

Core continuation: continuity and uniqueness

The same expression $1/(1-t)$ is also defined for $t > 1$. A continuation must be one real-valued differentiable function on a time interval containing the original trajectory and the later times. Supplied local theorem: a continuously differentiable rate law in y gives local uniqueness.

3 / prompt and solution

Check the formula's rate rule on $t > 1$. Then decide whether that branch continues the solution starting at 1. Prove your answer using the behavior near time 1.

Solution: The derivative identity still holds on $t > 1$, but those values form a separate negative branch. As $t \rightarrow 1$ from below, $1/(1-t) \rightarrow +\infty$. A differentiable real-valued continuation would be continuous and finite at $t=1$, which is impossible. Removing $t=1$ leaves a disconnected time domain, not a continuation interval.

3 / hints to offer only after exploration

1. A continuation has to include the intervening time.

4 / prompt and solution

Someone says, "The rate y^2 is finite at every finite y , so the solution must exist forever." Diagnose the claim. How can local uniqueness and finite-time escape both hold?

Solution: Finite rate at each fixed finite state is not a uniform bound on the rate as states grow. The exact solution becomes unbounded, while the rate y^2 is smooth and its y -derivative $2y$ is continuous. Local uniqueness therefore holds near every finite state, but provides only a local time interval; it does not prevent escape to arbitrarily large states by $t=1$.

Infinitely many doublings before the deadline

4 / hints to offer only after exploration

1. Separate a statement near one finite state from a statement for all future times.

Student page 3: Change the start

Continuation: classify positive, zero and negative starts

Use $y(0)=a$ and the candidate $y_a(t)=a/(1-at)$, wherever its denominator is nonzero. Consider forward time $t \geq 0$ only.

5 / prompt and solution

Choose one positive, one zero and one negative a . Verify the formula, find each maximal forward interval and describe its endpoint behavior. Then give the rule for every real a .

Solution: The derivative is $a^2/(1-at)^2 = y_a^2$ and $y_a(0)=a$. For $a > 0$ the maximal forward interval is $[0, 1/a)$, with blow-up to $+\infty$. For $a=0$ the solution is 0 for all $t \geq 0$. For $a < 0$ it exists for all $t \geq 0$ and approaches 0 from below. Local uniqueness for y^2 makes these the initial-value solutions; divergence prevents extending the positive case.

5 / hints to offer only after exploration

1. Look at the sign of a before solving $1-at=0$.

6 / prompt and solution

For a positive solution, differentiate $z=1/y$. Use its simple rule to recover the blow-up time. Compare with $y'=y$ from the same positive start: does every unbounded solution blow up in finite time?

Solution: For $y > 0$, $z' = -y'/y^2 = -1$, hence $z = 1/a - t$ and $y = a/(1-at)$ until $t = 1/a$. In contrast $y' = y$ gives $y = a e^t$, finite for every finite t and unbounded only as $t \rightarrow \infty$. Unbounded growth and finite-time blow-up are different claims.

6 / hints to offer only after exploration

1. The reciprocal turns the nonlinear rule into a constant rate.

Infinitely many doublings before the deadline

Optional facilitator extension 1

Gate: Calculus with real powers on positive values.

For $y' = y^p$ with $a > 0$ and $p > 1$, find and verify the finite blow-up time.

Solution: Differentiate y^{1-p} : its derivative is $(1-p)y^{-p}y' = 1-p$. Thus $y = [a^{1-p} - (p-1)t]^{-1/(p-1)}$ while the bracket is positive. Direct differentiation verifies the equation and initial value. Blow-up occurs at $T = a^{1-p}/(p-1)$. The law is locally Lipschitz on $y > 0$, so uniqueness holds there; the divergence prevents finite continuation through T .

Mathematics and prior use

A smooth locally Lipschitz rate law can have a unique local solution and still lose finite-valued existence in finite time. Reciprocal coordinates expose the mechanism exactly.

Prior use: New counterpart to GA-08: the waiting-time example loses uniqueness at zero, whereas this smooth rate law retains local uniqueness but loses global existence.

Source: Gerald Teschl, Ordinary Differential Equations and Dynamical Systems

§2.2, local uniqueness; Problem 2.9, printed p.42 (PDF p.53), $y' = y^2$ and maximal existence time.. The source supplies background definitions and theorem context. The worksheet instances, learner choices, elementary derivations and answer keys are independently developed. Evidence: Inspected Problem 2.9 and the local theorem context. Exact trajectories, all signs of the starting value and the $p > 1$ extension are independently derived. Checked 2026-09-25.

<https://www.mat.univie.ac.at/~gerald/ftp/book-ode/ode.pdf>

Can four samples fool the average?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising algebra investigation with a substantial calculus continuation. Keep the distinction between four samples and the entire circle visible throughout.

A complete quadratic classification, a counterexample to trusting one four-point test, and a circle-integral proof for calculus-ready learners.

Prerequisites and preparation

Core: Coordinates, signed squares and algebra for the finite identity. Circle integrals require trigonometry and calculus.

Extensions: Second partial derivatives for the Laplacian; the general harmonic mean-value theorem is background, not proved by four samples.

Reading and representation: Coordinates and short formulas; calculus notation only at the advanced gate.

Materials: Pencil, circle-and-grid diagrams, optional compass and calculator. Rotate a four-point cross on a circle; color is optional. Preparation: about 8 minutes.

5-10 minutes to handle the materials or choose examples; 15-20 explore page 1; 5 share conjectures; 15-20 develop page 2. Page 3 is a separate continuation when its tools are ready; allow another 30-45 minutes. Give one page at a time.

Launch and facilitation

Place four equally spaced points on a circle and evaluate $x^2 - y^2$ at each. Does their average know the value at the center? Move the circle and try to break your guess.

Student page 1: Move the sampling cross

Core: coordinates and signed squares

Let $h(x,y)=x^2-y^2$. Start with a circle centered at $(1,2)$ with radius 2. Use the four cardinal points, then choose a different center or radius. The average of four samples means their sum divided by 4.

1 / prompt and solution

Mark the four points and evaluate h there. Compare their average with h at the center. Choose a second circle and try to break the pattern.

Solution: The cardinal points are $(3,2), (-1,2), (1,4), (1,0)$, with values $5, -3, -15, 1$. Their average is $-3 = h(1,2)$. Every second circle also works for cardinal samples: opposite-point linear terms cancel and the $+r^2$ and $-r^2$ contributions cancel.

Can four samples fool the average?

1 / hints to offer only after exploration

1. Pair points on opposite sides before averaging.

2 / prompt and solution

Rotate the four-point cross. Write its displacement vectors as $(u,v), (-v,u), (-u,-v), (v,-u)$, where $u^2+v^2=r^2$. Prove the average is still the center value for every center and rotation.

Solution: At center (a,b) , averaging the displacements gives mean $dx=\text{mean } dy=0$; averaging dx^2 and dy^2 gives $(u^2+v^2)/2$ each. Expanding $(a+dx)^2-(b+dy)^2$ therefore leaves a^2-b^2 . The proof covers every u,v , so no trigonometric calculation is needed for the finite identity.

2 / hints to offer only after exploration

1. Expand first, then group linear and square terms.

Student page 2: Design a test, then fool it

Core continuation: quadratic algebra

Now $q(x,y)=A x^2+B xy+C y^2+D x+E y+K$. Use four quarter-turn-spaced samples on any circle, with any center and starting angle.

3 / prompt and solution

Which coefficients make every such four-sample average equal q at the center? Find a necessary and sufficient condition, not just examples.

Solution: The displacement means are 0, mean $dx^2=\text{mean } dy^2=r^2/2$ and mean $dx dy=0$. Expansion gives average $=q(a,b)+(A+C)r^2/2$. Thus $A+C=0$ is sufficient. It is necessary because the equality for any positive radius forces $A+C=0$. B,D,E,K are unrestricted.

3 / hints to offer only after exploration

1. Your previous expansion can be reused one term at a time.

4 / prompt and solution

The function $p(x,y)=x^2y^2$ passes the cardinal-point test on the unit circle centered at 0. Rotate the sample cross by 45° . Does it still pass? Explain what the first test did not prove.

Solution: At the cardinal points $p=0$, matching $p(0,0)=0$. At the four 45° -rotated points, $x^2=y^2=1/2$, so every p -value and their average equal $1/4$. The first test did not prove rotation independence or a full-circle average identity. p is quartic, outside the quadratic classification.

Can four samples fool the average?

4 / hints to offer only after exploration

1. At a 45° point, the two squared coordinates are equal and add to 1.

Student page 3: Average the whole circle

Calculus continuation: trigonometric integrals

The continuous circle mean is $(1/(2\pi)) \int_0^{2\pi} q(a+r \cos \theta, b+r \sin \theta) d\theta$. It averages every angle equally. This is different from checking four points.

5 / prompt and solution

Compute this mean for a general quadratic q and compare with your finite classification. Then calculate $p=x^2y^2$ on a radius- r circle centered at the origin.

Solution: The means of $\cos \theta$, $\sin \theta$ and $\cos \theta \sin \theta$ are 0; those of $\cos^2 \theta$ and $\sin^2 \theta$ are $1/2$. Thus the quadratic mean is $q(a,b)+(A+C)r^2/2$, so the same exact condition is $A+C=0$. For p , the mean is r^4 times the mean of $\cos^2 \theta \sin^2 \theta = (1-\cos 4\theta)/8$, giving $r^4/8$. On the unit circle it is $1/8$, different from both tested four-sample means.

5 / hints to offer only after exploration

1. Integrate the expansion term by term; use the double-angle identity for the quartic.

6 / prompt and solution

Define the Laplacian $\Delta q = q_{xx} + q_{yy}$. Relate it to the coefficient condition. State precisely which statements you proved, and which general claim about harmonic functions would need a further theorem.

Solution: For this quadratic $\Delta q = 2(A+C)$, so the finite and continuous mean identities for all circles hold exactly when $\Delta q = 0$. We proved the complete quadratic classification and a particular quartic counterexample. The theorem that every C^2 harmonic function has its center value as the mean on every circle whose closed disk lies in its open domain is broader and was not proved by these calculations.

6 / hints to offer only after exploration

1. A polynomial identity for degree two does not classify every smooth function.

Can four samples fool the average?

Worked diagrams / facilitator only

The quartic distinguishes three averages

$p=x^2y^2$ on the unit circle	Average
Four cardinal samples	0
Four samples rotated 45°	$1/4$
Every angle, equally weighted	$1/8$

A single finite sample pattern can miss the function between its points. For quadratics the symmetry calculation proves the exact formula; for this quartic the full-circle calculation uses $\cos^2\theta \sin^2\theta = (1 - \cos 4\theta)/8$.

Optional facilitator extension 1

Gate: Algebraic differences or first derivatives.

For $h=x^2-y^2$, prove directly that no point is a local maximum or local minimum. Compare with $-x^2-y^2$.

Solution: If $a \neq 0$ at (a,b) , a sufficiently small displacement along x with the sign of a increases h , and the opposite displacement decreases it. If $a=0, b \neq 0$, use y displacements instead. At $(0,0)$, x -displacements increase h and y -displacements decrease it. Thus no point is a local extremum. In contrast $-x^2-y^2$ has a strict global maximum at the origin; its Laplacian is -4 , not 0 .

Mathematics and prior use

Quadratic symmetry makes both finite and continuous averages depend on the trace $A+C$. A quartic counterexample exposes finite sampling's limit. The continuous result is an exact harmonic-function example.

Prior use: GA-05 introduced a finite network averaging mechanism. This is the deliberately separate continuous family and does not claim that network experiments prove harmonic-function theorems.

Source: John K. Hunter, Notes on Partial Differential Equations

§2.1, Theorems 2.1-2.2, printed pp.20-21, harmonic mean-value property.. The source supplies background definitions and theorem context. The worksheet instances, learner choices, elementary derivations and answer keys are independently developed. Evidence: Inspected C^2 /open-domain/contained-ball hypotheses. The quadratic classification and quartic sampling counterexample are independent exact calculations, not a proof of the general theorem. Checked 2026-09-25.

https://math.ucdavis.edu/~hunter/pdes/pde_notes.pdf

Will feedback find the point that exists?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising for algebra-ready learners. The shared fixed point of settling and cycling machines gives a compact surprise. Graph play alone is an entry, not the convergence proof.

A fixed-point existence proof and a complete family of examples separating existence, uniqueness and convergence.

Prerequisites and preparation

Core: Fractions, affine algebra and graph reasoning. Continuity and the intermediate value theorem are supplied.

Extensions: Geometric limits for convergence; completeness is discussed through an omitted-endpoint counterexample.

Reading and representation: Graph labels and a short iteration rule; no derivative notation is necessary.

Materials: Pencil, unit-square graphs, iteration record and optional calculator. Use arrows or a partner to feed outputs back into a rule. Preparation: about 7 minutes.

5-10 minutes to handle the materials or choose examples; 15-20 explore page 1; 5 share conjectures; 15-20 develop page 2. Page 3 is a separate continuation when its tools are ready; allow another 30-45 minutes. Give one page at a time.

Launch and facilitation

A machine turns a number between zero and one into another number in that interval. Can some input come out unchanged? If you repeatedly feed the output back in, must you find it?

Student page 1: Build a feedback experiment

Core: fractions and graphing

A machine takes x in $[0,1]$ and returns $f(x)$, also in $[0,1]$. Feedback repeatedly uses the output as the next input: $x_{(n+1)}=f(x_n)$. A fixed point is an input that comes out unchanged.

1 / prompt and solution

Choose a starting value and compare the machines $f(x)=(x+1)/3$ and $g(x)=1-x$. Record four updates. Find each machine's fixed points, then choose a start that makes their feedback behavior disagree.

Solution: Both have the unique fixed point $1/2$. From 0, f gives $1/3, 4/9, 13/27, 40/81$, approaching $1/2$; g gives $1, 0, 1, 0$. Any start other than $1/2$ produces the same settling-versus-two-cycle contrast. At $1/2$ both stay fixed.

Will feedback find the point that exists?

1 / hints to offer only after exploration

1. The equation $f(x)=x$ and the repeated process are different questions.

2 / prompt and solution

For f , derive an exact formula for the error after n updates from any start x_0 . How many updates guarantee error below $1/100$ for every start in $[0,1]$? Does g have a similar decreasing bound?

Solution: For f , $x_n - 1/2 = (x_0 - 1/2)/3^n$, so the largest absolute error is $1/(2 \cdot 3^n)$. Four updates give $1/162 < 1/100$; three allow $1/54$, so four are necessary and sufficient for this guarantee. For g the error changes sign but keeps its magnitude; it does not decrease unless initially zero.

2 / hints to offer only after exploration

1. Subtract the fixed point before iterating.

Student page 2: Prove existence without promising success

Core continuation: a supplied continuity theorem

Supplied theorem: if a continuous real function has values of opposite signs at two endpoints, it has a zero between them. A zero at an endpoint already counts. Now f is any continuous function from the closed interval $[0,1]$ into itself.

3 / prompt and solution

Try to draw a continuous graph inside the unit square that never meets the diagonal $y=x$. Then prove why no such graph exists, including the endpoint cases.

Solution: Let $d(x)=f(x)-x$. It is continuous, $d(0)=f(0)-0 \geq 0$, and $d(1)=f(1)-1 \leq 0$. If either is 0, that endpoint is fixed. Otherwise the signs are opposite and the supplied theorem gives a zero of d in $(0,1)$. A graph alone suggests but does not establish this proof.

3 / hints to offer only after exploration

1. Compare $f(x)$ with x at the two ends.

4 / prompt and solution

Give one continuous interval machine with many fixed points and one with a unique fixed point whose feedback need not converge. Which claims did the existence proof leave open?

Solution: The identity $f(x)=x$ fixes every point in $[0,1]$. Reflection $g(x)=1-x$ has the unique fixed point $1/2$ but cycles from any other start. The existence proof supplies neither uniqueness nor convergence of repeated feedback.

Will feedback find the point that exists?

4 / hints to offer only after exploration

1. Reuse a machine already tested if it meets the exact claim.

Student page 3: A whole family of machines

Continuation: powers, signs and limits

For $-1 \leq q \leq 1$, define $f_q(x) = 1/2 + q(x - 1/2)$. All these machines map $[0, 1]$ into itself. The parameter q can be chosen by the learner.

5 / prompt and solution

Classify the fixed points and feedback behavior for every allowed q . Explain exactly when errors alternate, shrink, stay unchanged or cycle. Include $q = -1, 0, 1$.

Solution: The error is $e_n = q^n e_0$. For $-1 < q < 1$ all starts converge to the unique fixed point $1/2$; $q=0$ reaches it in one update, negative q alternates signs for nonzero errors, and positive q preserves the side while shrinking. At $q=-1$, $1/2$ is uniquely fixed and every other start cycles with period two. At $q=1$ every point is fixed and feedback remains at its initial value.

5 / hints to offer only after exploration

1. First solve $(1-q)(x-1/2)=0$; then use the error recurrence.

6 / prompt and solution

Change the domain to the open interval $(0, 1)$ and use $h(x) = x/2$. It halves every distance. Does it have a fixed point in this domain? Explain why this does not contradict your closed-interval proof.

Solution: Solving $x = x/2$ gives $x=0$, excluded from $(0, 1)$. From any allowed start, $x_n = x_0/2^n$ tends to that omitted endpoint, so there is no fixed point in the domain. The earlier existence argument requires both endpoints in the domain; the general contraction theorem additionally requires a complete space, which $(0, 1)$ is not.

6 / hints to offer only after exploration

1. Check whether the limiting value belongs to the stated domain.

Will feedback find the point that exists?

Optional facilitator extension 1

Gate: Piecewise definitions and an exact counterexample.

Can discontinuity destroy fixed-point existence on the closed interval? Construct a self-map of $[0,1]$ with no fixed point.

Solution: Define $f(x)=1$ for $0 \leq x < 1/2$ and $f(x)=0$ for $1/2 \leq x \leq 1$. In the first region $f(x) > x$; in the second $f(x) < x$, including $x=1/2$. Thus there is no fixed point, and the jump at $1/2$ breaks continuity. All outputs still lie in $[0,1]$.

Mathematics and prior use

The interval fixed-point theorem is a direct topological existence result. Exact affine error recurrences reveal the extra mechanism needed for iterative convergence and the role of the domain.

Prior use: New fixed-point synthesis. GA-02 and GA-28 also use continuity; GA-10 studies iteration through symbolic orbits. Return visits should foreground a different question, not repeat the same calculation.

Source: Jiří Lebl, Basic Analysis I

§3.3, intermediate value theorem; §7.6, Theorem 7.6.2, printed p.296; Exercises 7.6.4 and 7.6.6, pp.299-300 (version 6.3).. The source supplies background definitions and theorem context. The worksheet instances, learner choices, elementary derivations and answer keys are independently developed.

Evidence: Inspected the complete-space contraction hypotheses and exercises. Affine-family classification, quantitative errors and discontinuous counterexample are independently derived. Checked 2026-09-25.

<https://www.jirka.org/ra/realanal.pdf>